

Dong-A Univ. (ISPL)



동아대학교
DONG-A UNIVERSITY

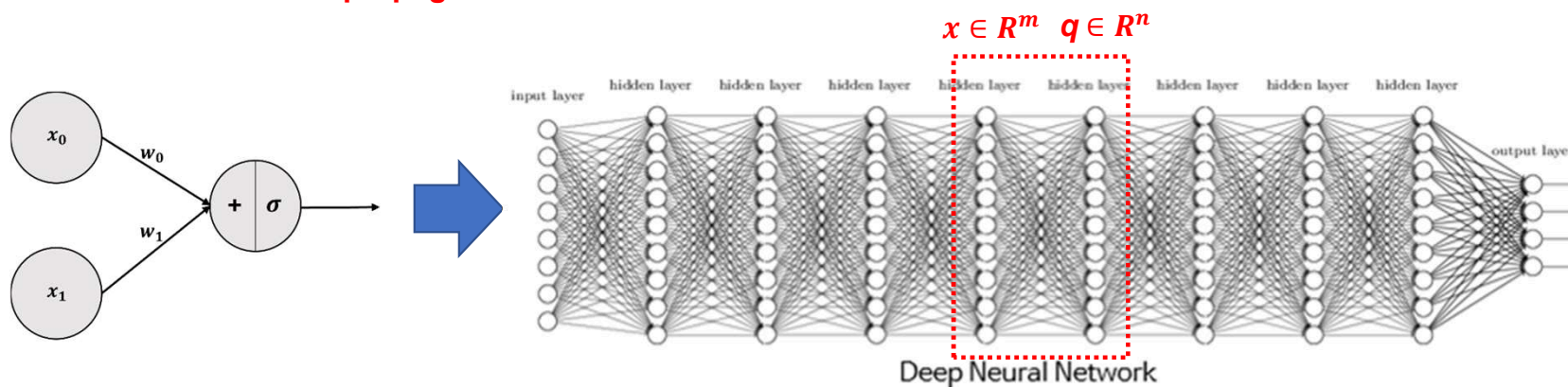
Backpropagation (역전파)

컴퓨터공학부
2024년 1학기 인공지능

Review: MLP by Linear Algebra

- SLP → MLP (One of the Deep Neural Network: DNN)

→ 행렬에 대한 Forward propagation



$$W \in R^{m \times n} = \begin{pmatrix} w_{11} & \cdots & w_{1n} \\ \vdots & \ddots & \vdots \\ w_{m1} & \cdots & w_{mn} \end{pmatrix}, \quad W^T \in R^{n \times m} = \begin{pmatrix} w_{11} & \cdots & w_{m1} \\ \vdots & \ddots & \vdots \\ w_{1n} & \cdots & w_{mn} \end{pmatrix}, \quad x \in R^m = \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix}, \quad b \in R^n = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$$

$$q = \sigma(W^T \cdot x + b) = \sigma \left\{ \begin{pmatrix} W_{1,1}x_1 + \cdots + W_{m,1}x_m \\ \vdots \\ W_{1,n}x_1 + \cdots + W_{m,n}x_m \end{pmatrix} + \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \right\} = \begin{pmatrix} q_1 \\ \vdots \\ q_n \end{pmatrix} \in R^n$$

Introduction

- Backpropagation(역전파)
- 모험가 이야기: 어떤 모험가는 여행을 하며 가장 깊고 낮은 골짜기를 찾아가려 한다.
 - ✓ 조건 1: 지도를 보지 않을 것
 - ✓ 조건 2: 눈가리개를 쓰는 것
 - 어떻게 모험가는 가장 깊고 낮은 골짜기를 찾아 갈 수 있을까?



GD Method (역전파를 위한 최적화: Optimization for Backpropagation)

Gradient Descent Method (GD Method / 경사하강법)

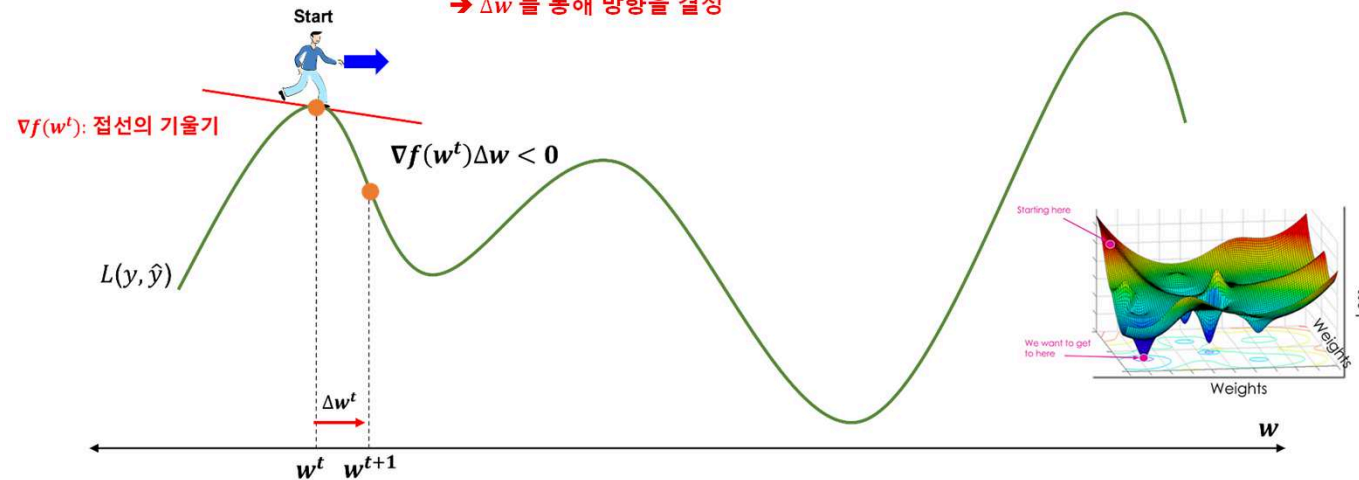
학습률(learning rate, step size)

$$w^{t+1} \leftarrow w^t - \mu \nabla f(w^t), \text{ where } \nabla f(w^t) \Delta w < 0$$

탐색 방향

하강법에서 접선의 기울기와 w 는 항상 다른 부호를 가짐

→ Δw 를 통해 방향을 결정



Contents

1. Introduction

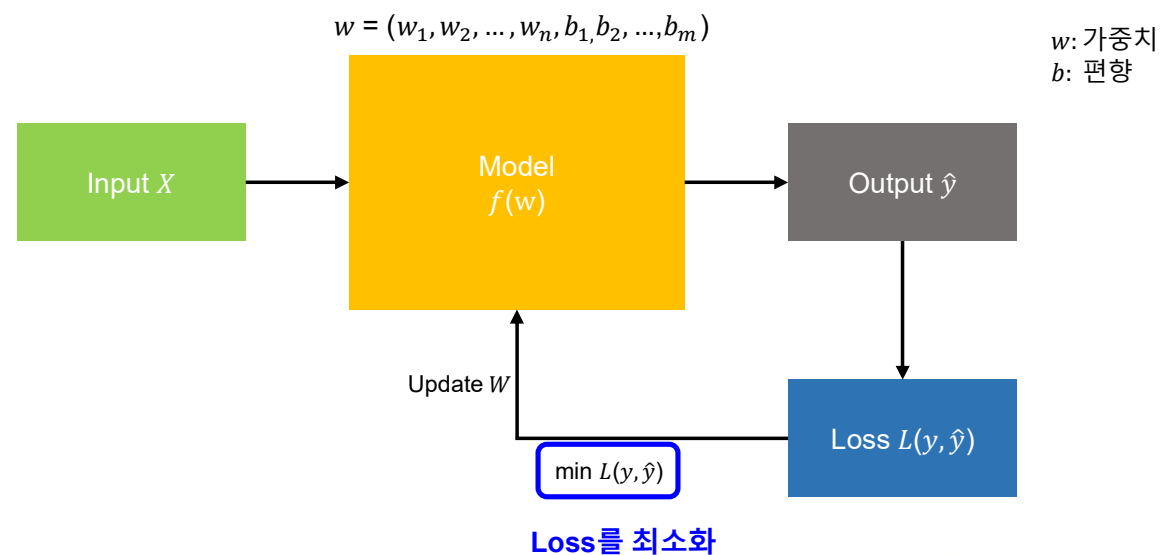
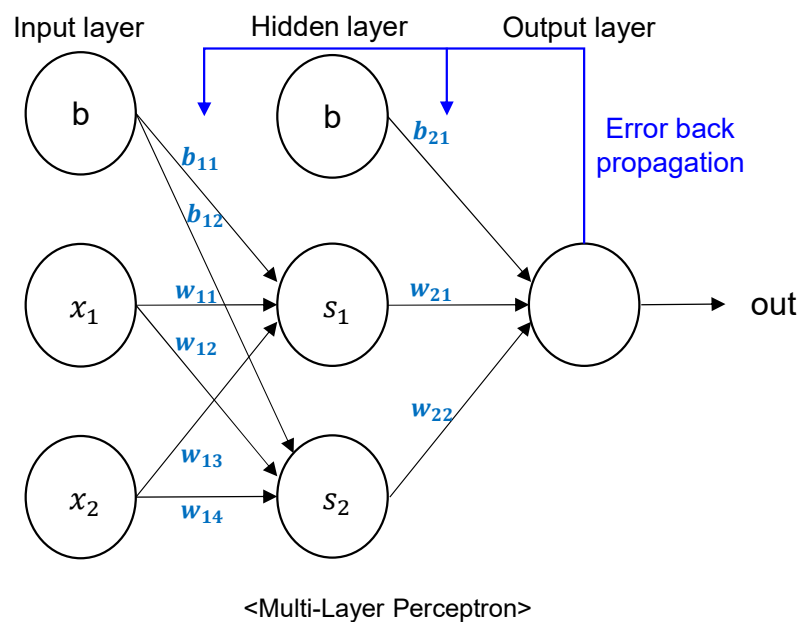
2. Concept of Backpropagation

: Gradient Descent(GD) Method (경사하강법)



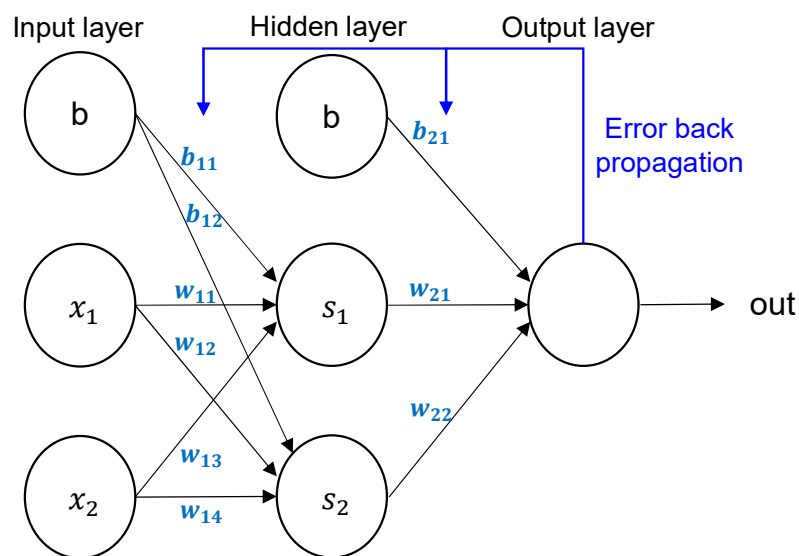
Introduction

- Backpropagation: 계산 결과와 정답의 오차를 구해서 오차에 관여하는 노드 값들의 가중치와 편향을 수정하는데, **오차가 작아지는 방향으로 반복해서 수정하는 기법**



Introduction

- Backpropagation: 계산 결과와 정답의 오차를 구해서 오차에 관여하는 노드 값들의 가중치와 편향을 수정하는데, 오차가 작아지는 방향으로 반복해서 수정하는 기법



<Multi-Layer Perceptron>

- ✓ Parameter (w , b) 업데이트: per Mini-batch or Epoch (Iteration)
- ✓ Iteration 수가 늘어나면 정확성을 증가됨 → 시간多
- ✓ Iteration 수가 줄어들면 정확성을 떨어짐 → 시간小

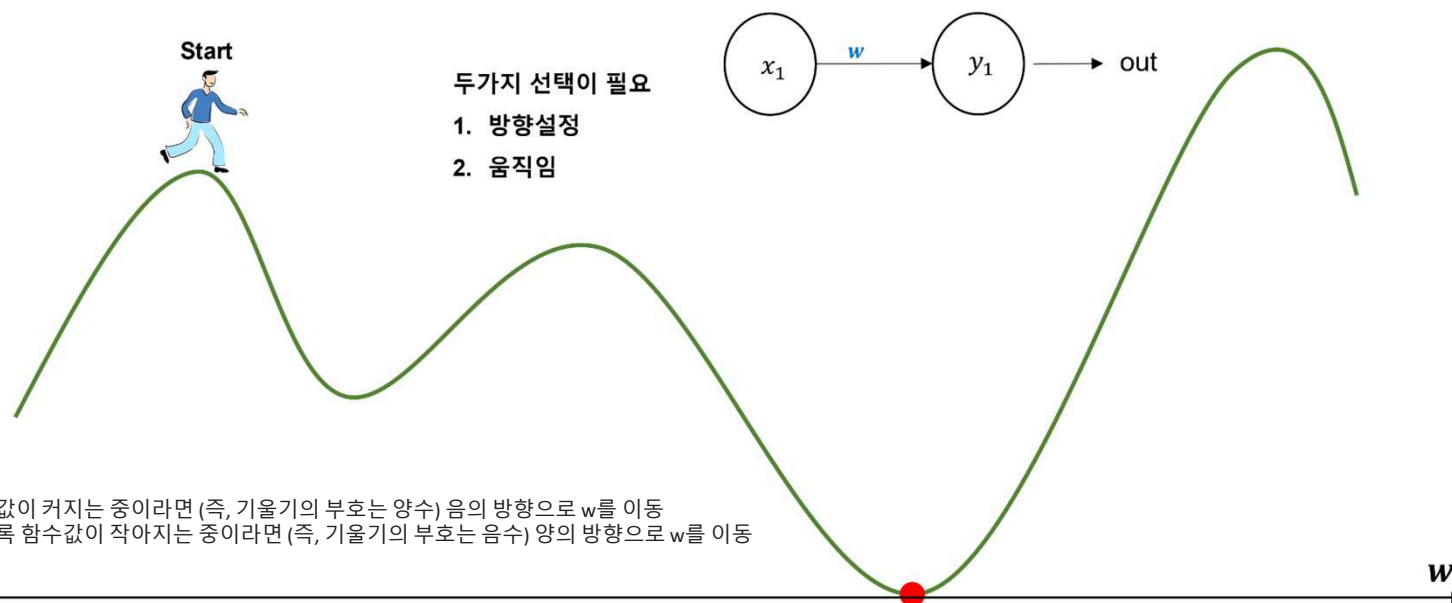
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➤ 어떻게 모험가는 가장 깊고 낮은 골짜기를 찾아 갈 수 있을까?



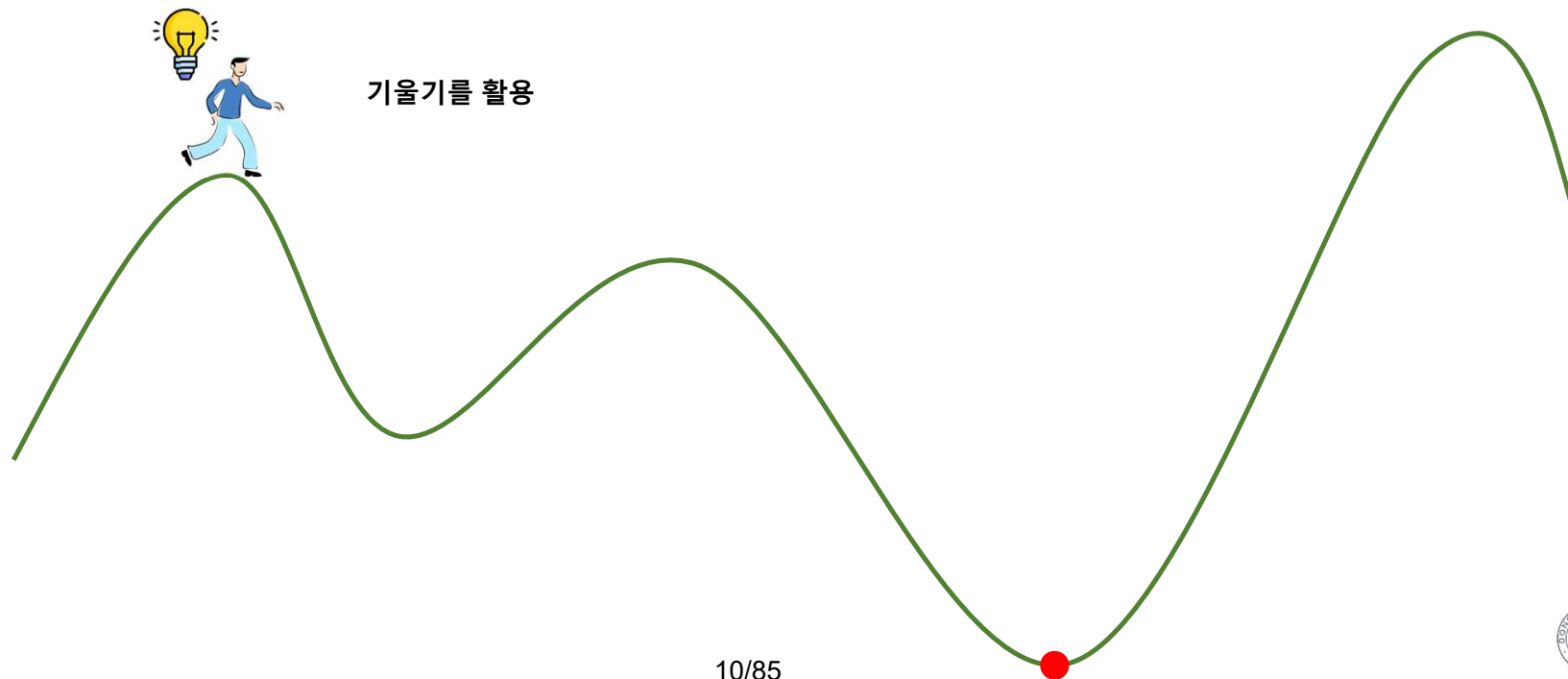
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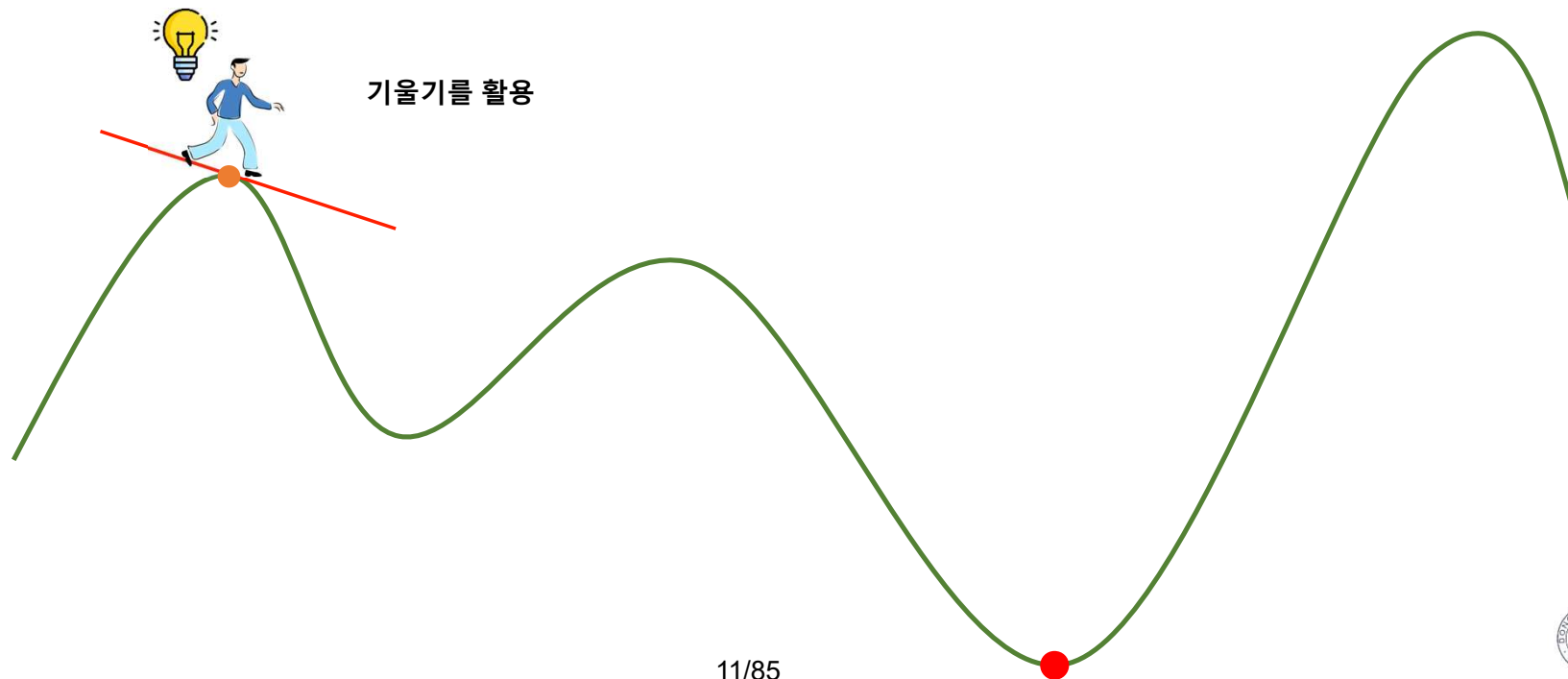
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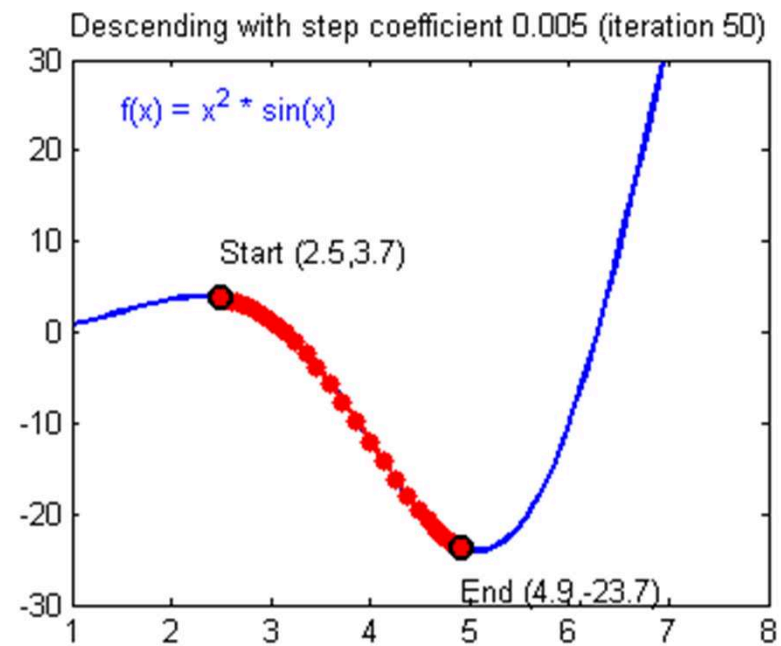
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이동거리에 사용할 값을 gradient의 크기와 비례하는 factor를 이용하면 현재 xx의 값이 극소값에서 멀 때는 많이 이동하고, 극소값에 가까워졌을 때는 조금씩 이동할 수 있게 된다.

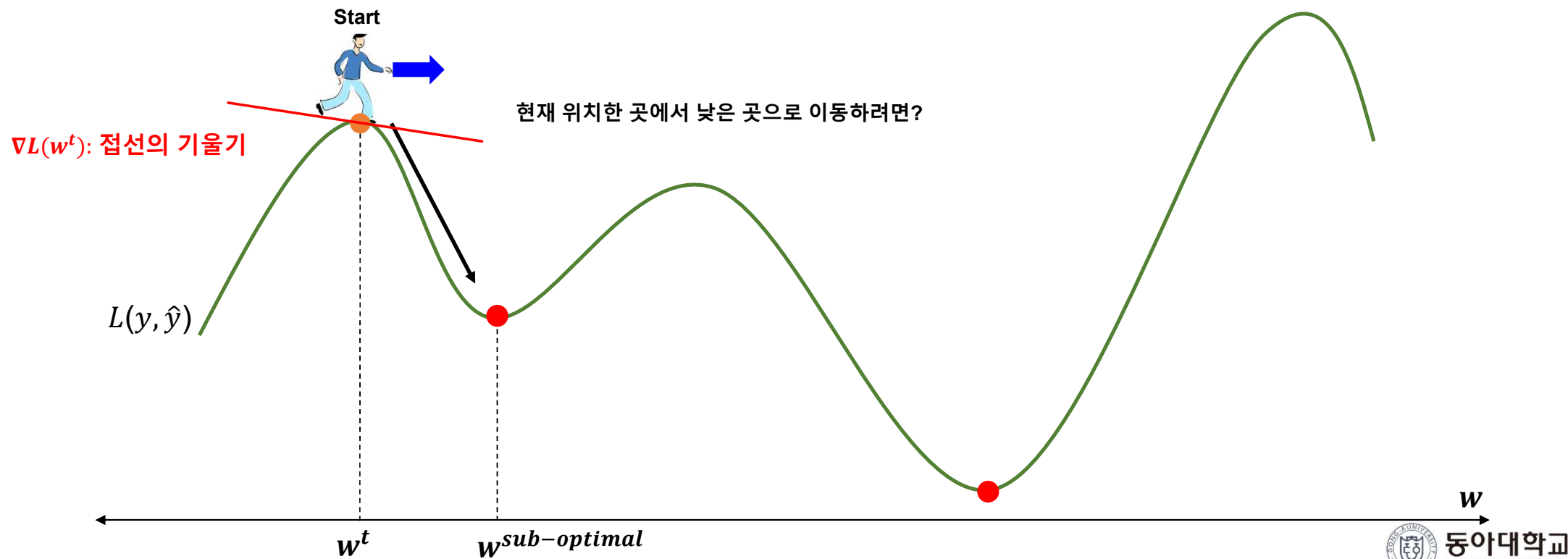
Contents

1. Introduction
2. Derivative Review (미분/편미분/체인룰)
3. Concept of Backpropagation
: Gradient Descent(GD) Method (경사하강법)



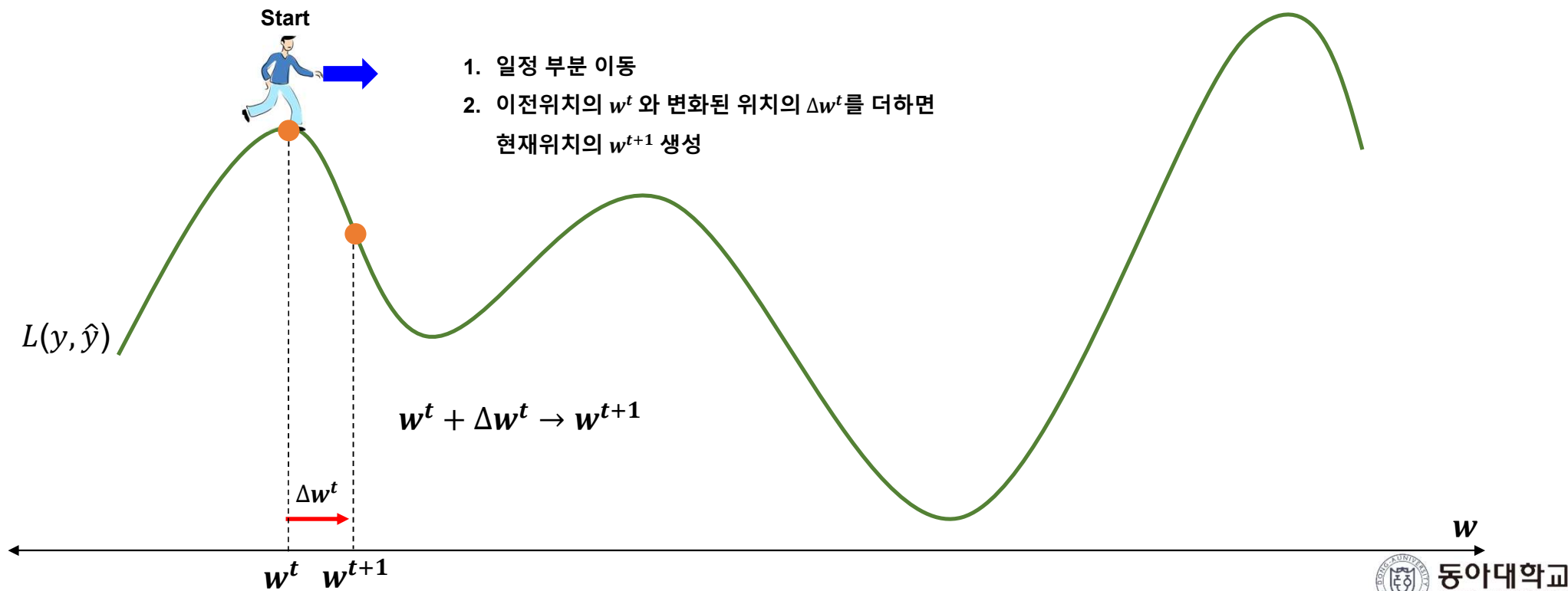
Descent Method(하강법)

- 주어진 어떤 지점에서부터 오차가 더 작은 곳으로 이동하려는 방법



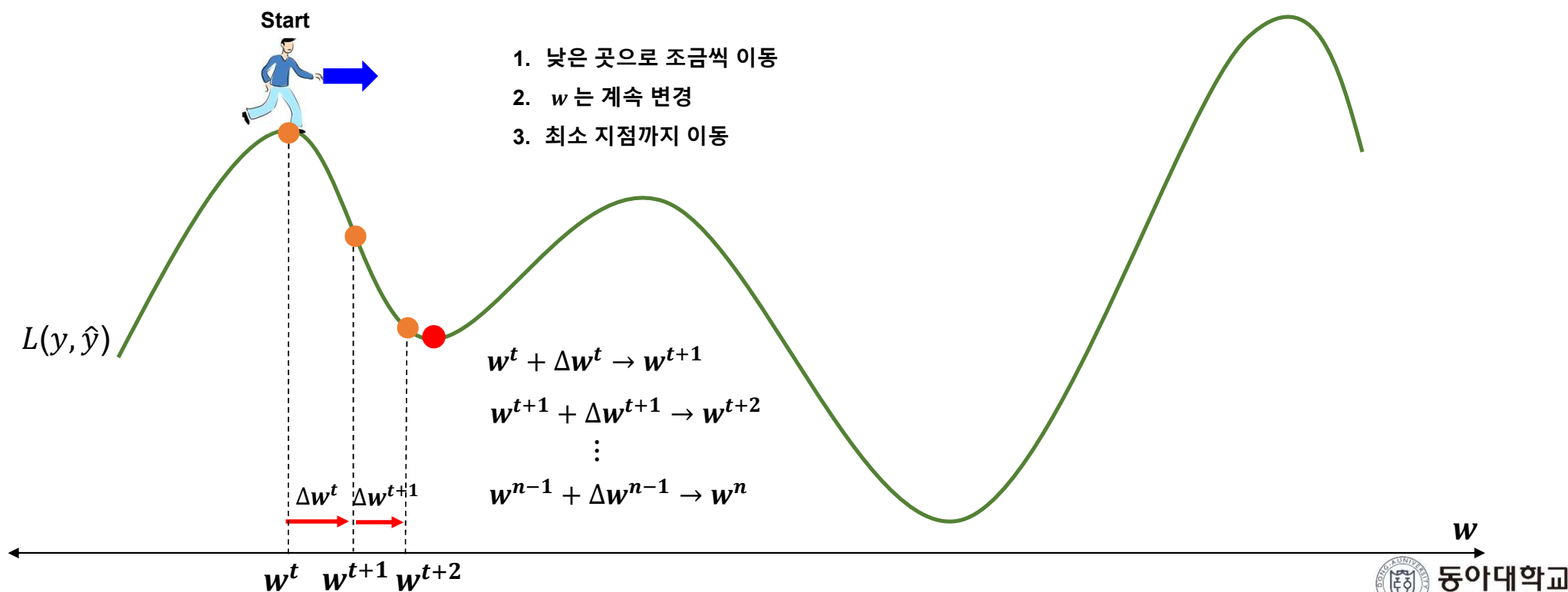
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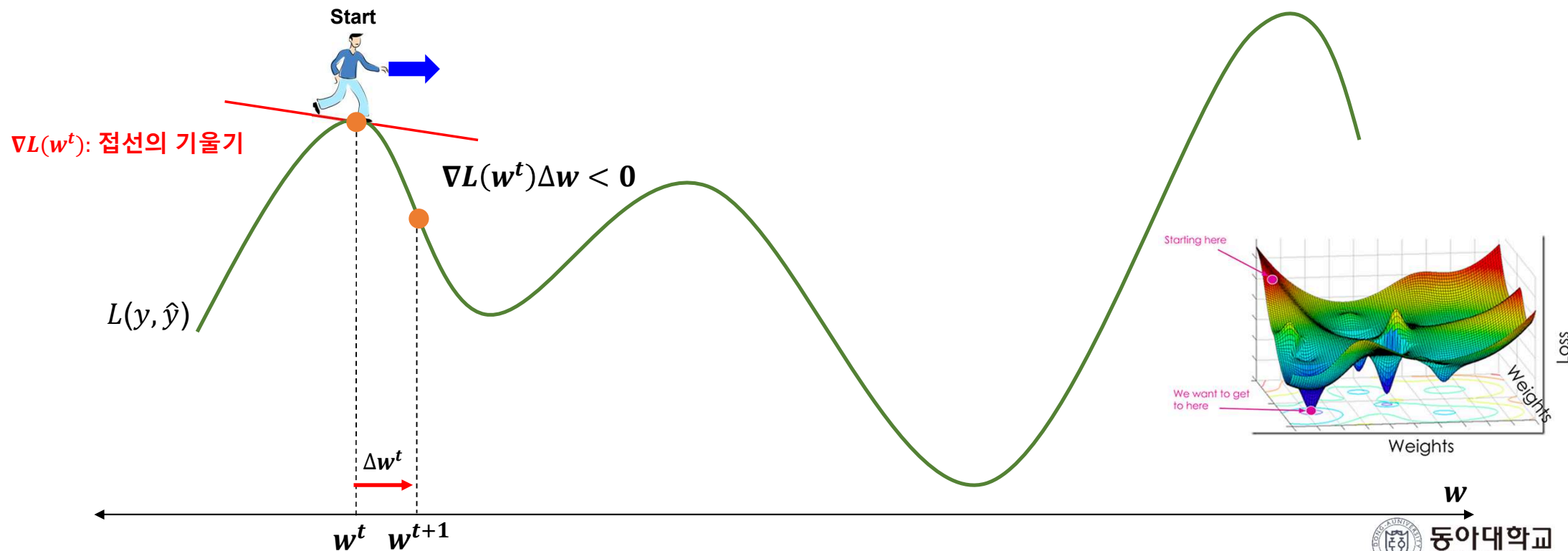
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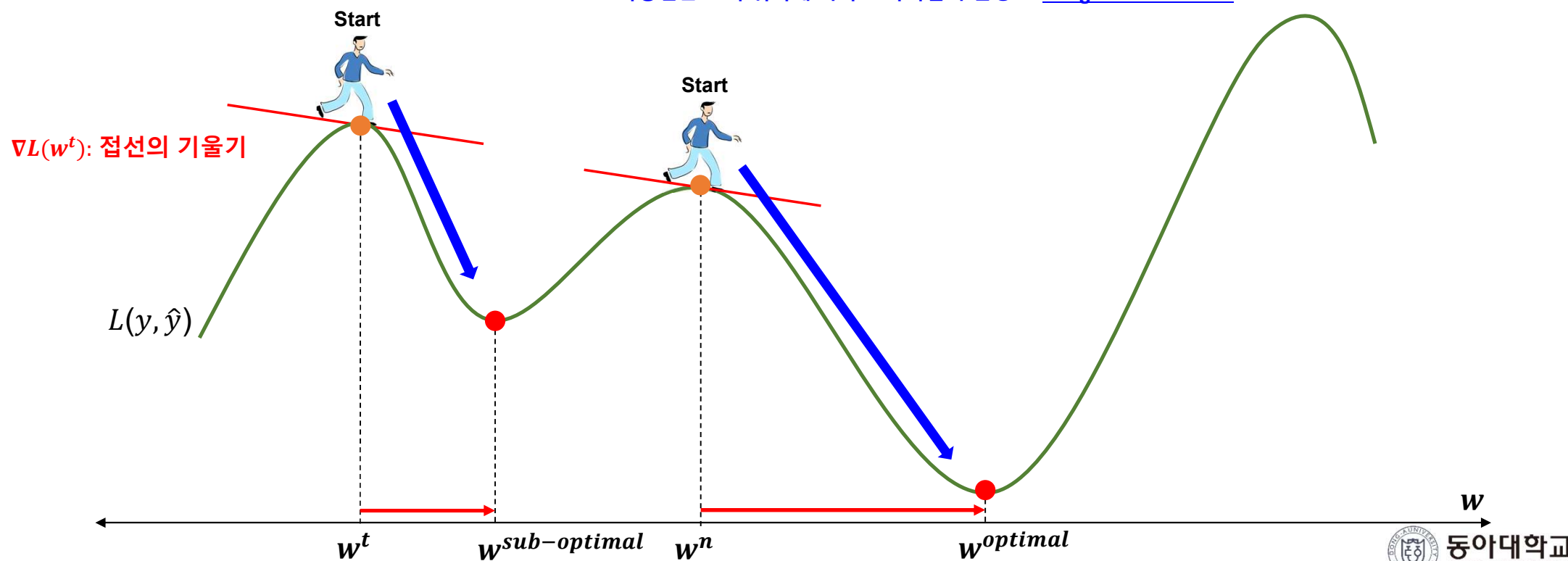


Descent Method(하강법)

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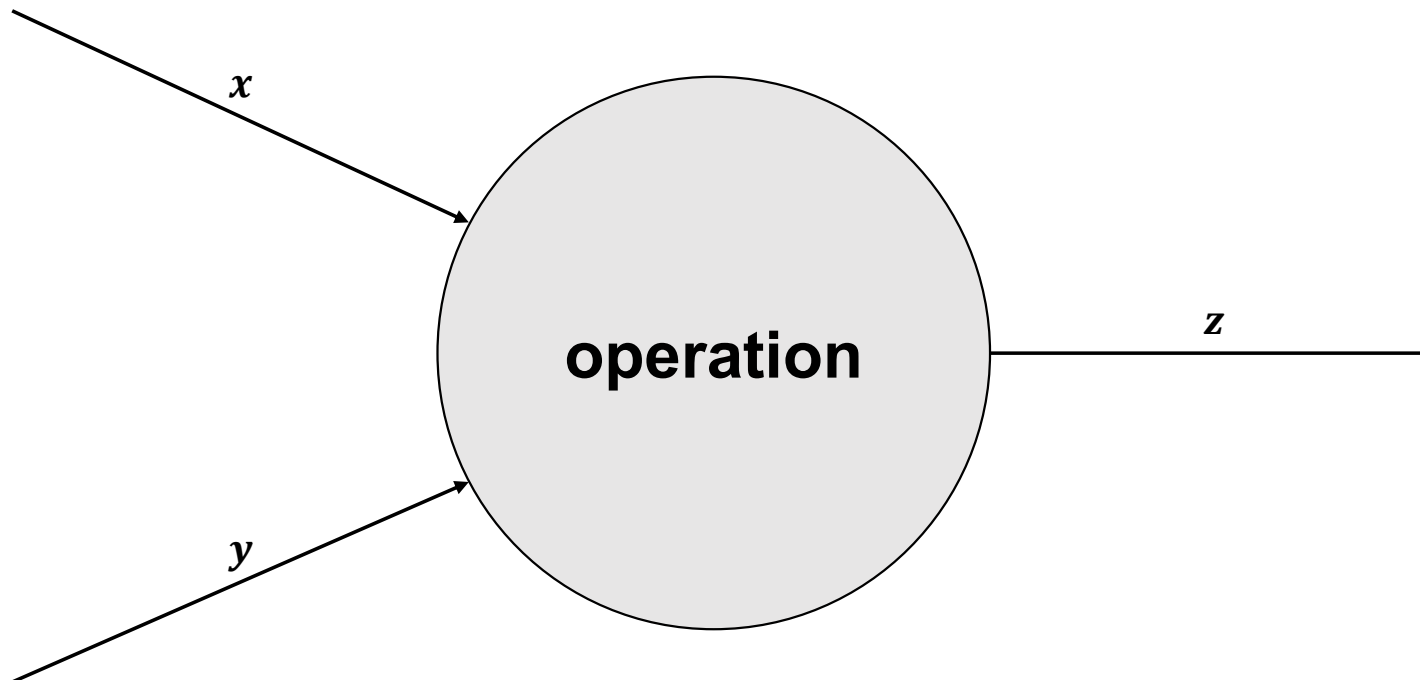
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2. 하강법은 초기 위치에 따라 도착지점이 변경 $\rightarrow$ [Weight Initialization](#)



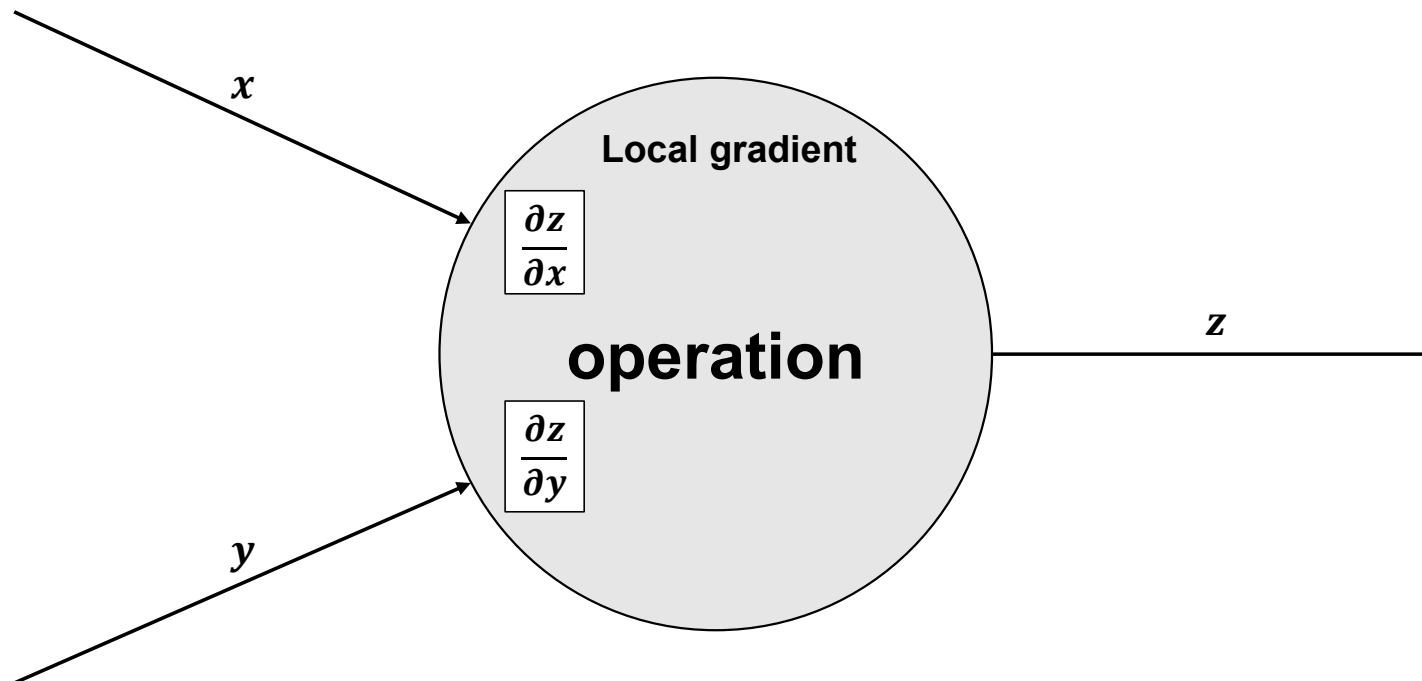
Backpropagation 동작원리

- Chain rule



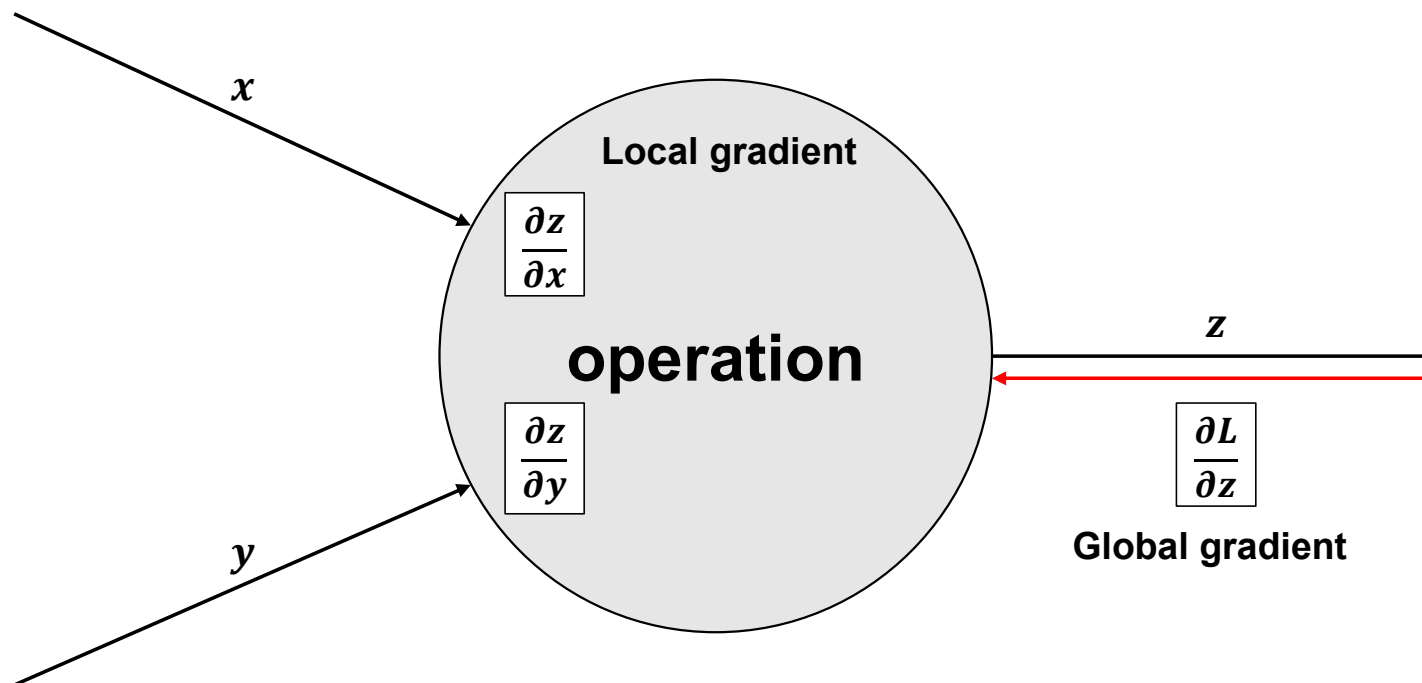
Backpropagation 동작원리

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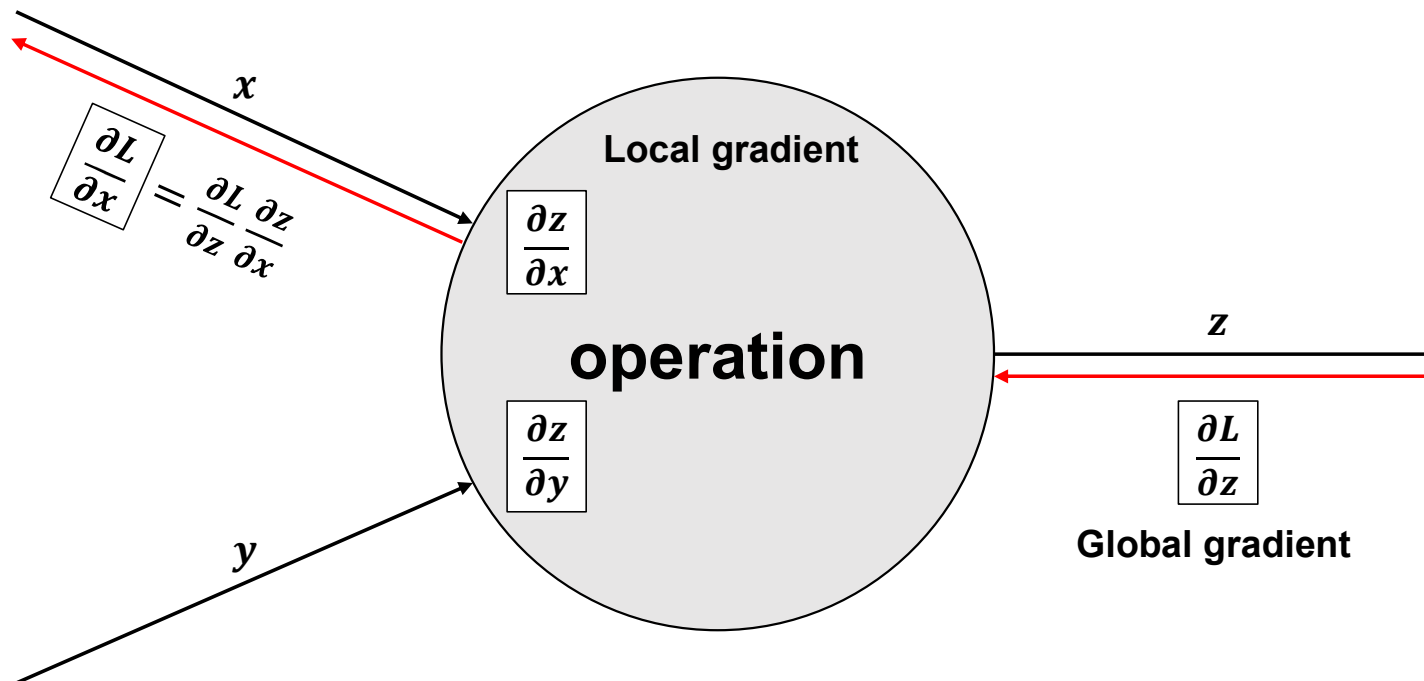
Backpropagation 동작원리

▪ Chain rule



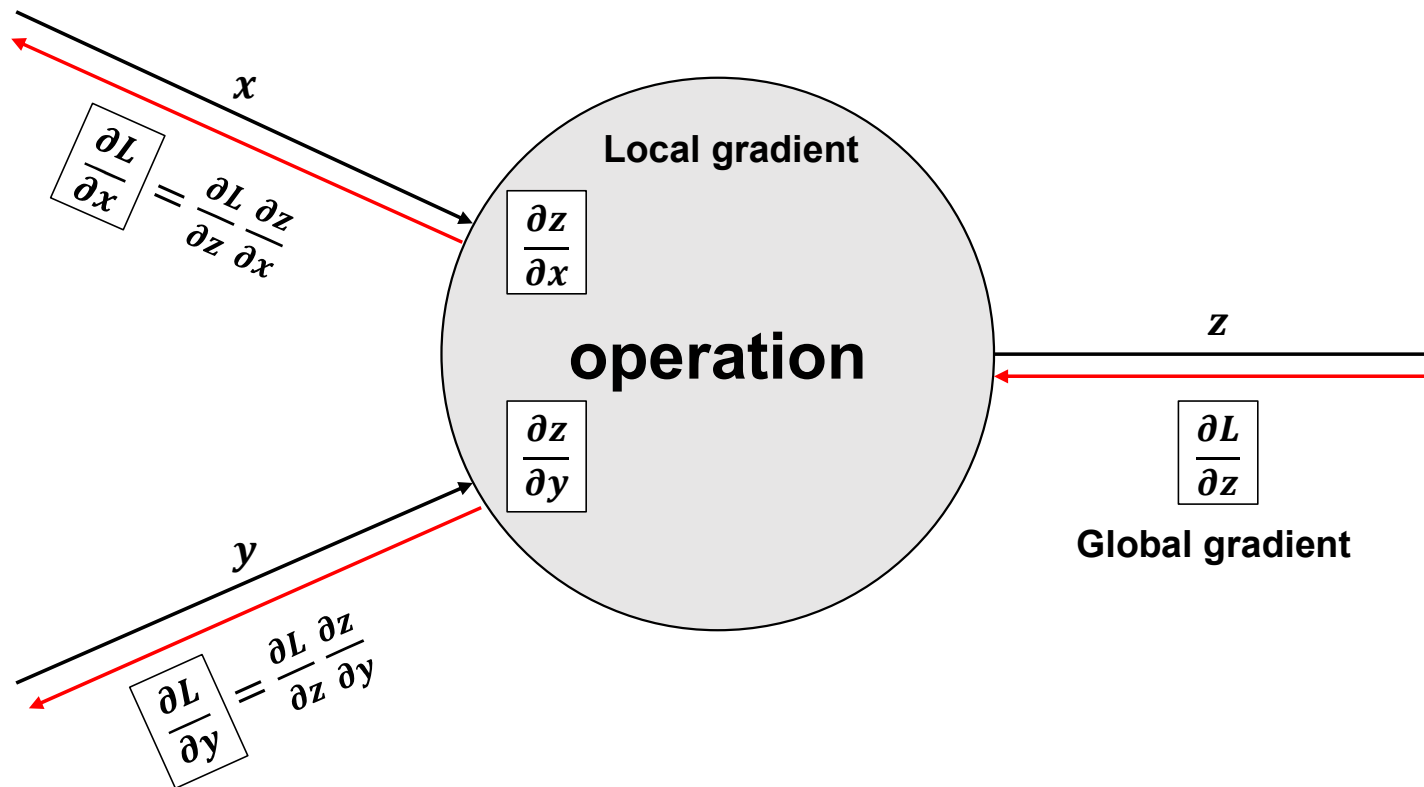
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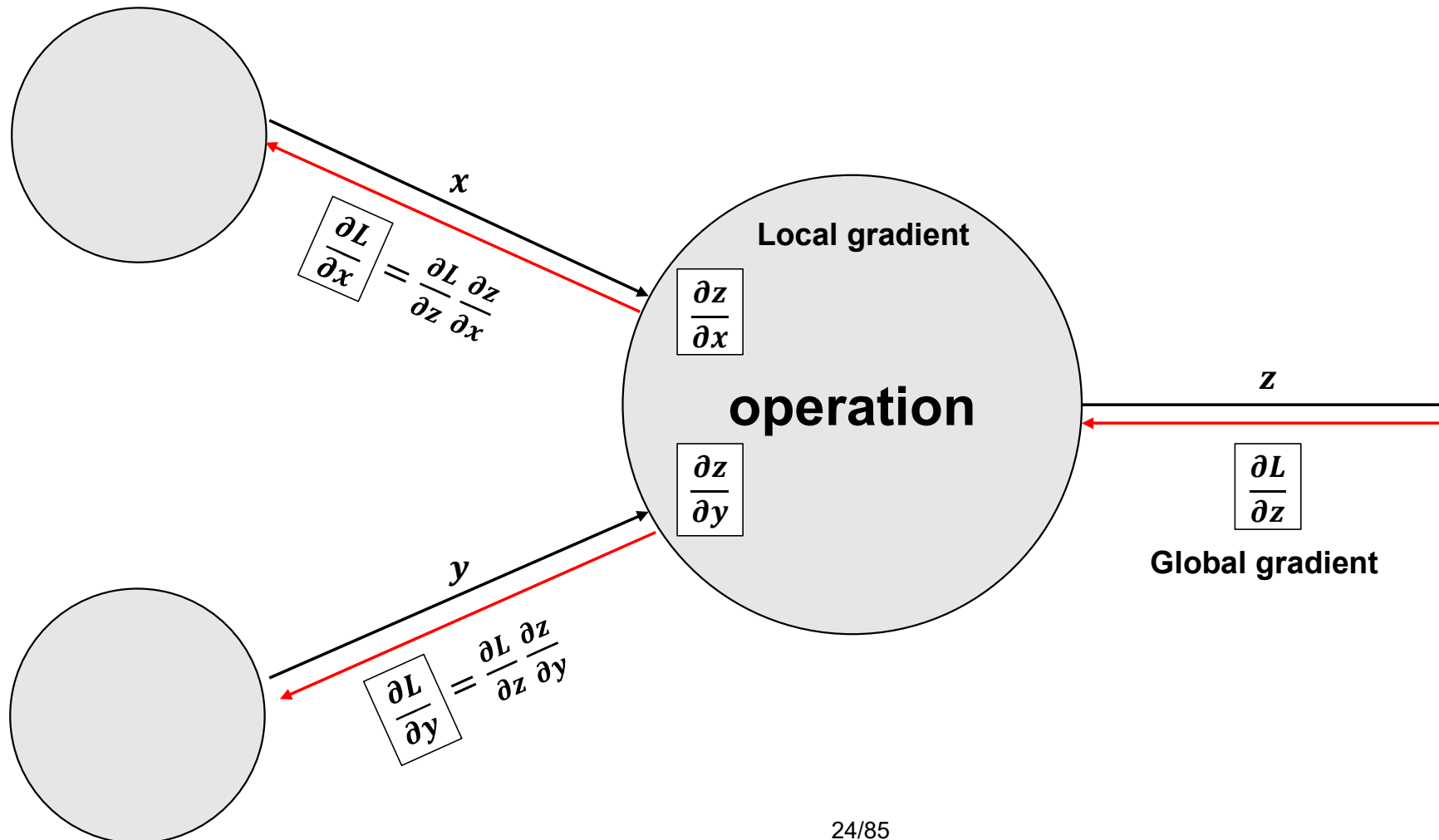


Backpropagation 동작원리

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Backpropagation 동작원리



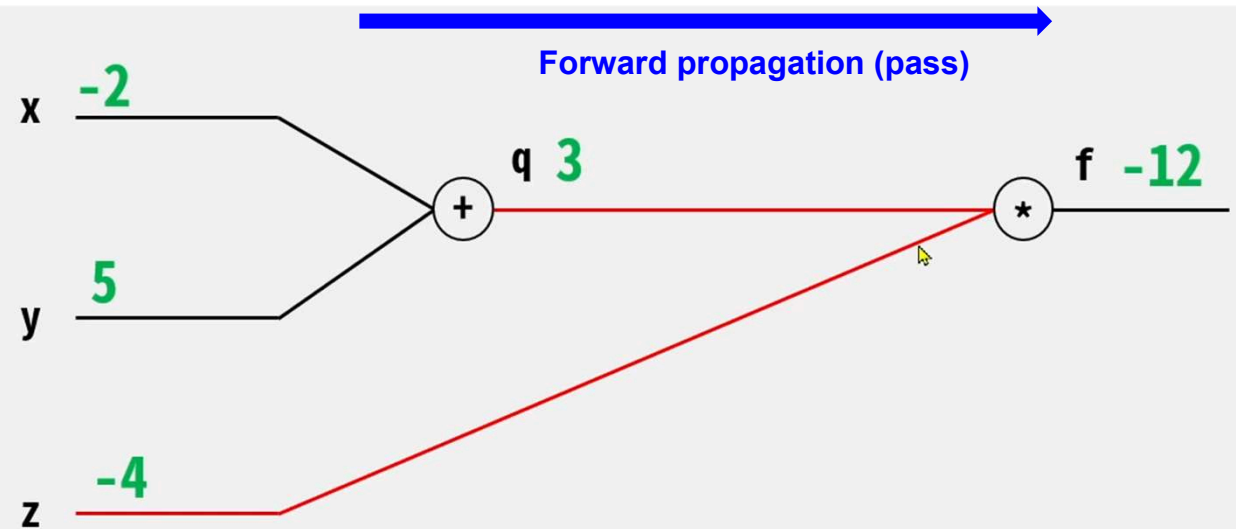
Backpropagation 동작원리

$$f(x, y, z) = (x + y)z$$

$$q = x + y$$

$$f = qz$$

$$x=-2, y=5, z=-4$$



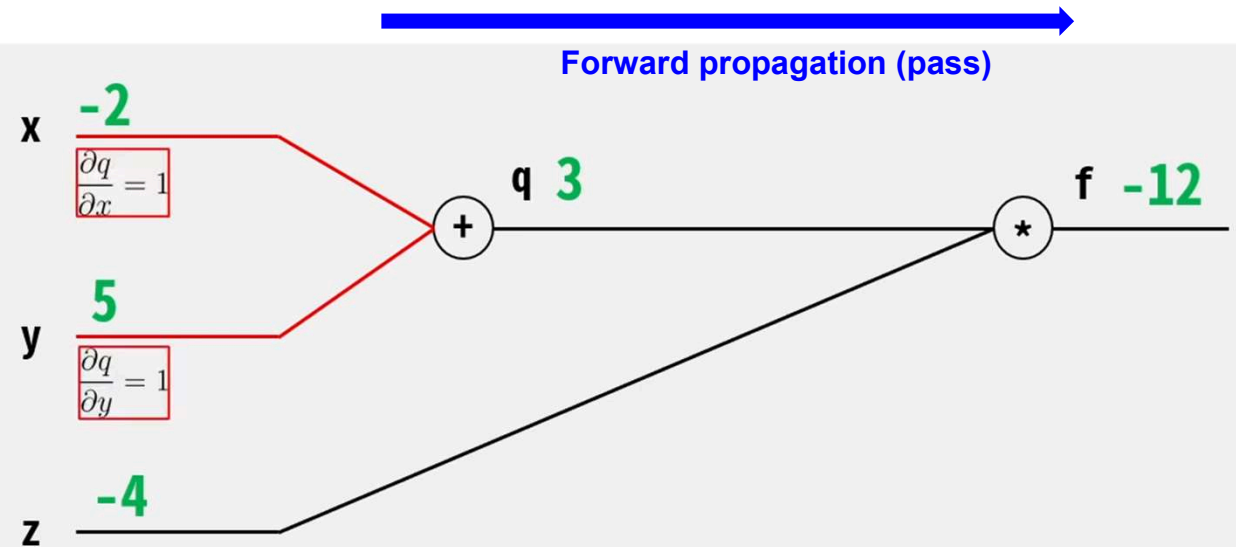
Backpropagation 동작원리

$$f(x, y, z) = (x + y)z$$

$$x = -2, y = 5, z = -4$$

$$q = x + y$$

$$\frac{\partial q}{\partial x} = 1 \quad \frac{\partial q}{\partial y} = 1$$



Backpropagation 동작원리

$$f(x, y, z) = (x + y)z$$

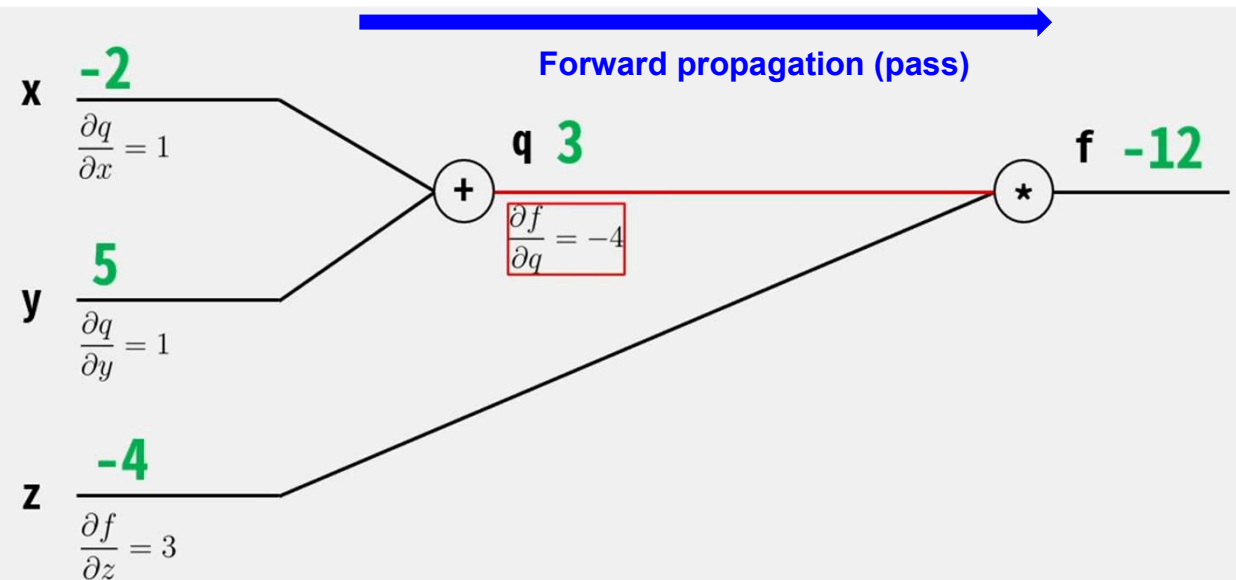
$$x=-2, y=5, z=-4$$

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$$\frac{\partial f}{\partial q} = z = -4$$

$$\frac{\partial f}{\partial z} = q = (x + y) = (-2 + 5) = 3$$



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$$f(x, y, z) = (x + y)z$$

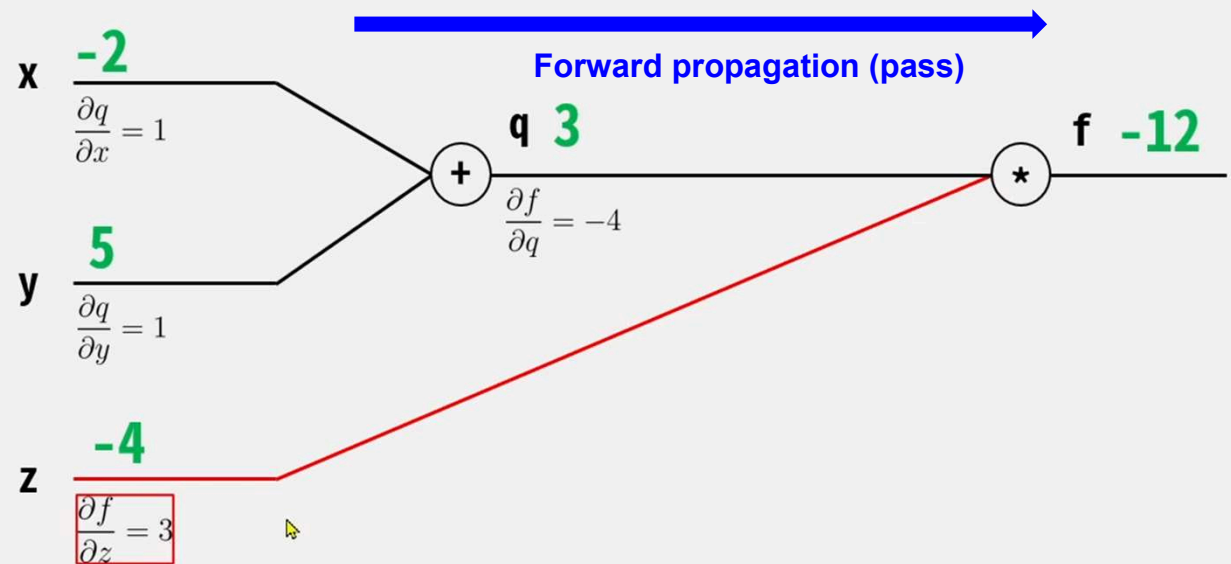
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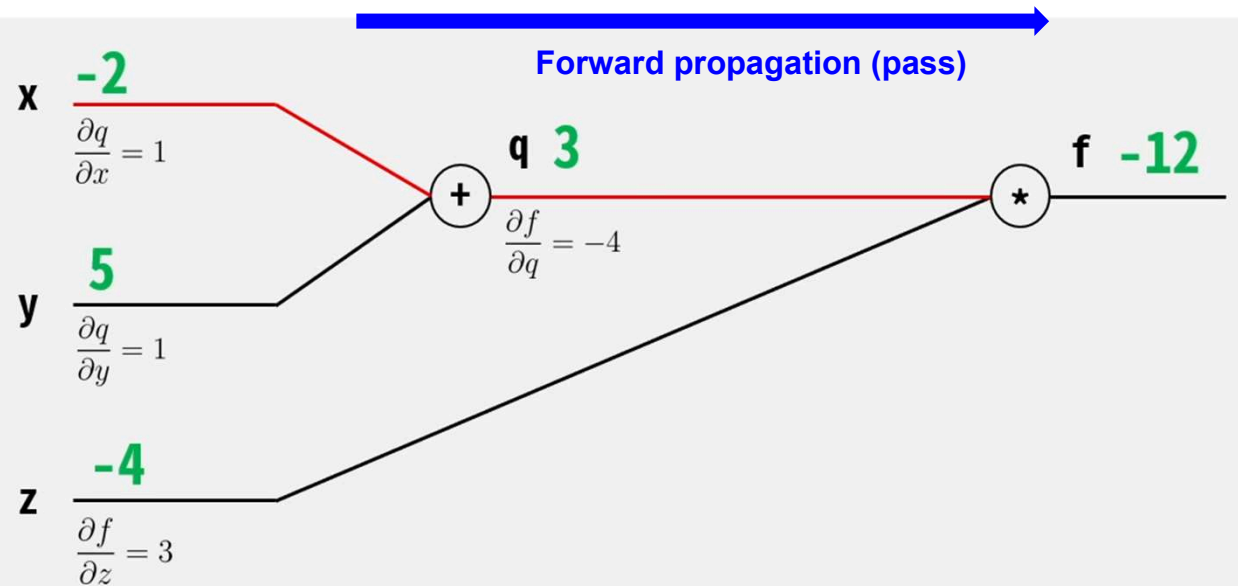
$$f(x, y, z) = (x + y)z$$

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$$\frac{\partial f}{\partial x} = ?$$



Backpropagation 동작원리

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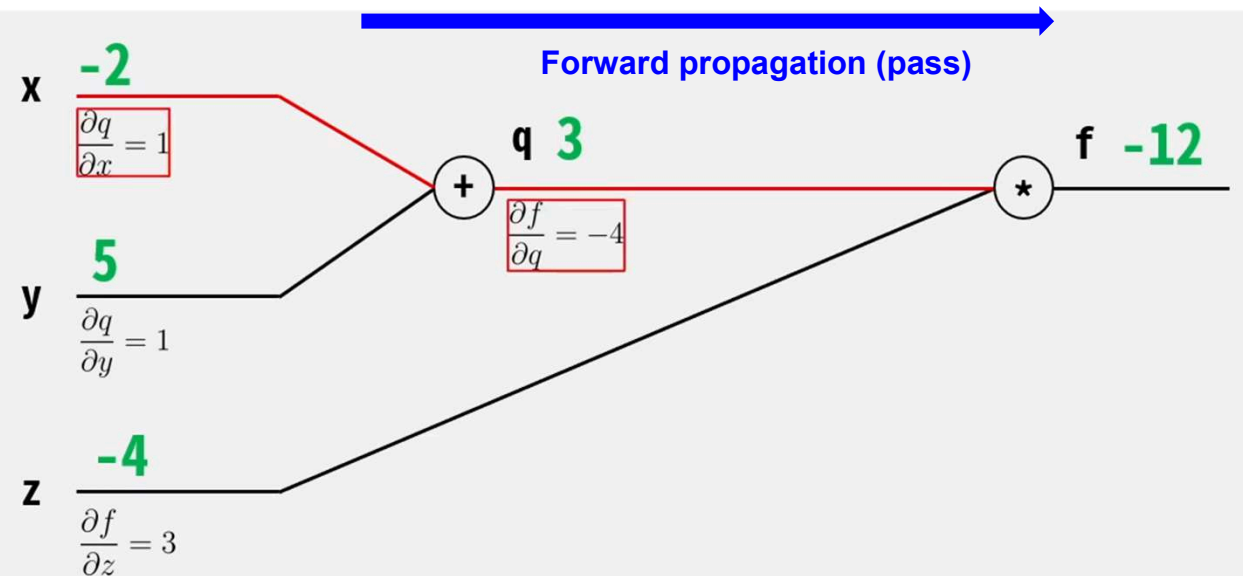
$$x=-2, y=5, z=-4$$

$$q = x + y$$

$$f = qz$$

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial q} \frac{\partial q}{\partial x}$$

Chain Rule



Backpropagation 동작원리

$$f(x, y, z) = (x + y)z$$

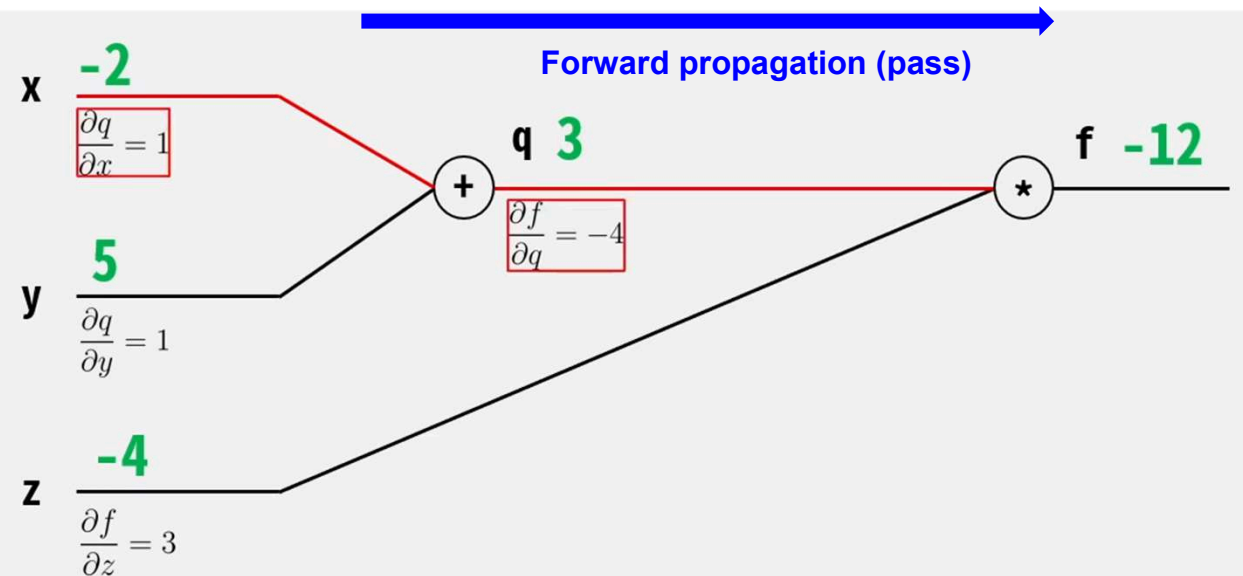
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Chain Rule

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial q} \frac{\partial q}{\partial x} = (-4)(1) = -4$$



Backpropagation 동작원리

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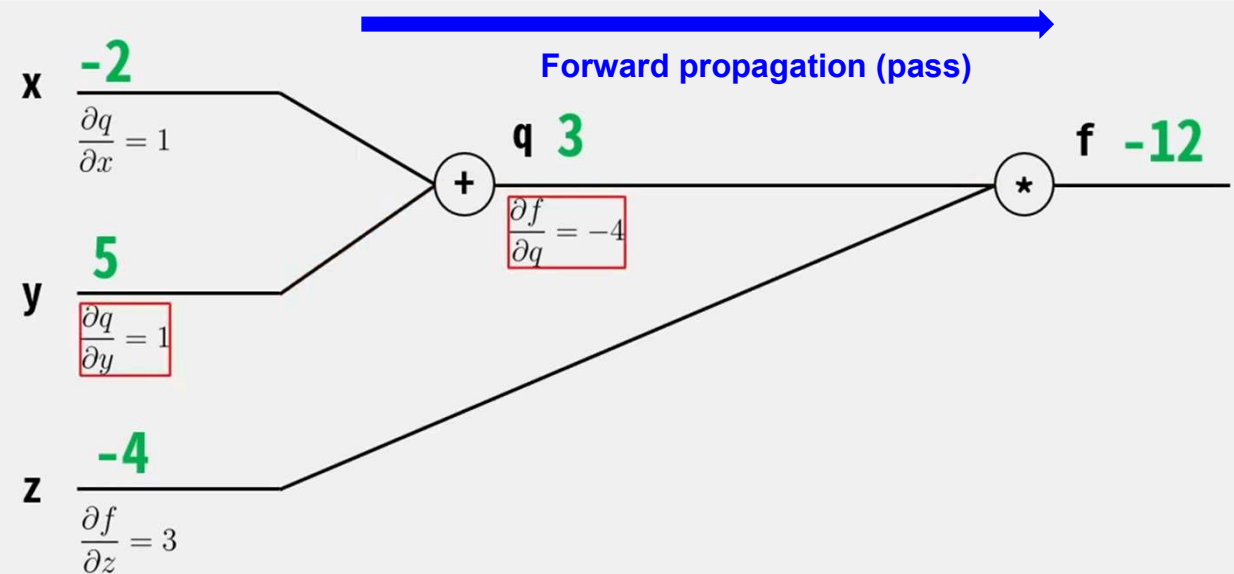
$$q = x + y$$

$$f = qz$$

Chain Rule

$$\frac{\partial f}{\partial x} = -4$$

$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial q} \frac{\partial q}{\partial y} = (-4)(1) = -4$$



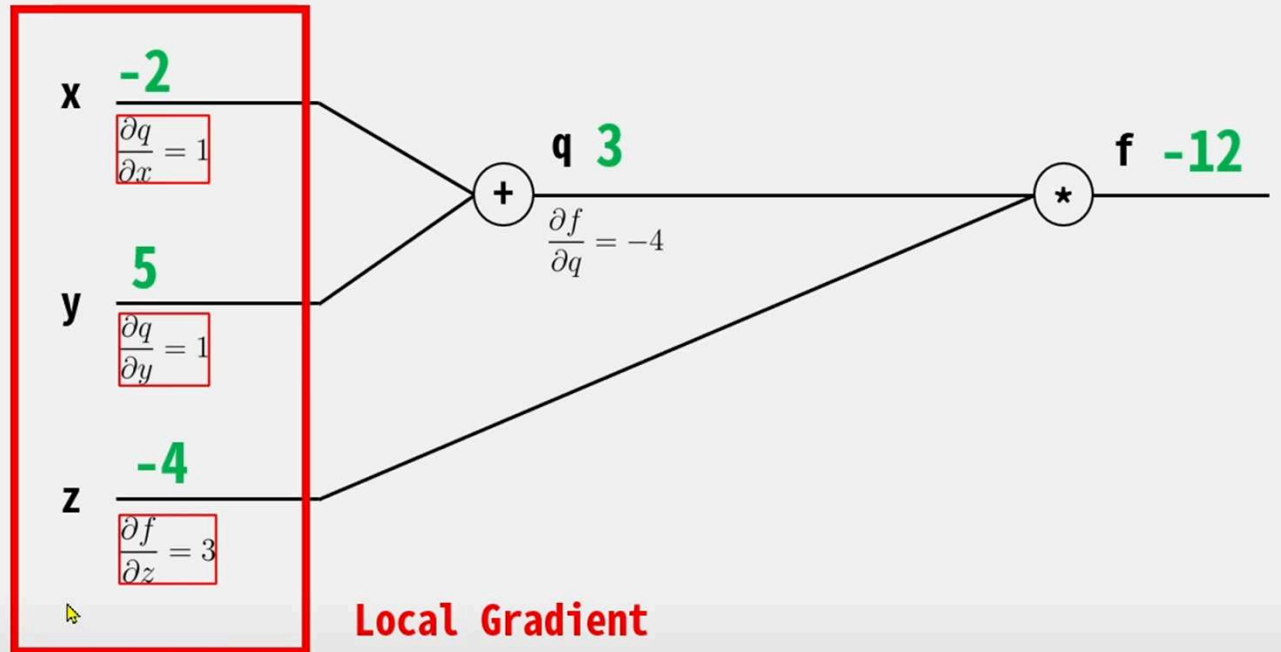
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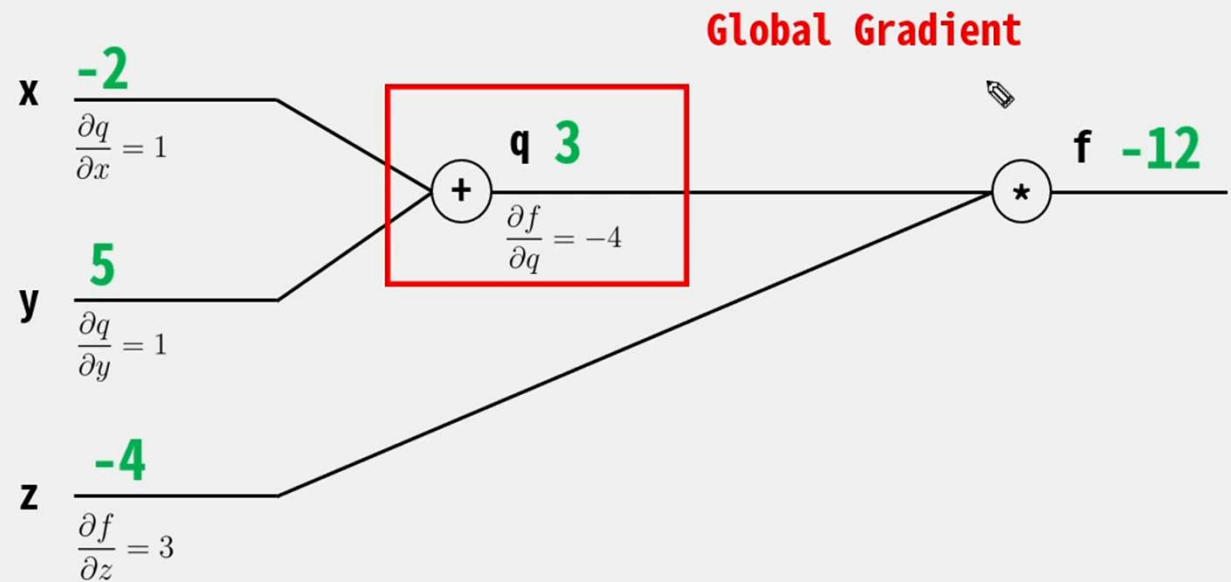
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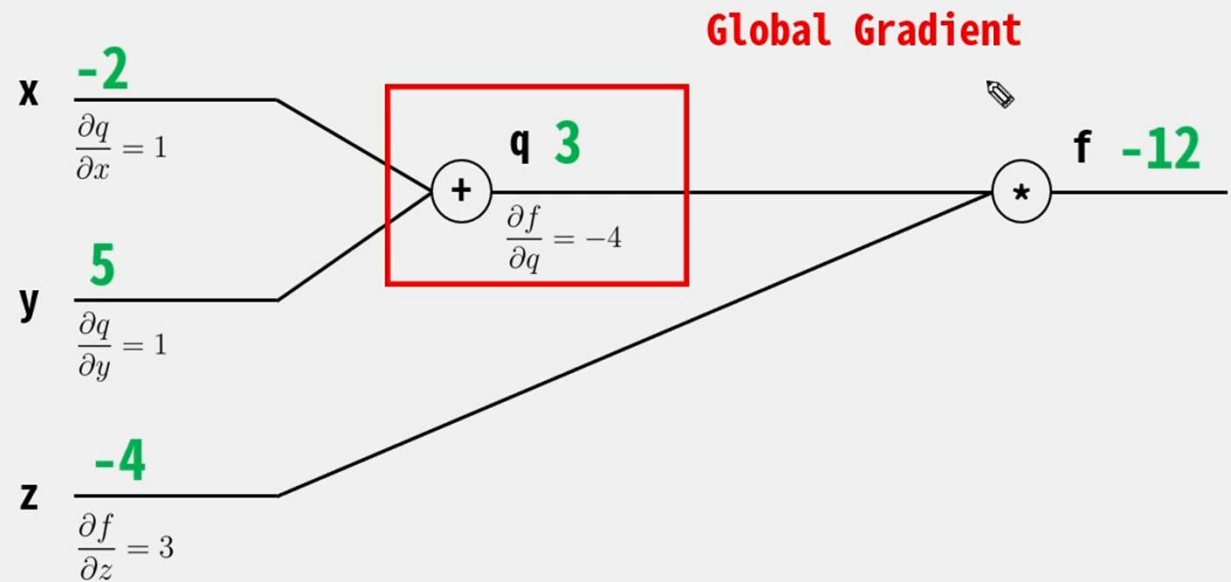
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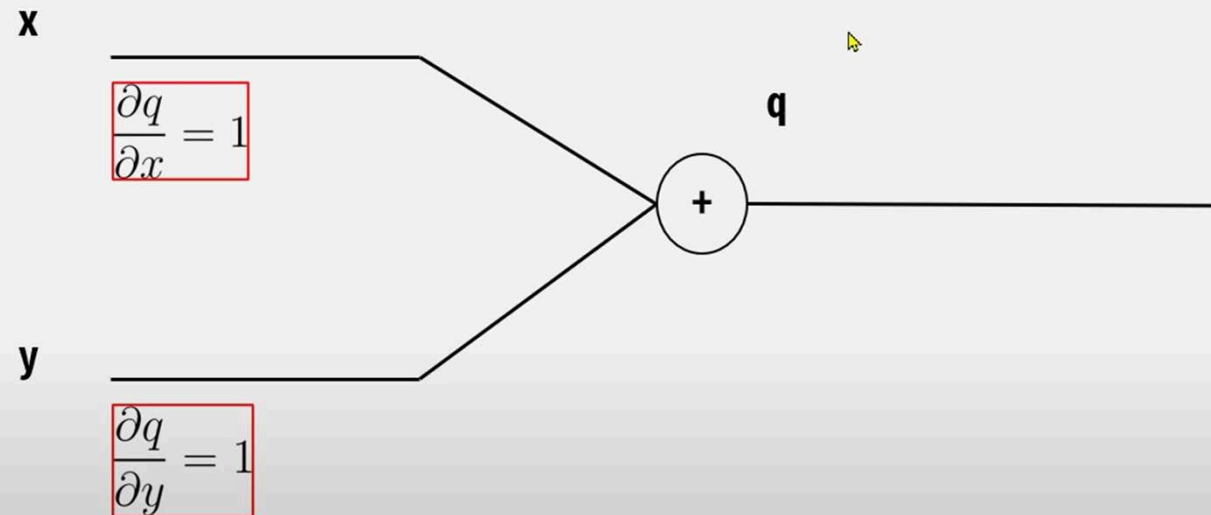
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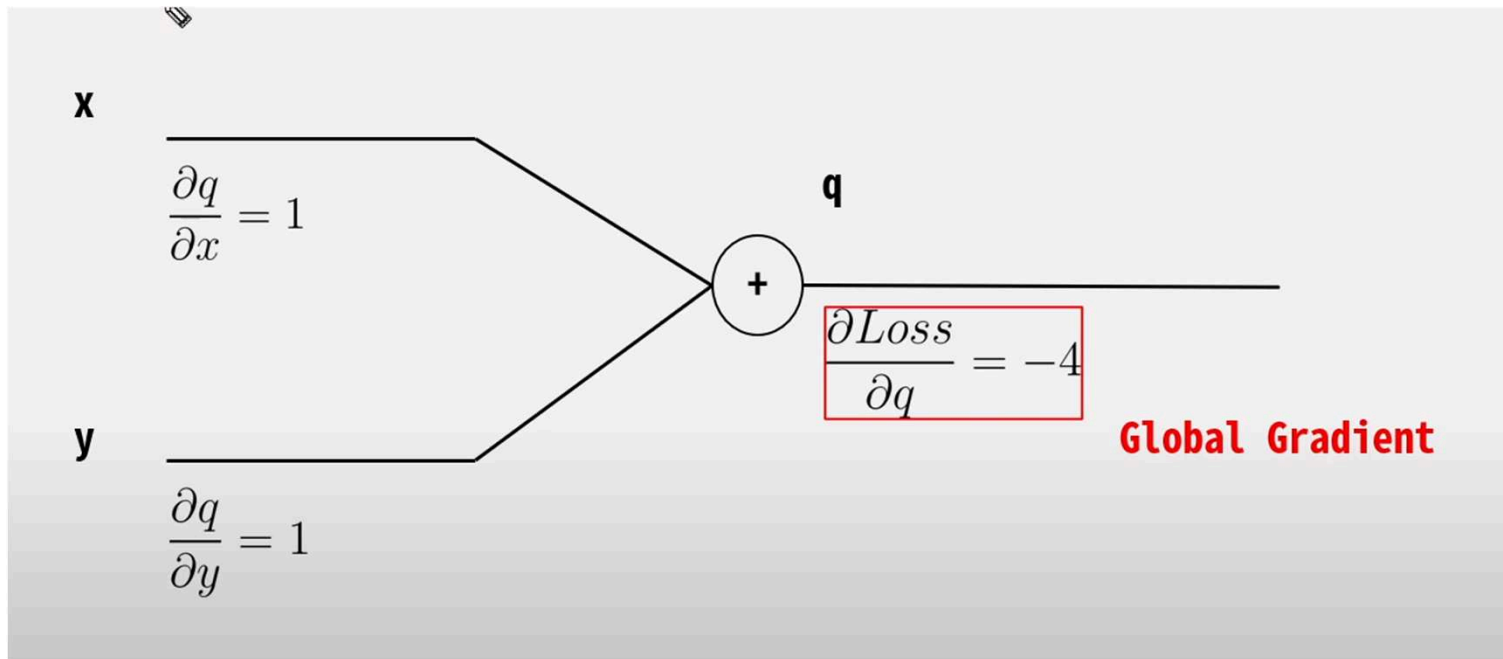
$$f = qz$$

$$\frac{\partial f}{\partial q} = z \quad \frac{\partial f}{\partial z} = q$$

Forward Pass 시 우리는 **Local Gradient**를 미리 구하여 저장할 수 있다!

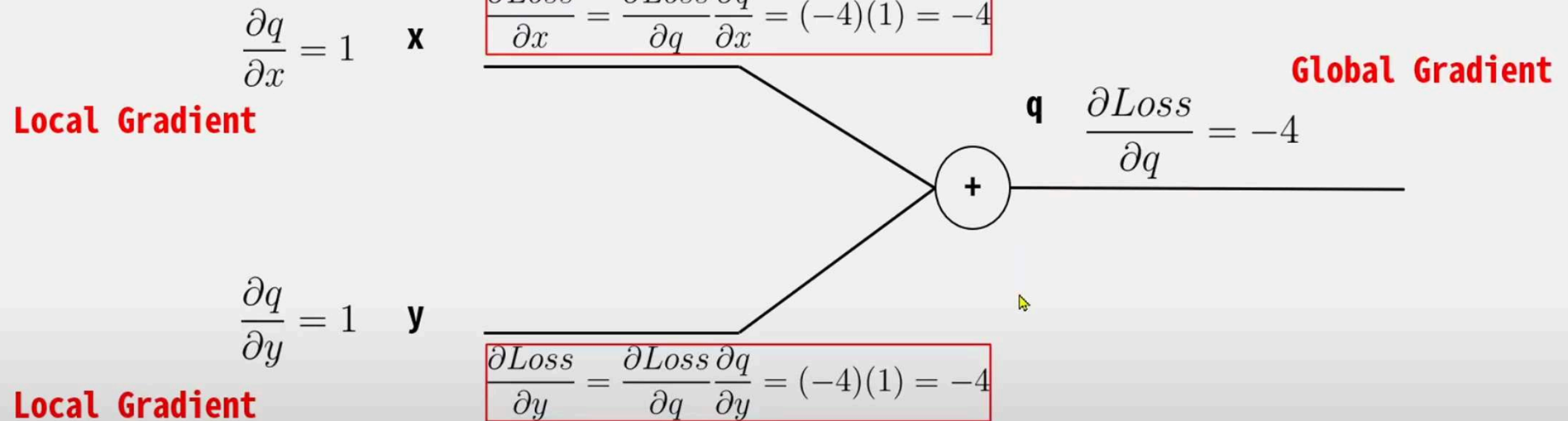


Backpropagation 동작원리



Backpropagation 동작원리

Chain Rule 을 활용하여
Local Gradient * Global Gradient를 곱하여 계산한다.

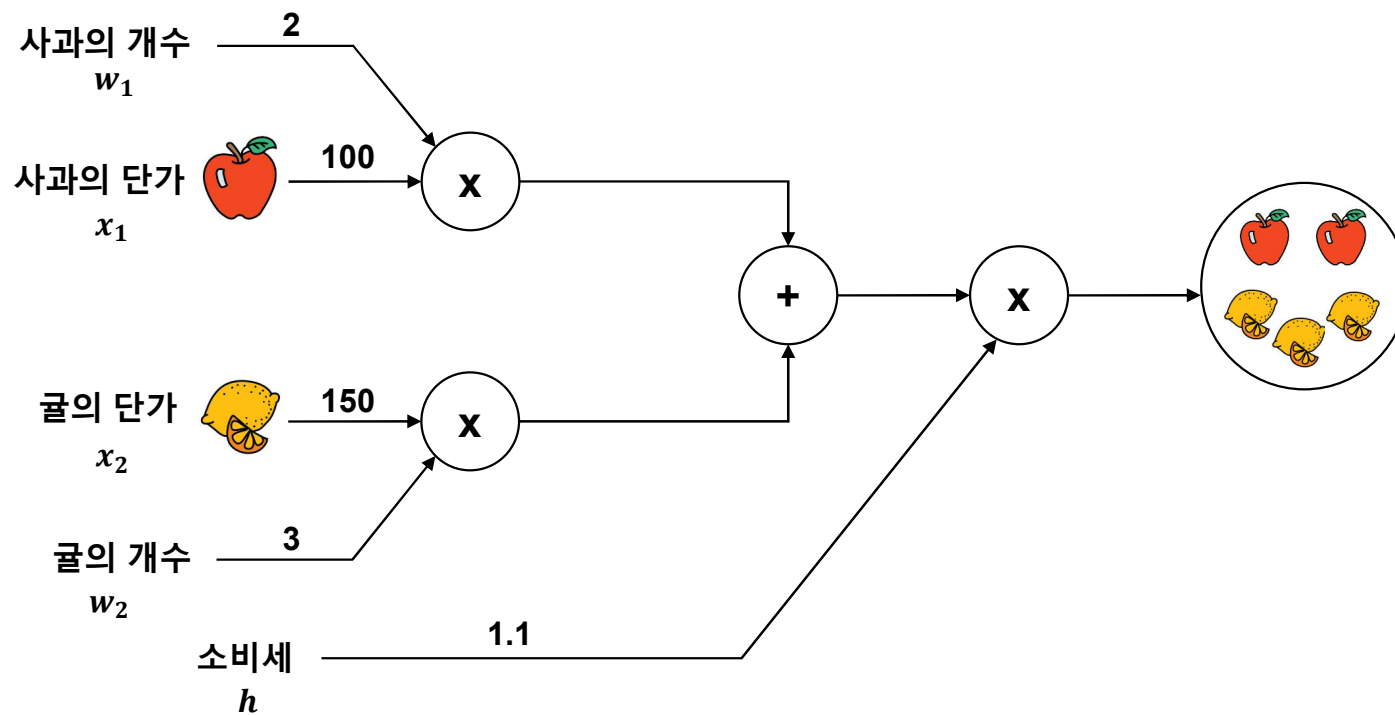


Backpropagation 동작원리

- 아무리 깊고 복잡한 층으로 구성되어 있다 하더라도 **Chain Rule**을 활용하여 미분 값을 얻어낼 수 있다.
- Forward Pass시 **Local Gradient**를 미리 계산하여 저장해 둔다.
- 저장해둔 **Local Gradient**와 **Global Gradient**를 Backward Pass시 곱하여 최종 미분 값을 얻는다.

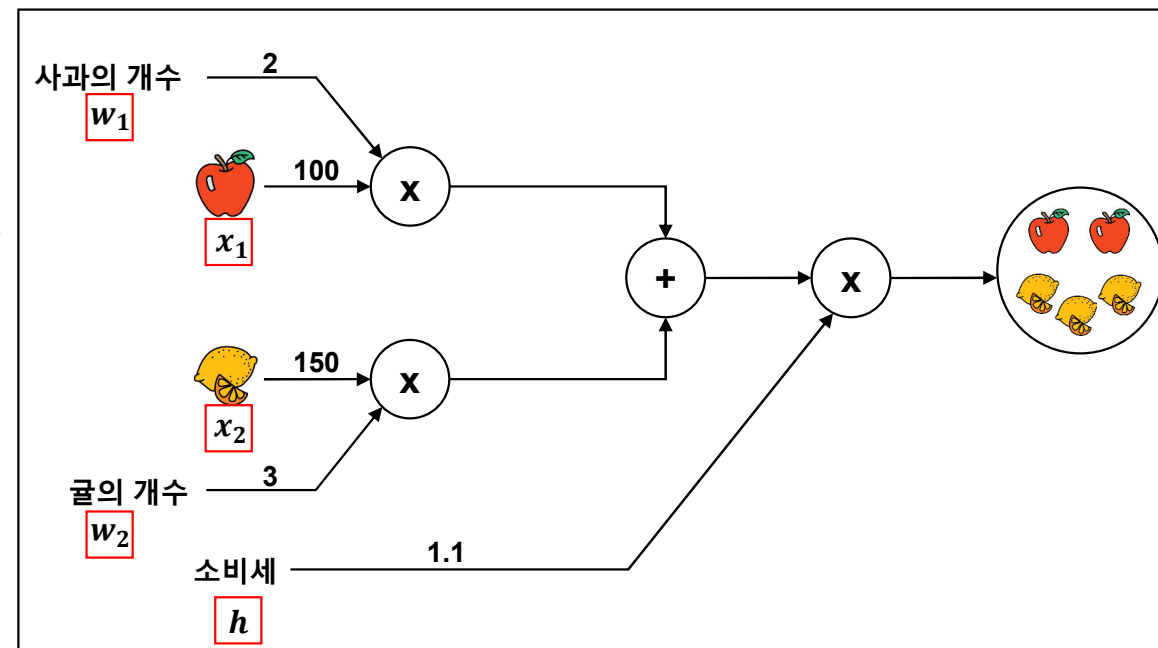
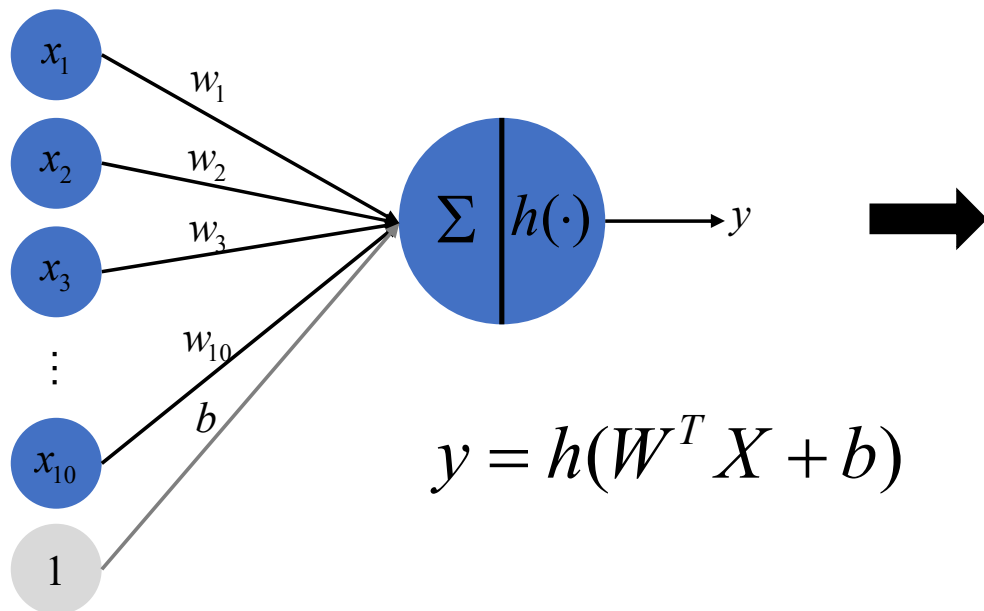
Backpropagation 동작원리

- 예제 1: 슈퍼에서 사과를 2개, 귤을 3개 구매 시 지불금액은?
 - ✓ 사과는 1개 100원, 귤은 1개 150원
 - ✓ 소비세 10%



Backpropagation 동작원리

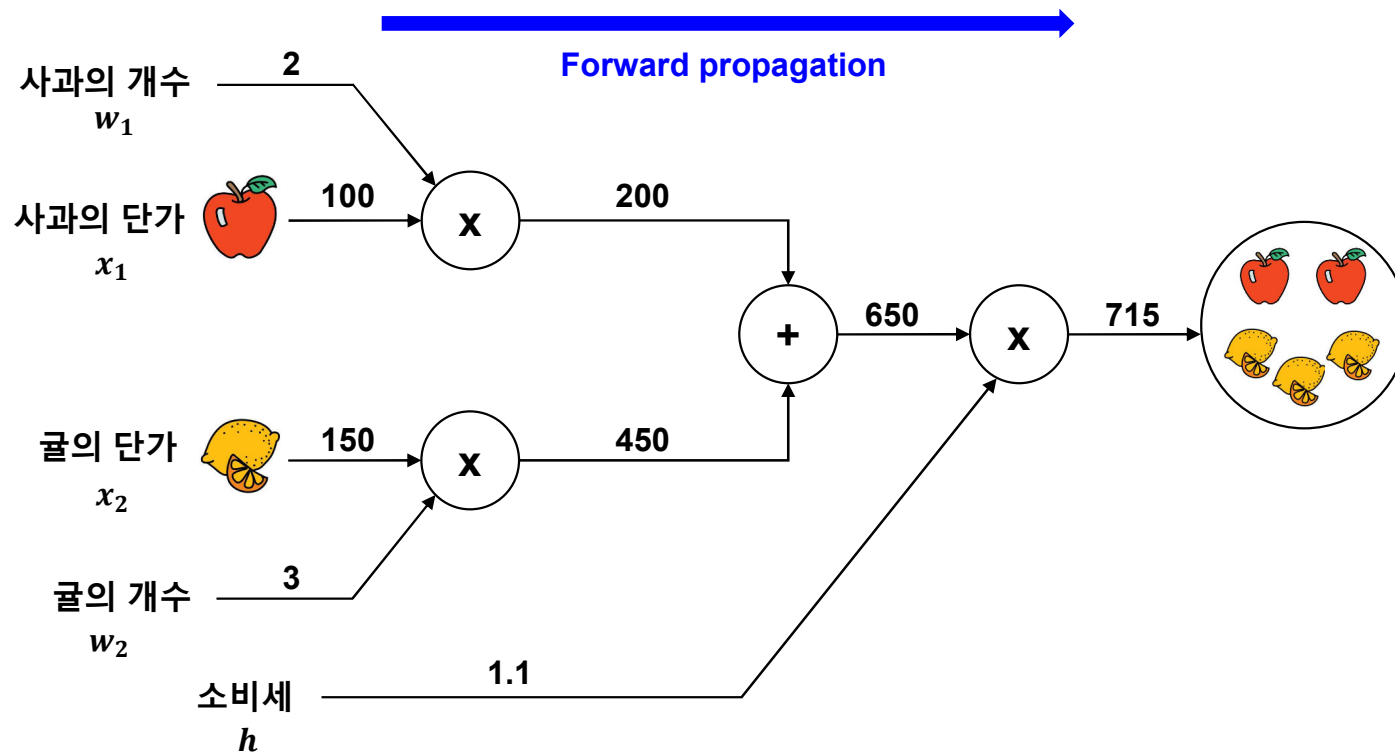
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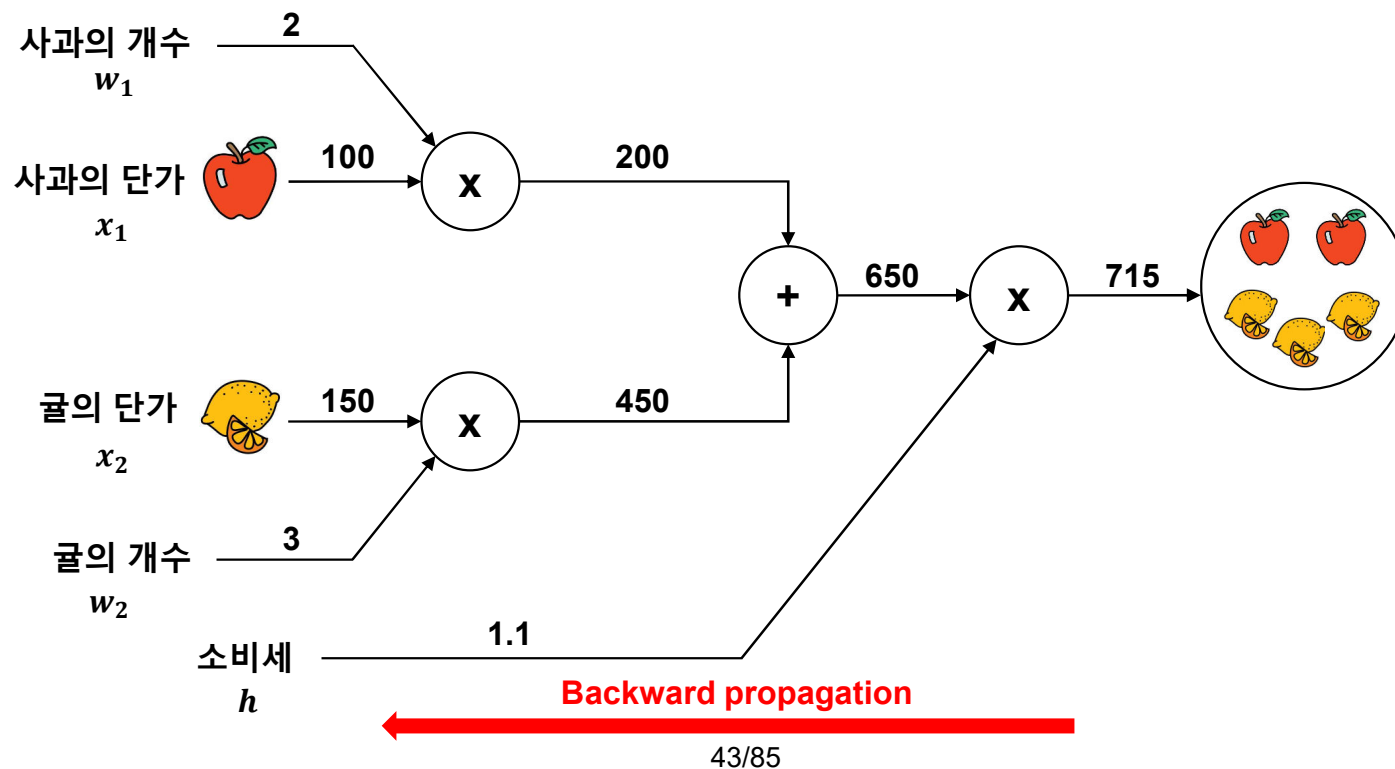
■ 예제 1: 슈퍼에서 사과를 2개, 귤을 3개 구매 시 지불금액은?

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Backpropagation 동작원리

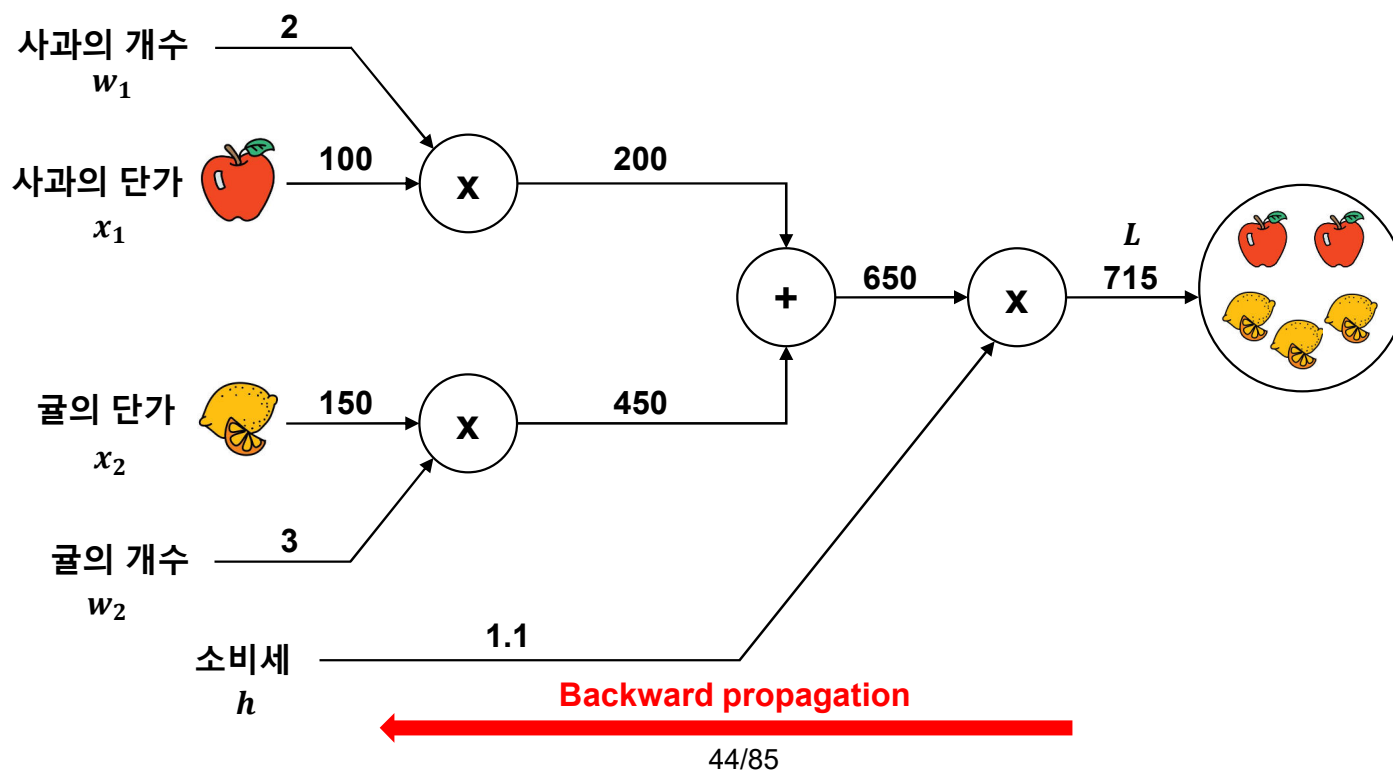
- 예제2: 사과 개수가 변하면 최종 금액에 어떤 영향을 끼칠까?



Backpropagation 동작원리

■ 예제2: 사과 개수가 변하면 최종 금액에 어떤 영향을 끼칠까?

- ✓ 사과 개수: w_1
 ✓ 지불 금액: L
- $\longrightarrow \frac{\partial L}{\partial w_1} \longrightarrow$ 사과 개수가 증가했을 때 지불 금액이 얼마나 증가하는지 표시

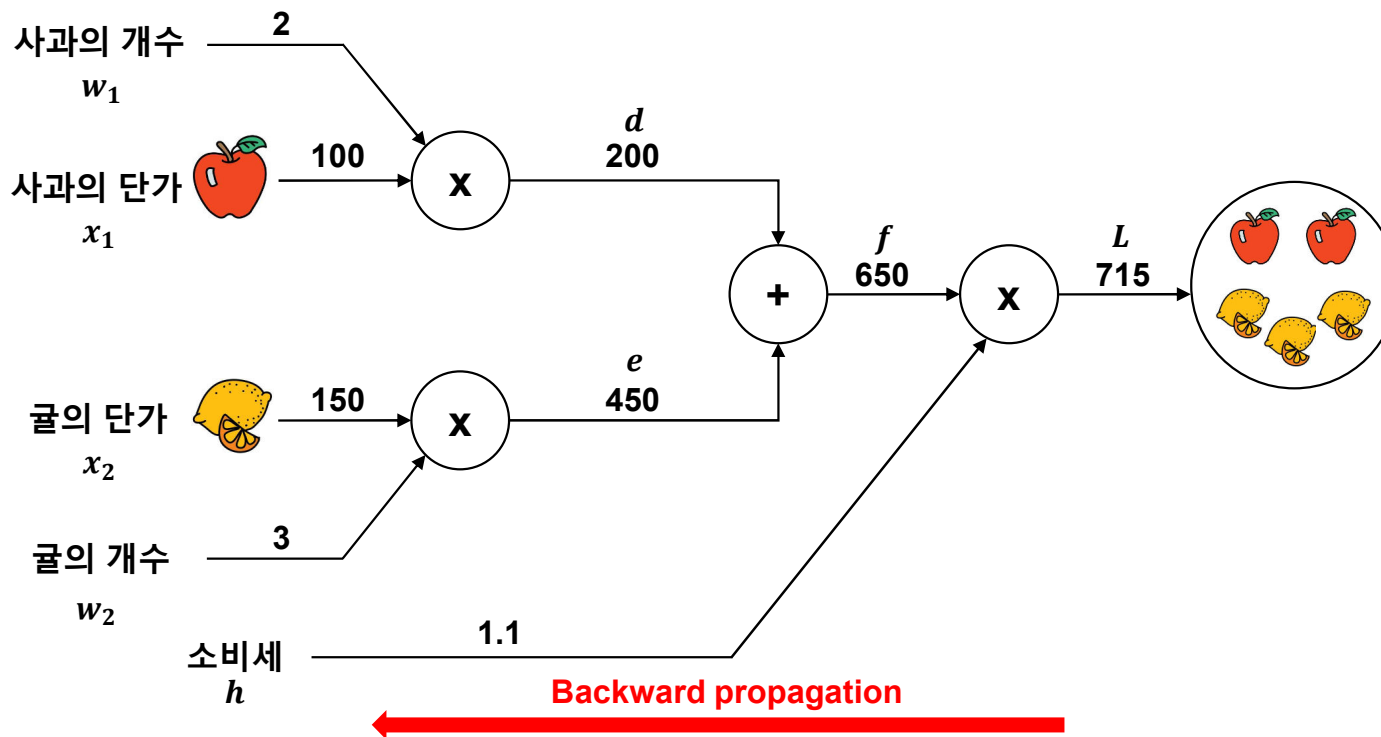


Backpropagation 동작원리

■ 예제2: 사과 개수가 변하면 최종 금액에 어떤 영향을 끼칠까?

✓ 사과 개수: w_1
 ✓ 지불 금액: L

$\longrightarrow \frac{\partial L}{\partial w_1} \longrightarrow$ 사과 개수가 증가했을 때 지불 금액이 얼마나 증가하는지 표시



$$L(w_1, w_2, x_1, x_2) = c(w_1x_1 + w_2x_2)$$

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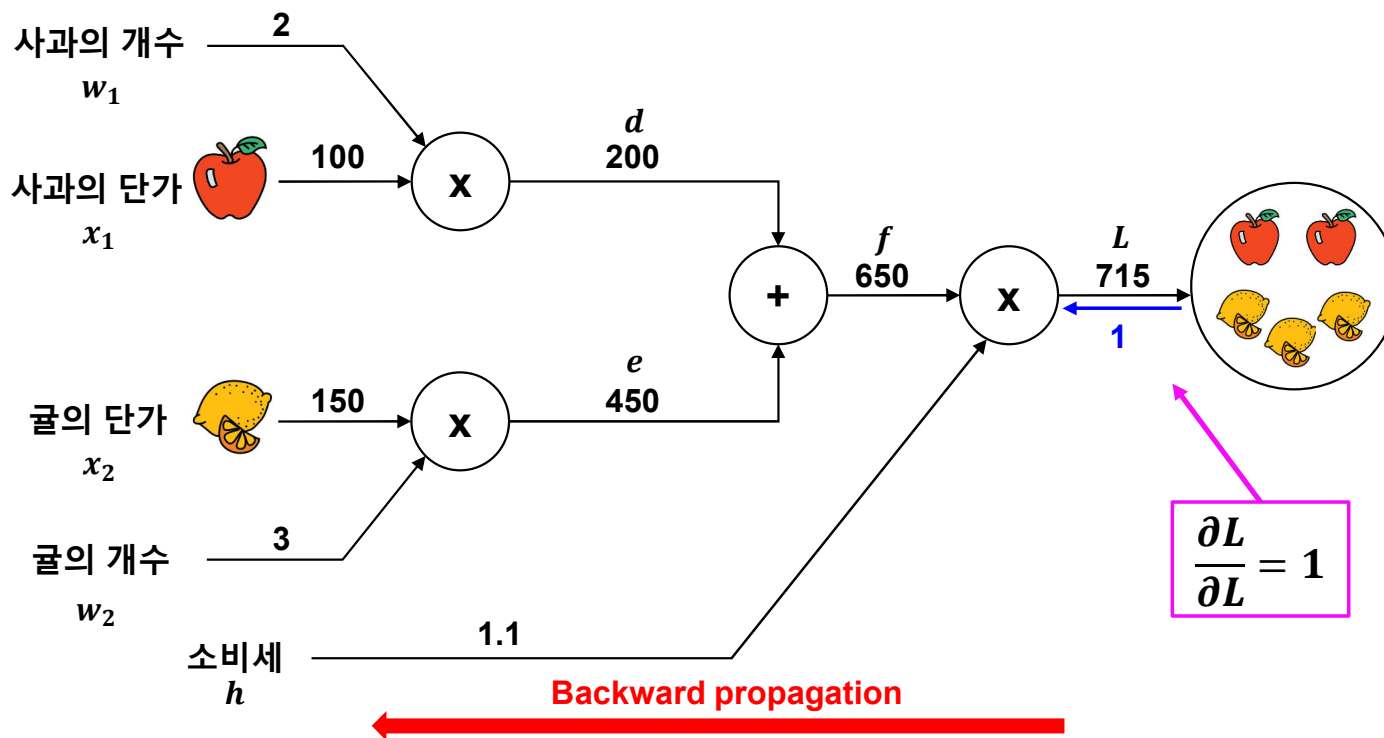
$$L = hf \quad \frac{\partial L}{\partial h} = f, \quad \frac{\partial L}{\partial f} = h$$

Backpropagation 동작원리

■ 예제2: 사과 개수가 변하면 최종 금액에 어떤 영향을 끼칠까?

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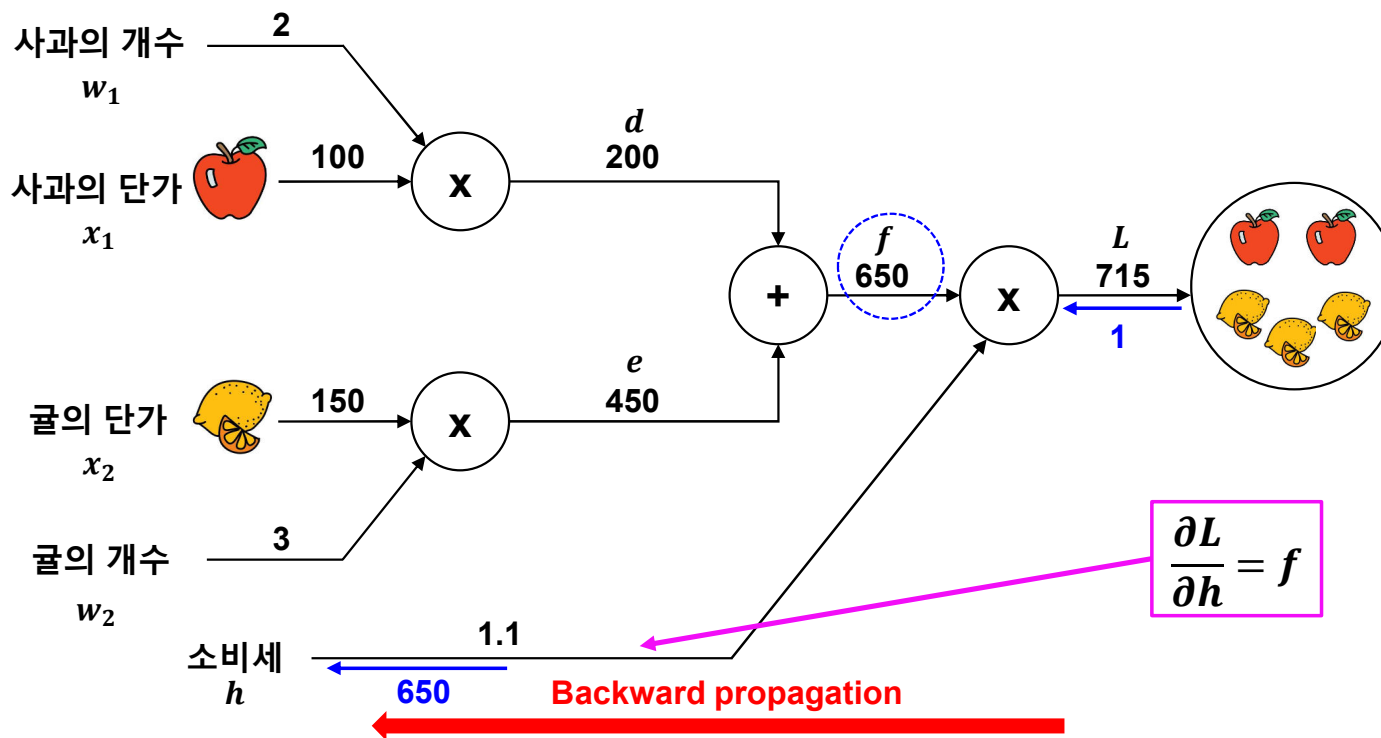
$$\frac{\partial L}{\partial L} = 1$$

Backpropagation 동작원리

■ 예제2: 사과 개수가 변하면 최종 금액에 어떤 영향을 끼칠까?

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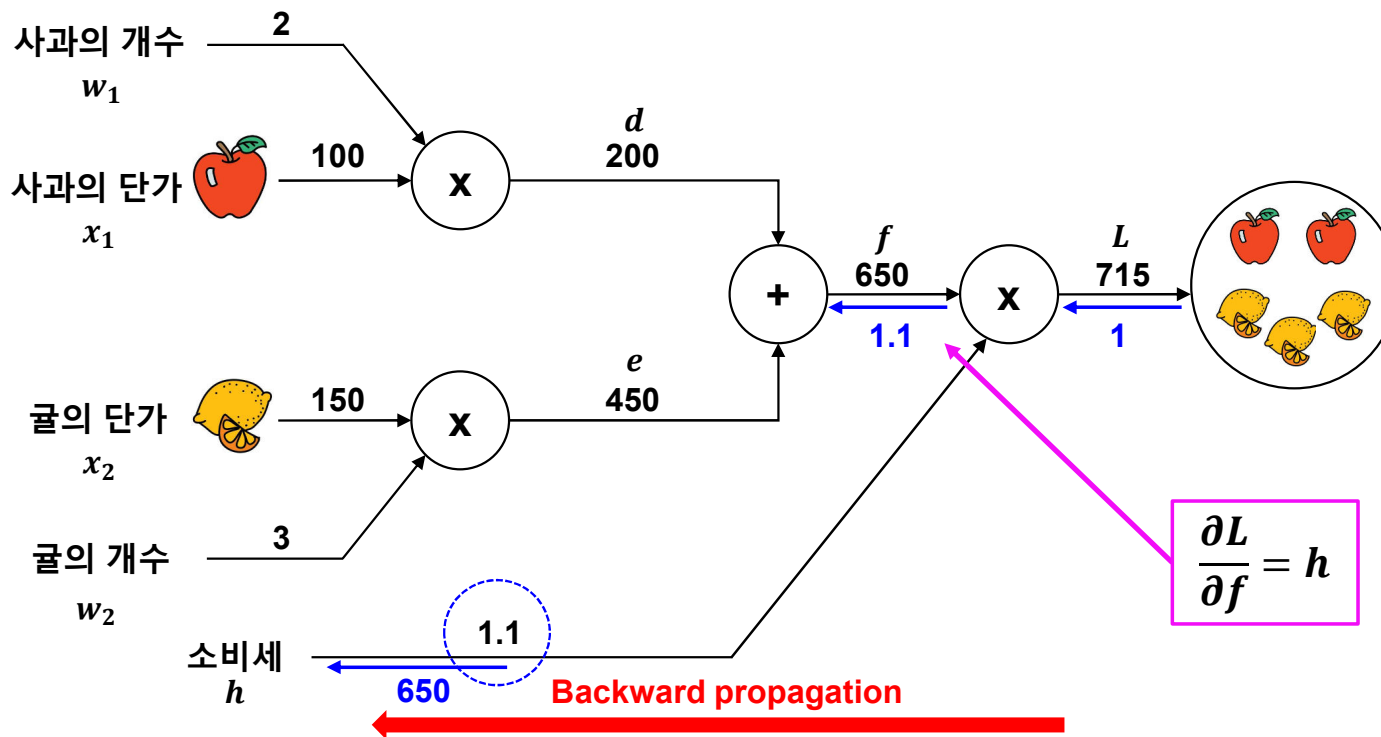
$$L = hf \quad \frac{\partial L}{\partial h} = f, \quad \frac{\partial L}{\partial f} = h$$

Backpropagation 동작원리

■ 예제2: 사과 개수가 변하면 최종 금액에 어떤 영향을 끼칠까?

✓ 사과 개수: w_1
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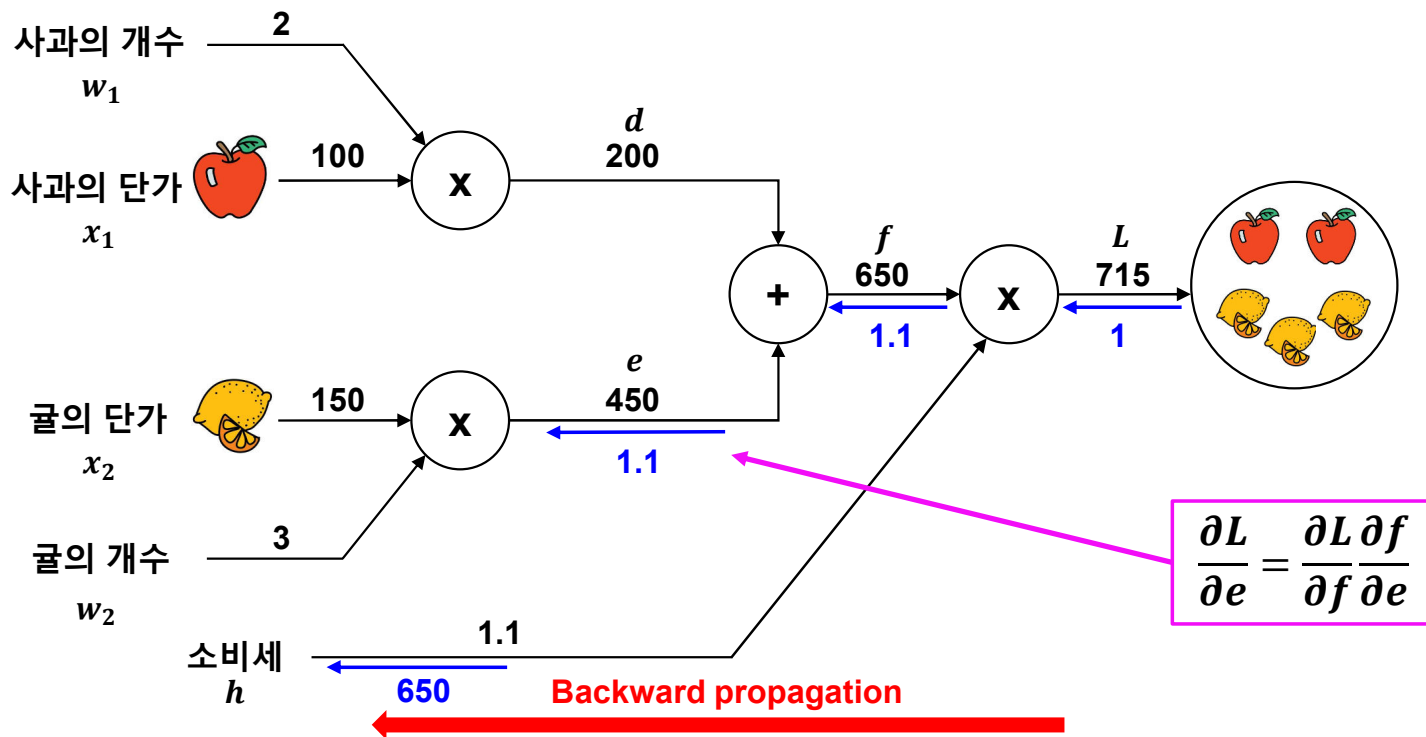
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Backpropagation 동작원리

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$\longrightarrow \frac{\partial L}{\partial w_1} \longrightarrow$ 사과 개수가 증가했을 때 지불 금액이 얼마나 증가하는지 표시



$$\frac{\partial L}{\partial e} = \frac{\partial L}{\partial f} \frac{\partial f}{\partial e}$$

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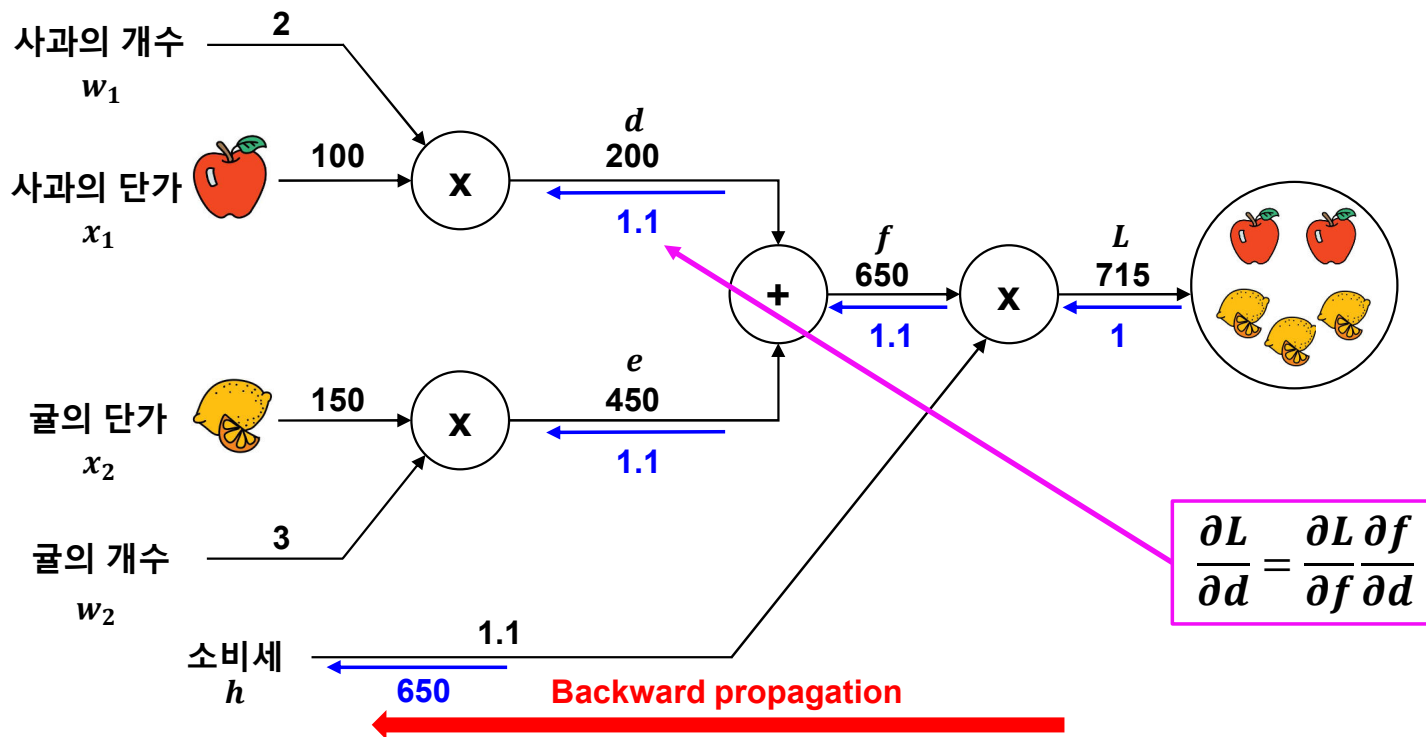
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Backpropagation 동작원리

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$$\frac{\partial L}{\partial d} = \frac{\partial L}{\partial f} \frac{\partial f}{\partial d}$$

$$L(w_1, w_2, x_1, x_2) = c(w_1x + w_2x_2)$$

$$d = w_1x_1 \quad \frac{\partial d}{\partial w_1} = x_1, \quad \frac{\partial d}{\partial x_1} = w_1$$

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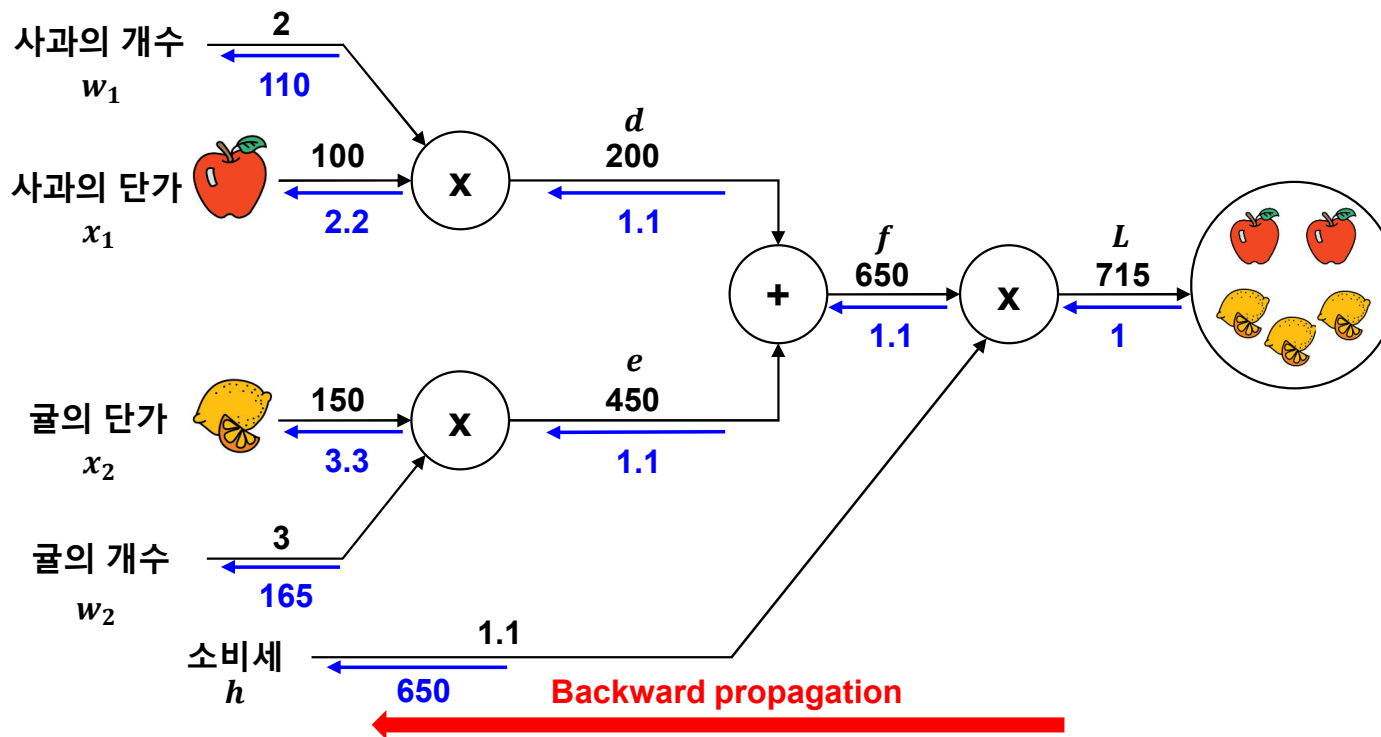
$$f = d + e \quad \frac{\partial f}{\partial d} = 1, \quad \frac{\partial f}{\partial e} = 1$$

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Backpropagation 동작원리

■ 예제2: 사과 개수가 변하면 최종 금액에 어떤 영향을 끼칠까?

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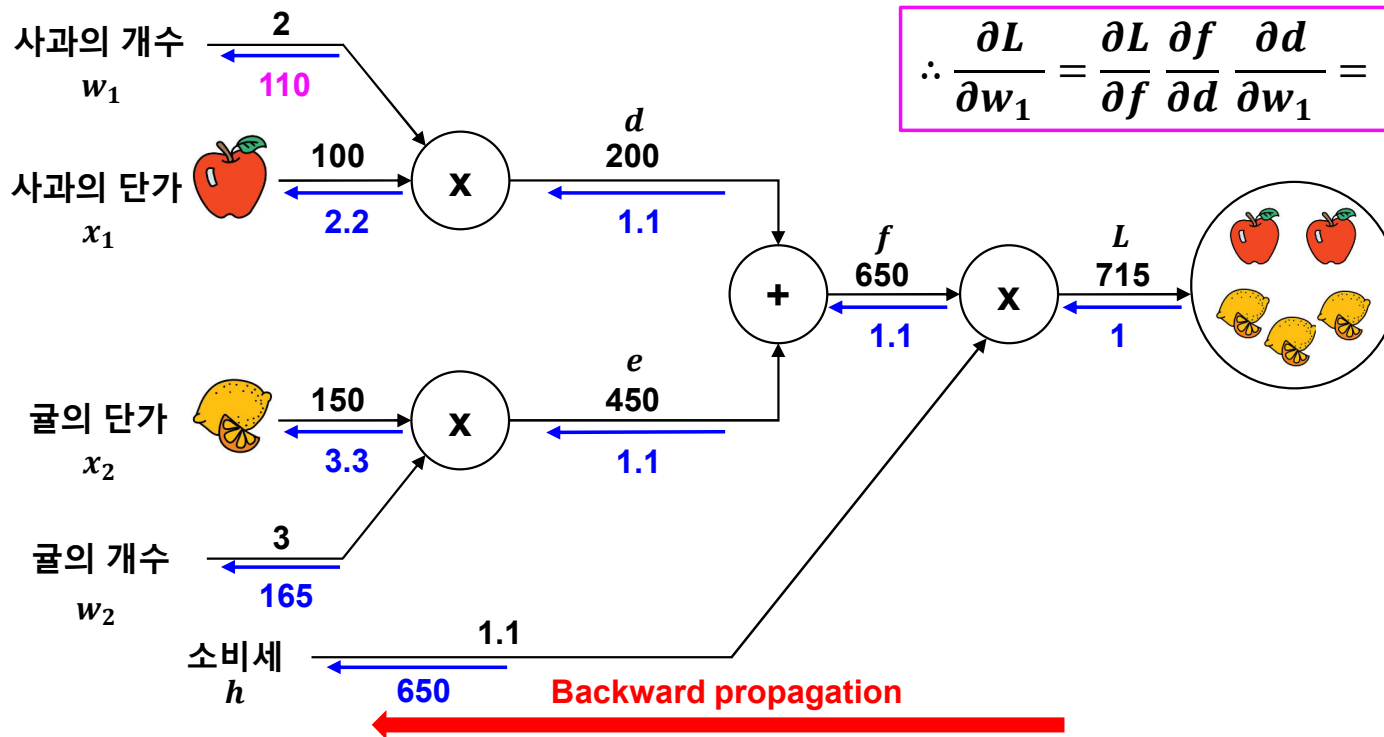
Backpropagation 동작원리

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Chain rule

$$\therefore \frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial f} \frac{\partial f}{\partial d} \frac{\partial d}{\partial w_1} = 110$$



$$L(w_1, w_2, x_1, x_2) = c(w_1 x_1 + w_2 x_2)$$

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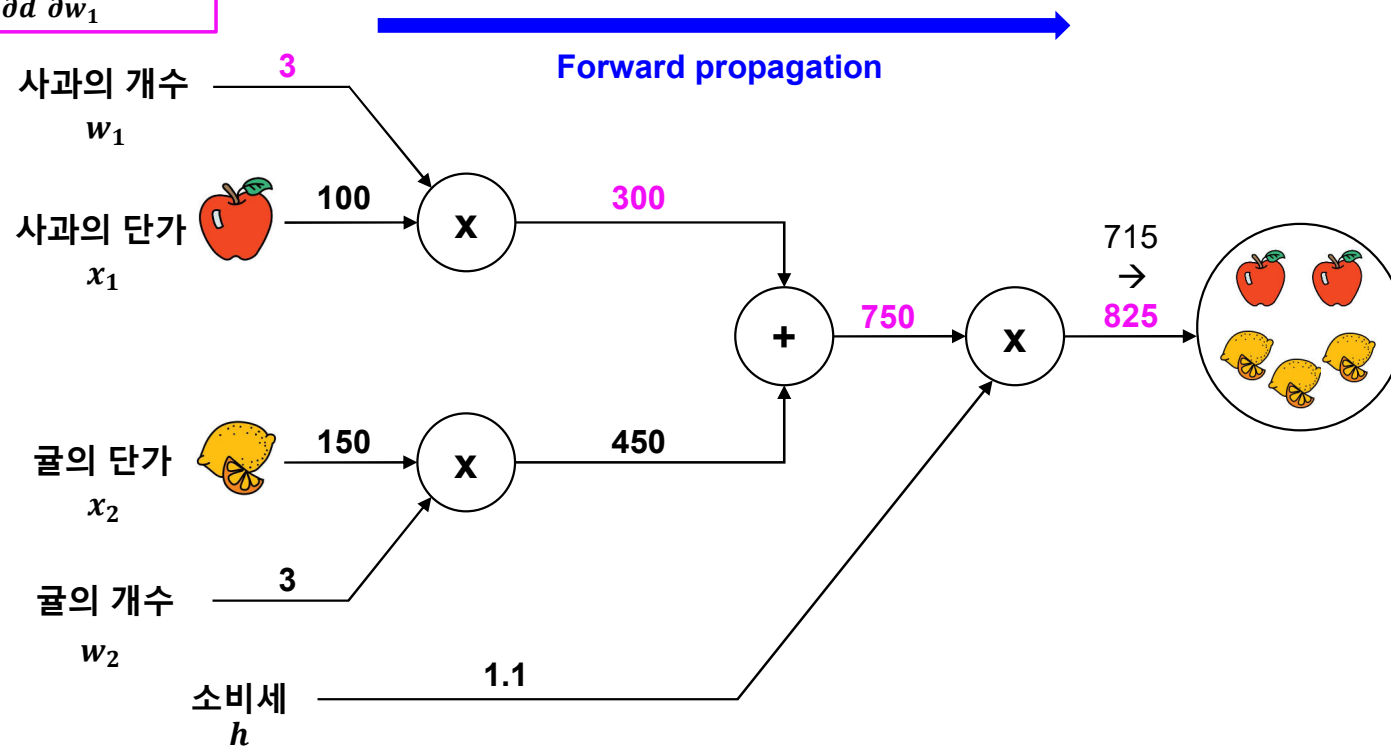
$$L = hf \quad \frac{\partial L}{\partial h} = f, \quad \frac{\partial L}{\partial f} = h$$

Backpropagation 동작원리

- Weight Update (a: 사과의 개수)

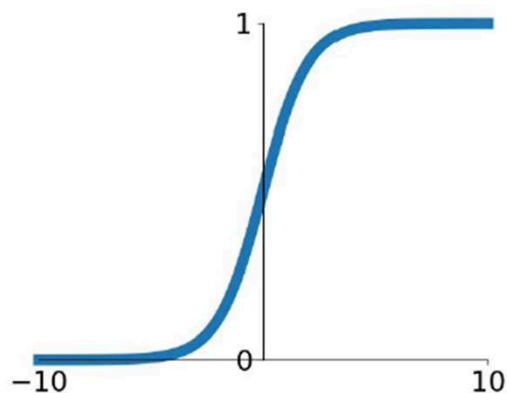
Chain rule

$$\therefore \frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial f} \frac{\partial f}{\partial d} \frac{\partial d}{\partial w_1} = 110$$



Backpropagation 동작원리

- Sigmoid 계층의 backpropagation

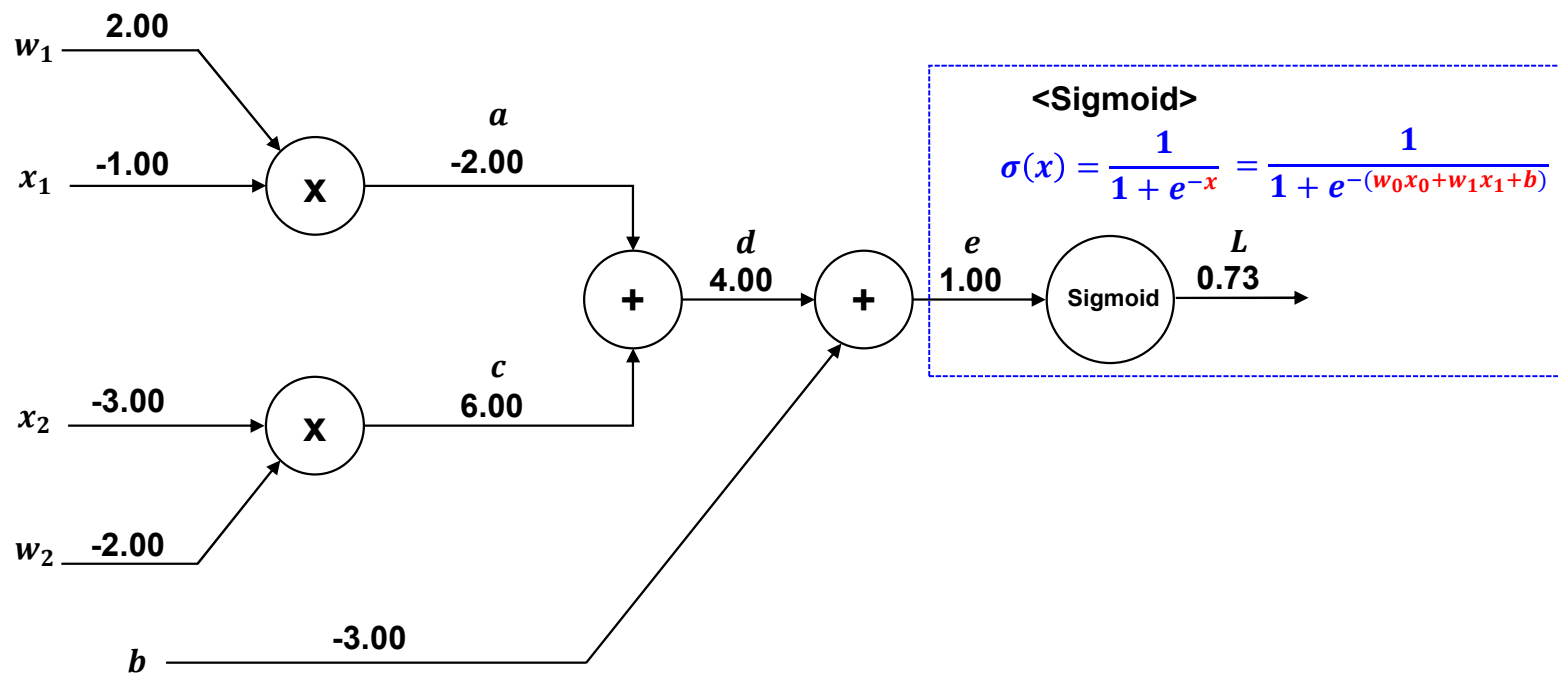
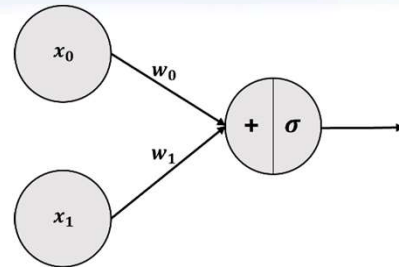


$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

$$\begin{aligned}
 \frac{d}{dx} \text{sigmoid}(x) &= \frac{d}{dx} (1 + e^{-x})^{-1} \\
 &= (-1) \frac{1}{(1 + e^{-x})^2} \frac{d}{dx} (1 + e^{-x}) \\
 &= (-1) \frac{1}{(1 + e^{-x})^2} (0 + e^{-x}) \frac{d}{dx} (-x) \\
 &= (-1) \frac{1}{(1 + e^{-x})^2} e^{-x} (-1) \\
 &= \frac{e^{-x}}{(1 + e^{-x})^2} \\
 &= \frac{1 + e^{-x} - 1}{(1 + e^{-x})^2} \\
 &= \frac{(1 + e^{-x})}{(1 + e^{-x})^2} - \frac{1}{(1 + e^{-x})^2} \\
 &= \frac{1}{1 + e^{-x}} - \frac{1}{(1 + e^{-x})^2} \\
 &= \frac{1}{1 + e^{-x}} \left(1 - \frac{1}{1 + e^{-x}} \right) \\
 &= \text{sigmoid}(x) (1 - \text{sigmoid}(x))
 \end{aligned}$$

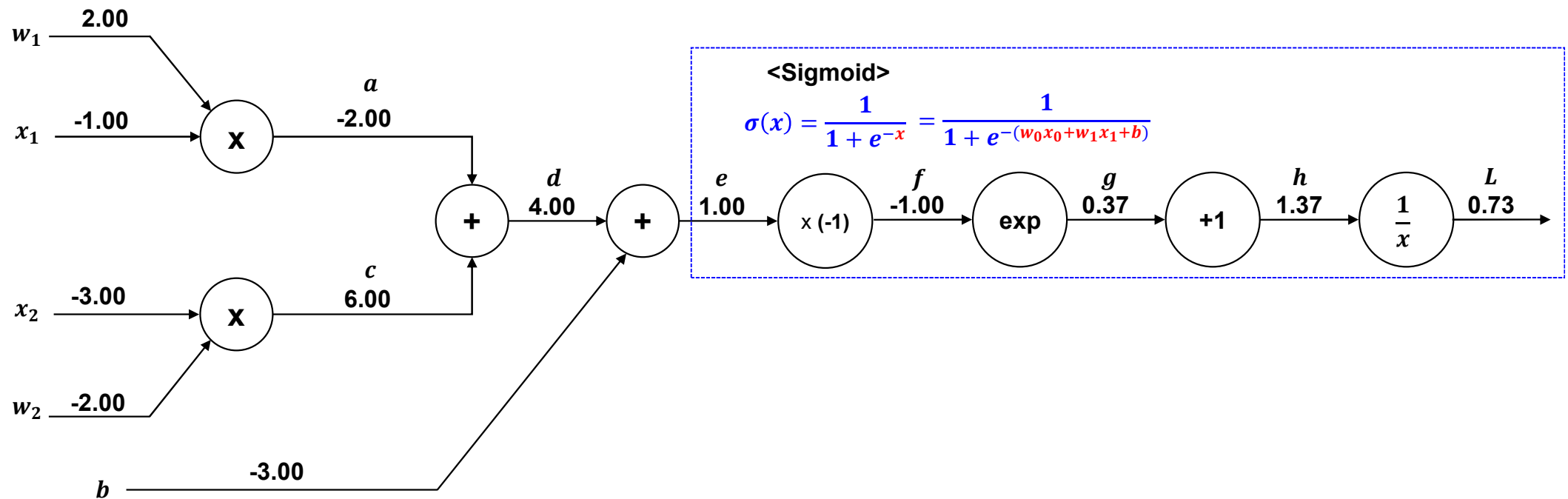
Backpropagation 동작원리

- Sigmoid 계층의 backpropagation



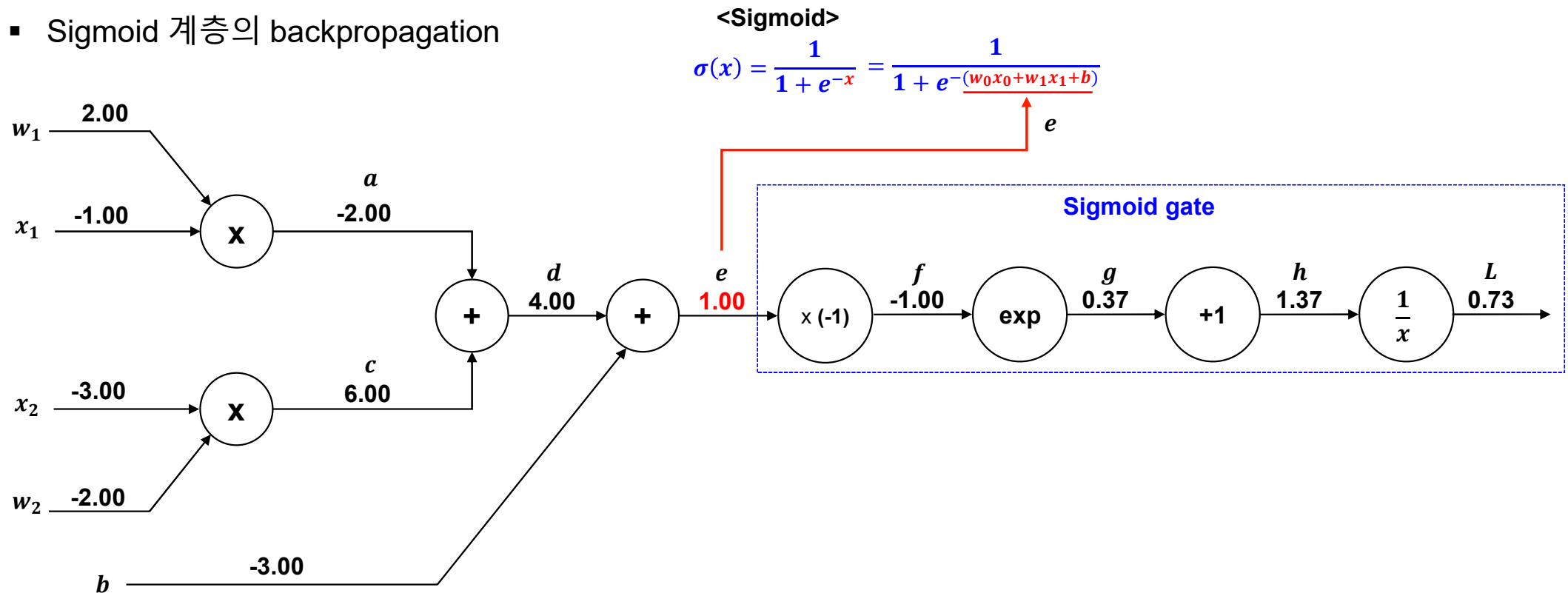
Backpropagation 동작원리

- Sigmoid 계층의 backpropagation



Backpropagation 동작원리

- Sigmoid 계층의 backpropagation

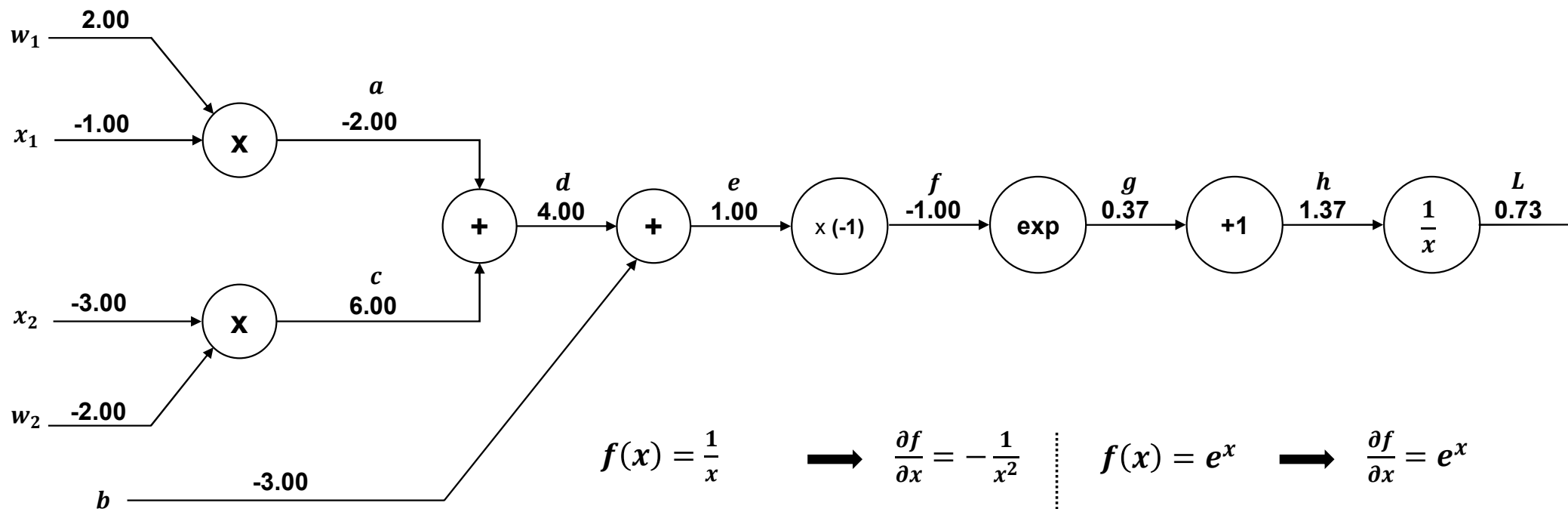


Backpropagation 동작원리

- Sigmoid 계층의 backpropagation

<Sigmoid>

$$\sigma(x) = \frac{1}{1 + e^{-x}} = \frac{1}{1 + e^{-(w_0x_0 + w_1x_1 + b)}}$$



$$f(x) = \frac{1}{x} \longrightarrow \frac{\partial f}{\partial x} = -\frac{1}{x^2}$$

$$f(x) = x + c \longrightarrow \frac{\partial f}{\partial x} = 1$$

$$f(x) = e^x \longrightarrow \frac{\partial f}{\partial x} = e^x$$

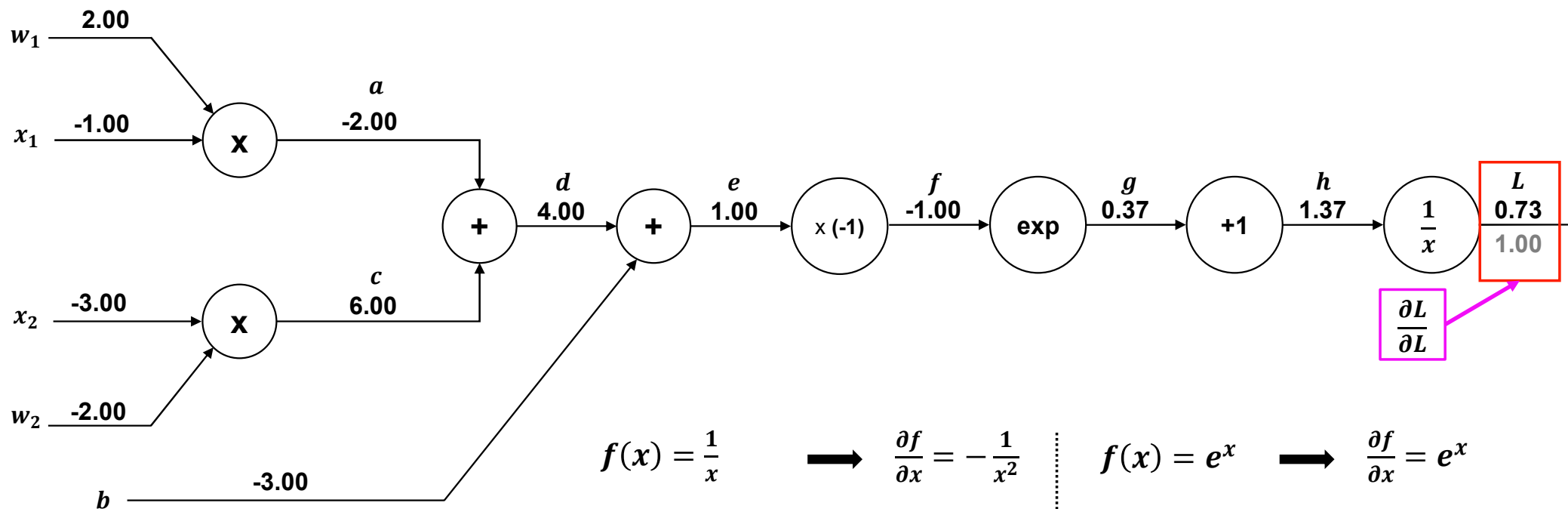
$$f(x) = \alpha x \longrightarrow \frac{\partial f}{\partial x} = \alpha$$

Backpropagation 동작원리

- Sigmoid 계층의 backpropagation

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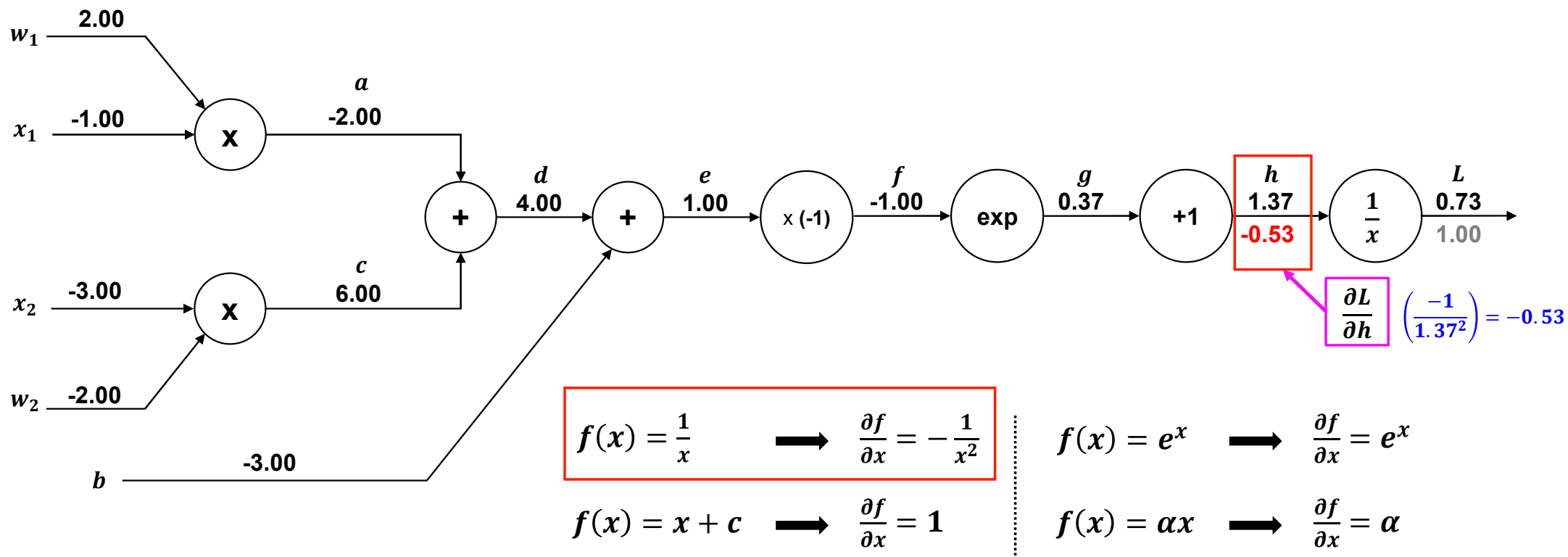
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Backpropagation 동작원리

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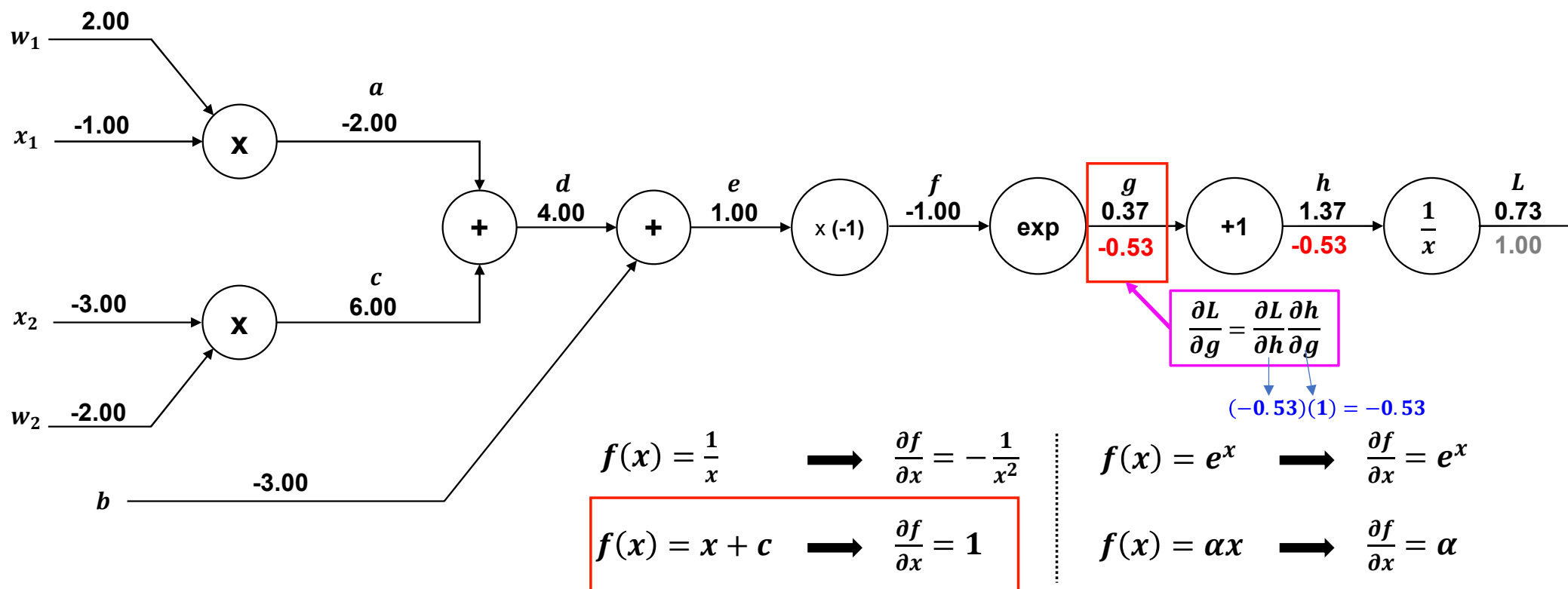


Backpropagation 동작원리

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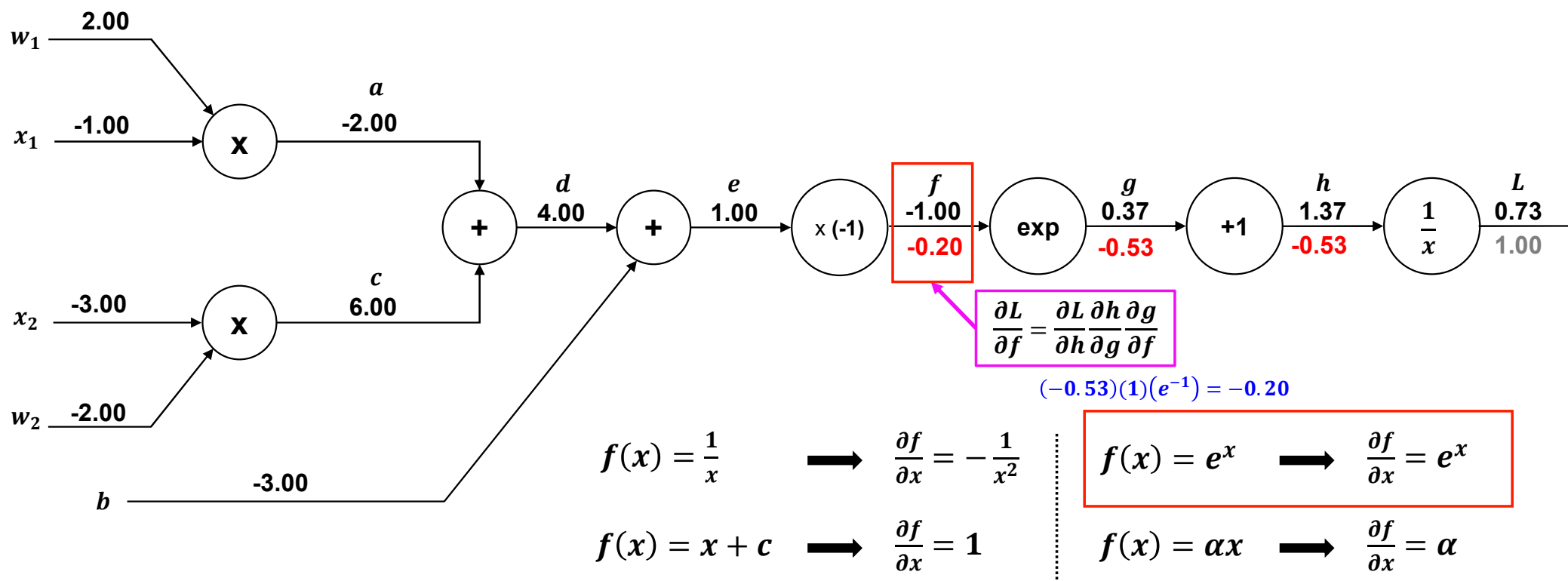


Backpropagation 동작원리

- Sigmoid 계층의 backpropagation

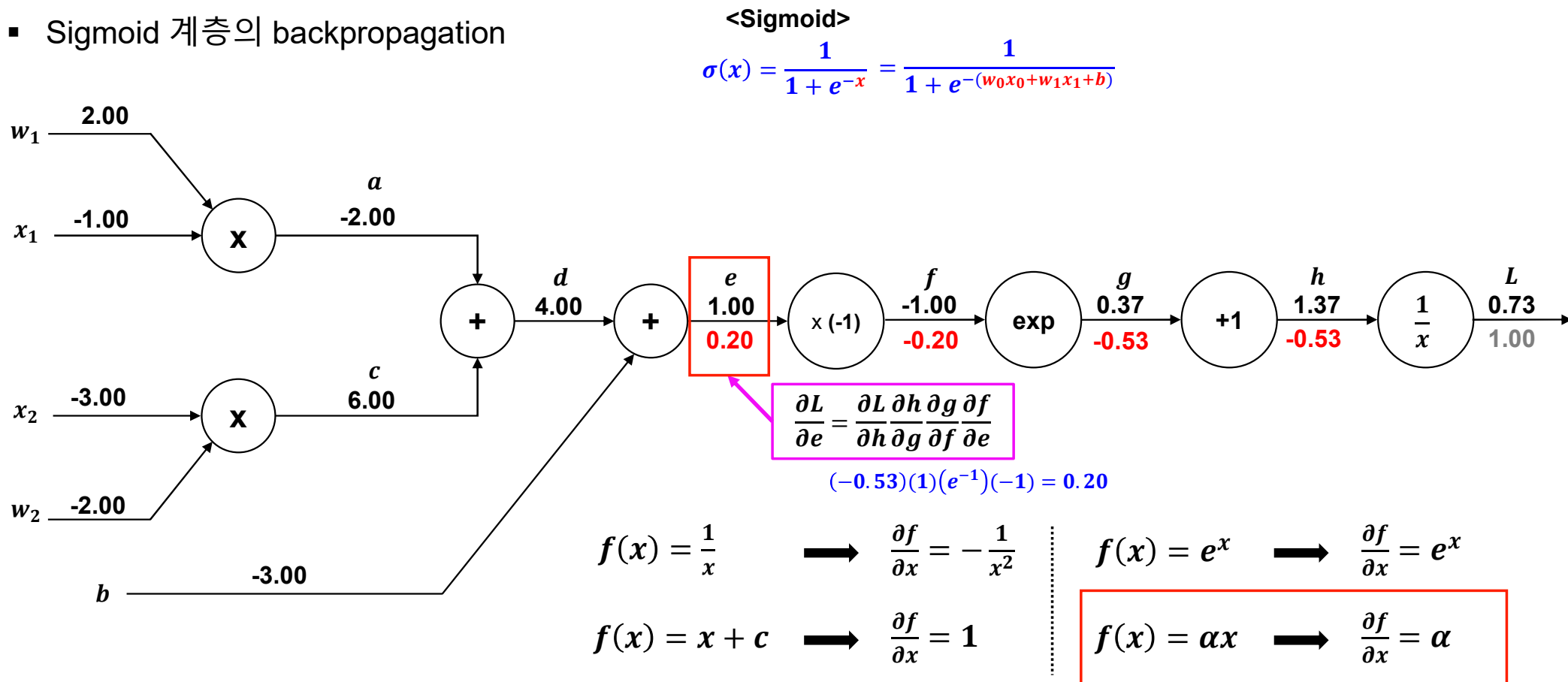
<Sigmoid>

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Backpropagation 동작원리

- Sigmoid 계층의 backpropagation

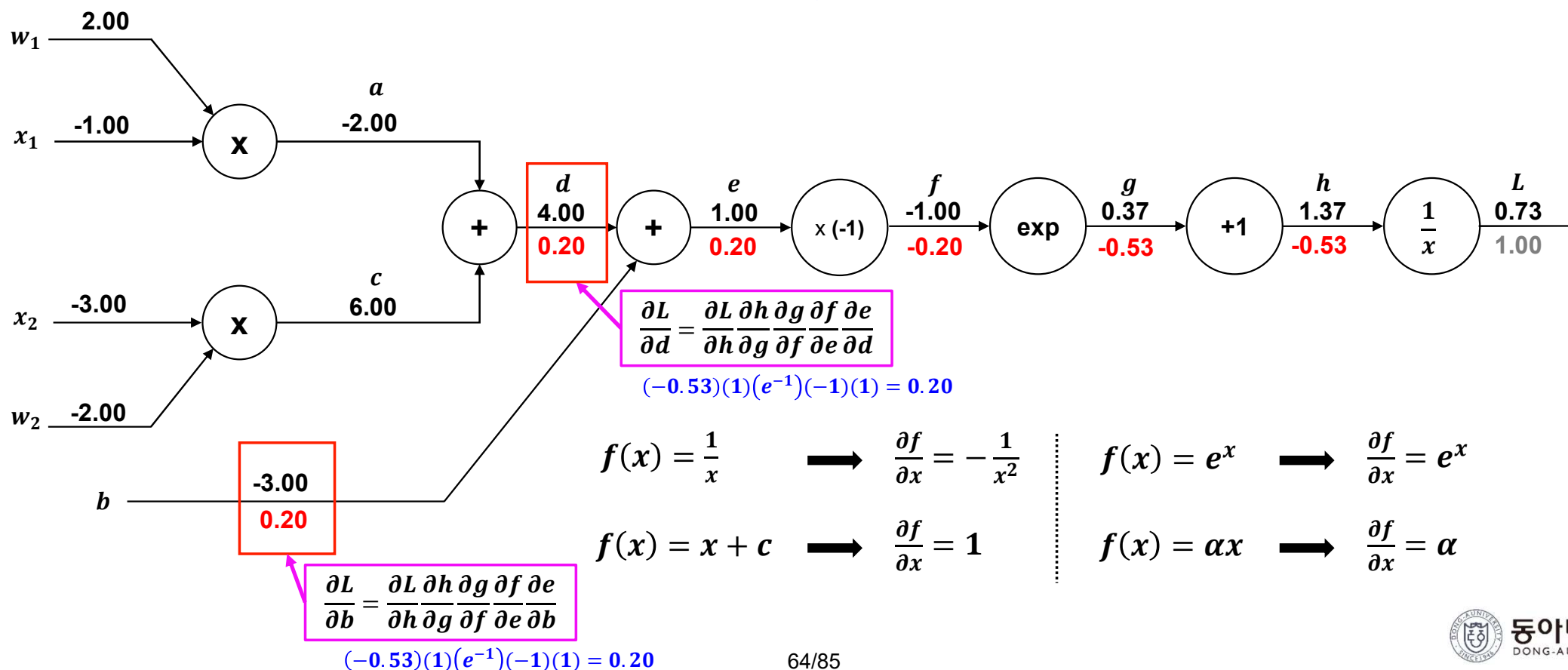


Backpropagation 동작원리

- Sigmoid 계층의 backpropagation

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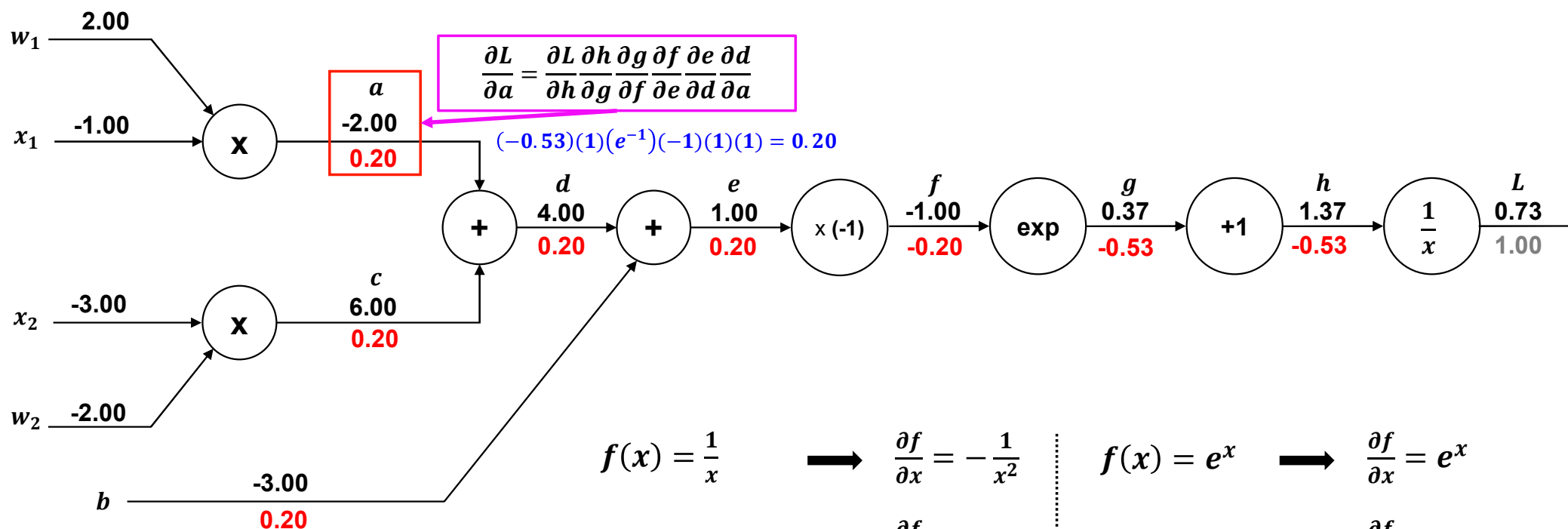


Backpropagation 동작원리

- Sigmoid 계층의 backpropagation

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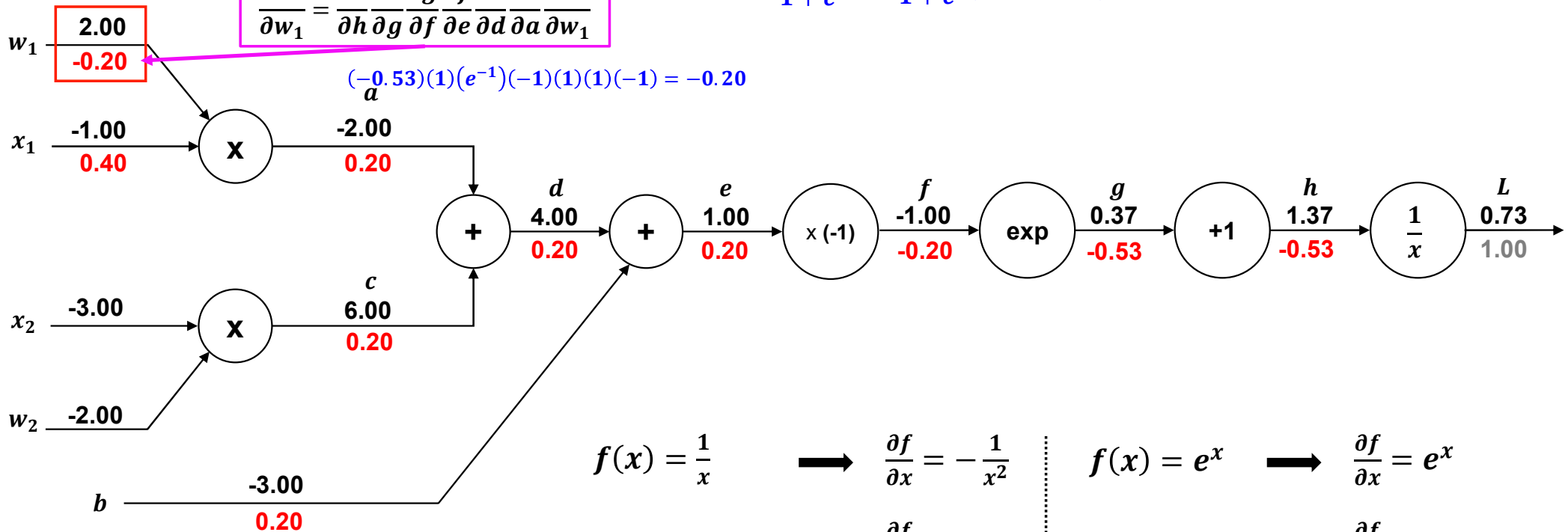
Backpropagation 동작원리

■ Sigmoid 계층의 backpropagation

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial h} \frac{\partial h}{\partial g} \frac{\partial g}{\partial f} \frac{\partial f}{\partial e} \frac{\partial e}{\partial d} \frac{\partial d}{\partial a} \frac{\partial a}{\partial w_1}$$

<Sigmoid>

$$\sigma(x) = \frac{1}{1 + e^{-x}} = \frac{1}{1 + e^{-(w_0x_0 + w_1x_1 + b)}}$$



$$(-0.53)(1)(e^{-1})(-1)(1)(1)(-1) = -0.20$$

$$f(x) = \frac{1}{x} \longrightarrow \frac{\partial f}{\partial x} = -\frac{1}{x^2}$$

$$f(x) = x + c \longrightarrow \frac{\partial f}{\partial x} = 1$$

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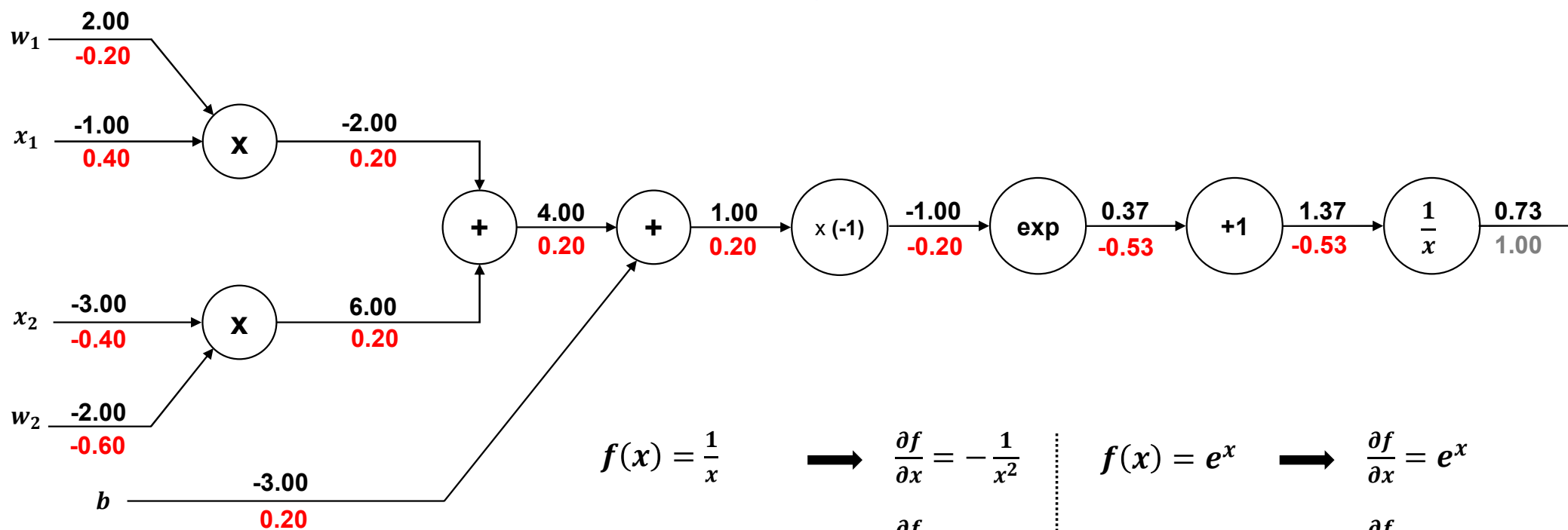
$$f(x) = \alpha x \longrightarrow \frac{\partial f}{\partial x} = \alpha$$

Backpropagation 동작원리

- Sigmoid 계층의 backpropagation

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$$f(x) = e^x \quad \longrightarrow \quad \frac{\partial f}{\partial x} = e^x$$

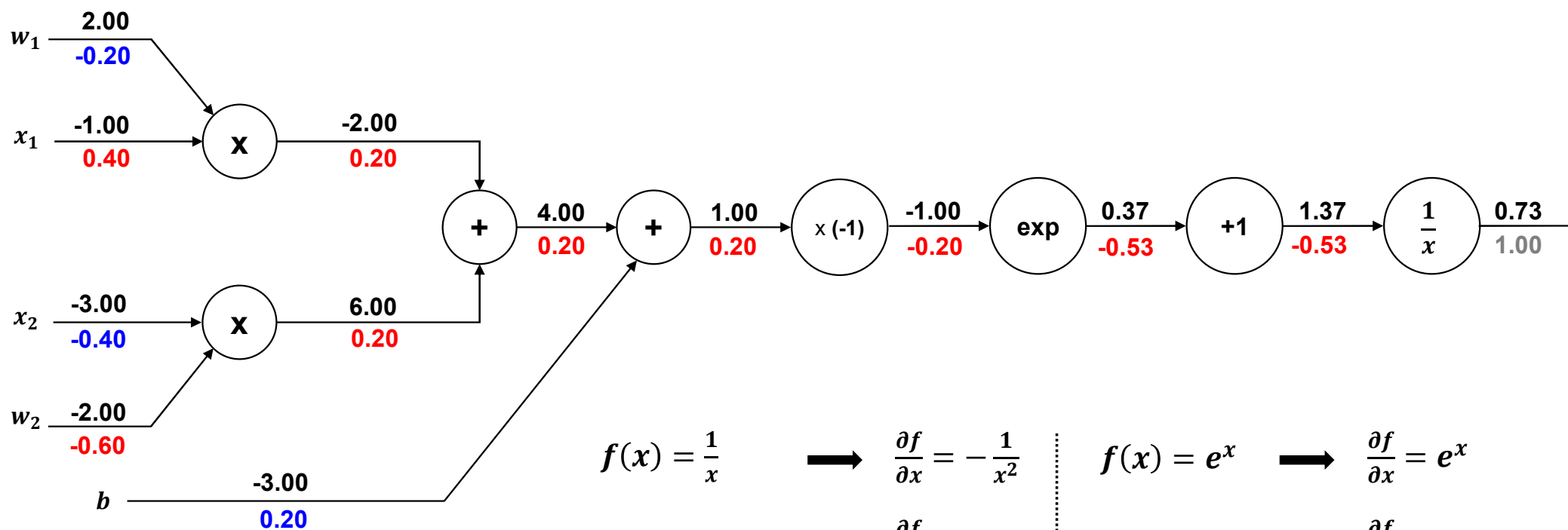
$$f(x) = \alpha x \quad \longrightarrow \quad \frac{\partial f}{\partial x} = \alpha$$

Backpropagation 동작원리

- Sigmoid 계층의 backpropagation

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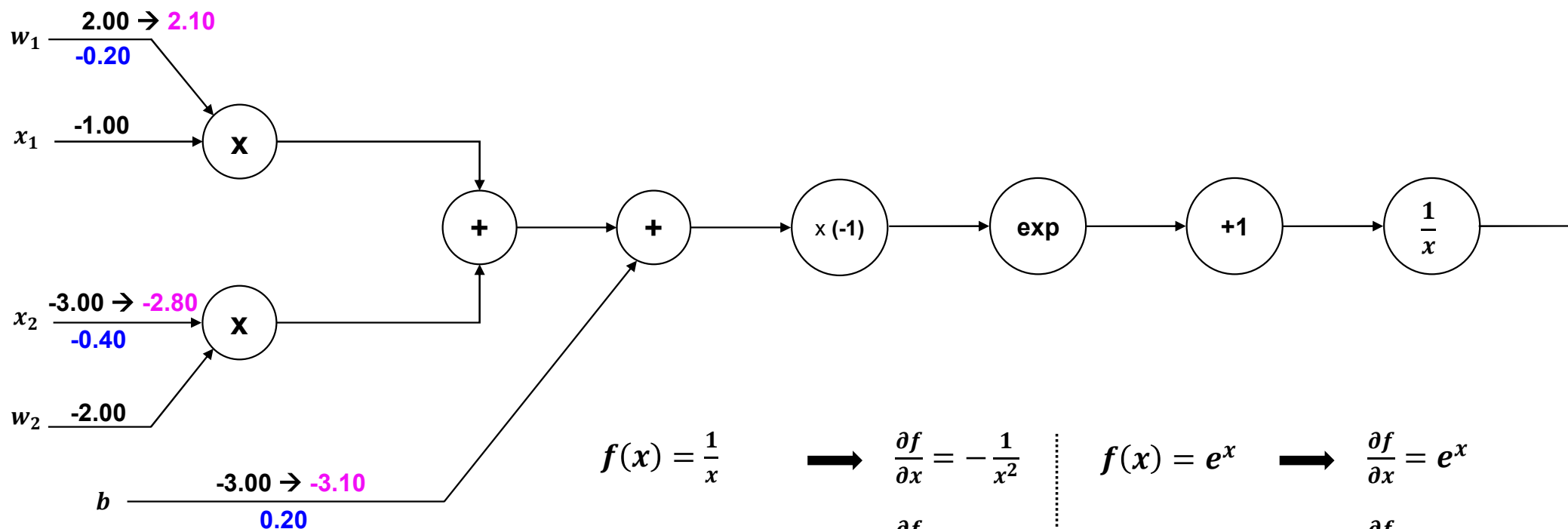
$$f(x) = e^x \longrightarrow \frac{\partial f}{\partial x} = e^x$$

$$f(x) = \alpha x \longrightarrow \frac{\partial f}{\partial x} = \alpha$$

$$w^{t+1} \leftarrow w^t - \mu \nabla f(w^t)$$

Backpropagation 동작원리

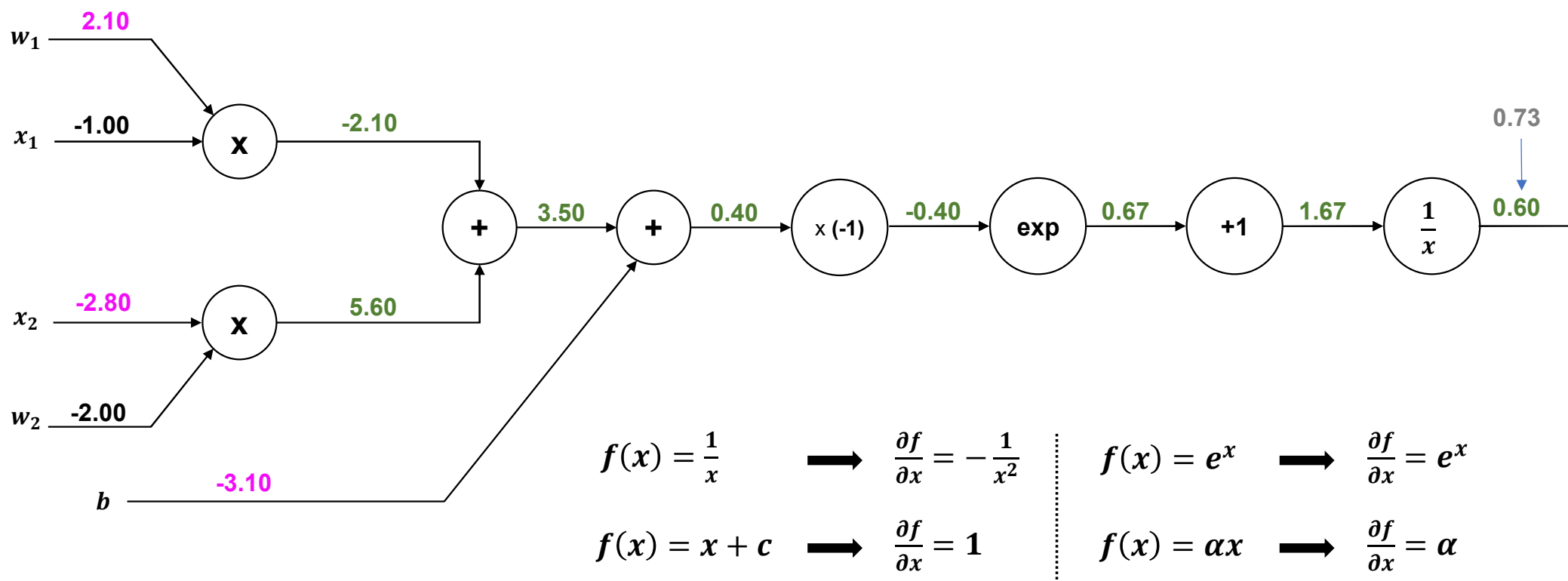
- Sigmoid 계층의 backpropagation (Ex., Learning Rate = 0.5)



$$w^{t+1} \leftarrow w^t - \mu \nabla f(w^t)$$

Backpropagation 동작원리

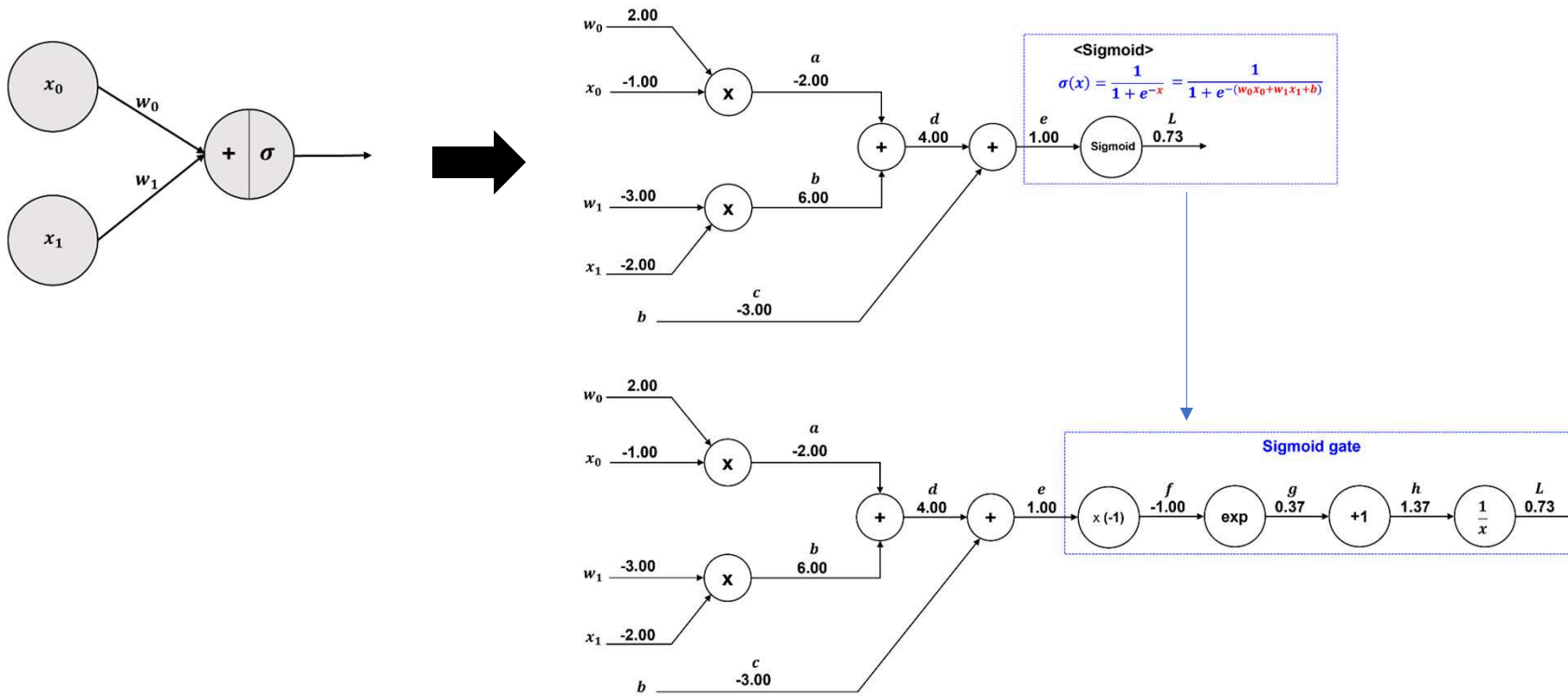
- Sigmoid 계층의 backpropagation (Ex., Learning Rate = 0.5)



Backpropagation 동작원리

- Single Layer Perceptron (SLP)

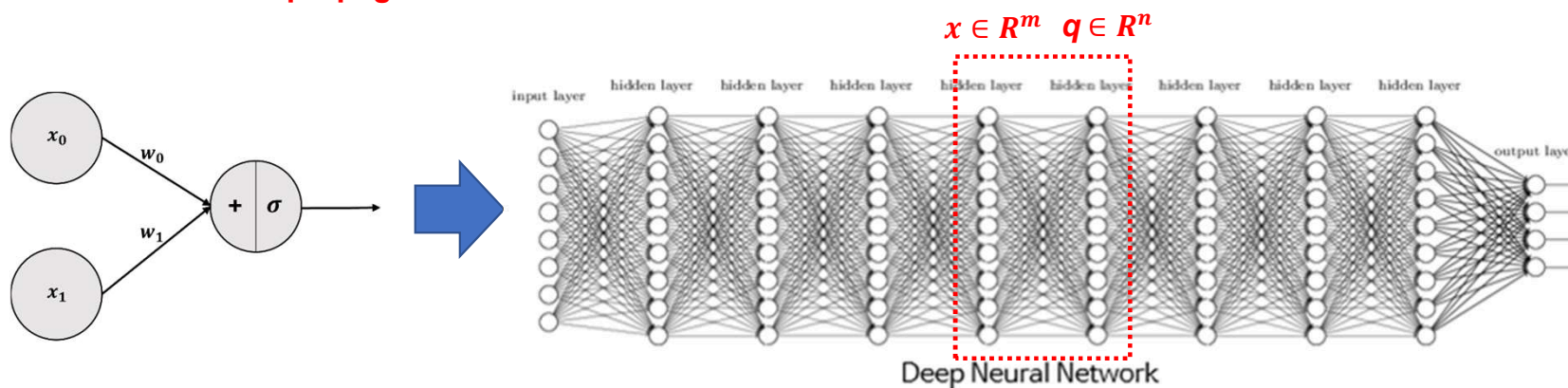
Forward propagation (pass)



Backpropagation 동작원리

■ SLP → Deep Neural Network (MLP)

→ 행렬에 대한 Backpropagation



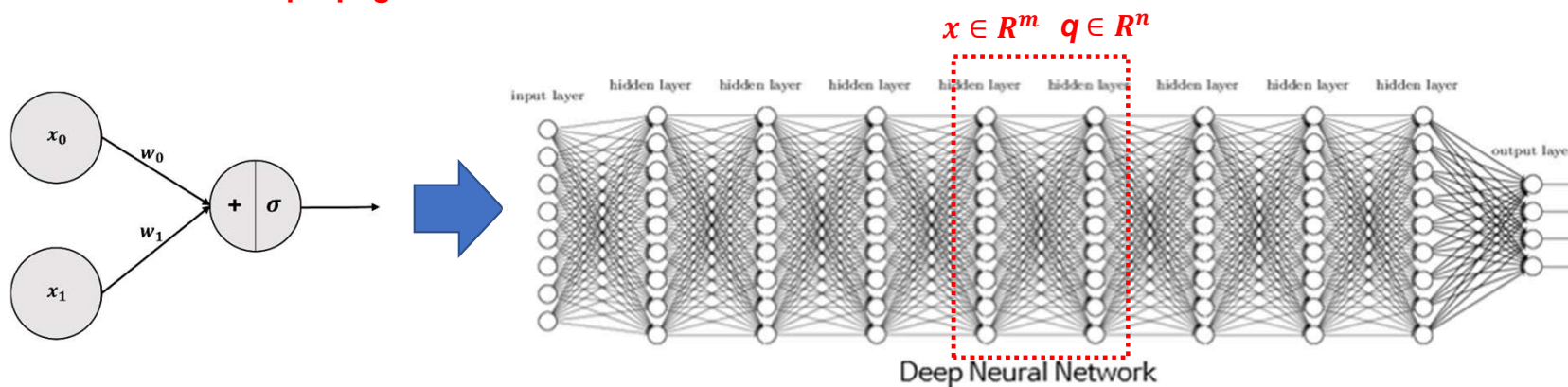
$$W \in R^{m \times n} = \begin{pmatrix} w_{11} & \cdots & w_{1n} \\ \vdots & \ddots & \vdots \\ w_{m1} & \cdots & w_{mn} \end{pmatrix}, \quad W^T \in R^{n \times m} = \begin{pmatrix} w_{11} & \cdots & w_{m1} \\ \vdots & \ddots & \vdots \\ w_{1n} & \cdots & w_{mn} \end{pmatrix}, \quad x \in R^m = \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix}, \quad b \in R^n = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$$

$$q = \sigma(W^T \cdot x + b) = \sigma \left\{ \begin{pmatrix} W_{1,1}x_1 + \cdots + W_{m,1}x_m \\ \vdots \\ W_{1,n}x_1 + \cdots + W_{m,n}x_m \end{pmatrix} + \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \right\} = \begin{pmatrix} q_1 \\ \vdots \\ q_n \end{pmatrix} \in R^n$$

Backpropagation 동작원리

- SLP → Deep Neural Network (MLP)

→ 행렬에 대한 Backpropagation



$$W \in R^{m \times n} = \begin{pmatrix} w_{11} & \cdots & w_{1n} \\ \vdots & \ddots & \vdots \\ w_{m1} & \cdots & w_{mn} \end{pmatrix}, \quad W^T \in R^{n \times m} = \begin{pmatrix} w_{11} & \cdots & w_{m1} \\ \vdots & \ddots & \vdots \\ w_{1n} & \cdots & w_{mn} \end{pmatrix}, \quad x \in R^m = \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix}, \quad b \in R^n = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$$

$$y = \sigma(W^T \cdot x + b) = \sigma \left\{ \begin{pmatrix} W_{1,1}x_1 + \cdots + W_{m,1}x_m \\ \vdots \\ W_{1,n}x_1 + \cdots + W_{m,n}x_m \end{pmatrix} + \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \right\} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$$

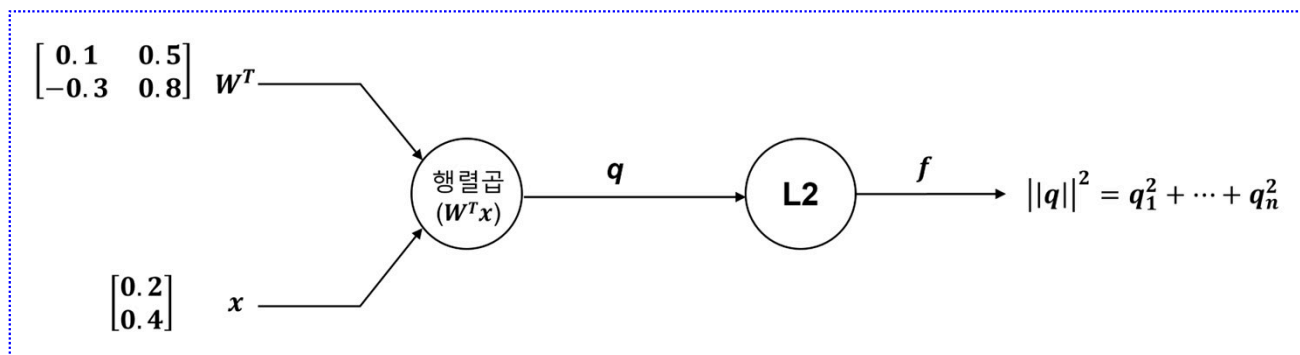
Backpropagation 동작원리

- 행렬의 Backpropagation (Example)

$$q = \sigma(W^T \cdot x + b) = \sigma \left\{ \begin{pmatrix} W_{1,1}x_1 + \cdots + W_{m,1}x_m \\ \vdots \\ W_{1,n}x_1 + \cdots + W_{m,n}x_m \end{pmatrix} + \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \right\} = \begin{pmatrix} q_1 \\ \vdots \\ q_n \end{pmatrix}$$

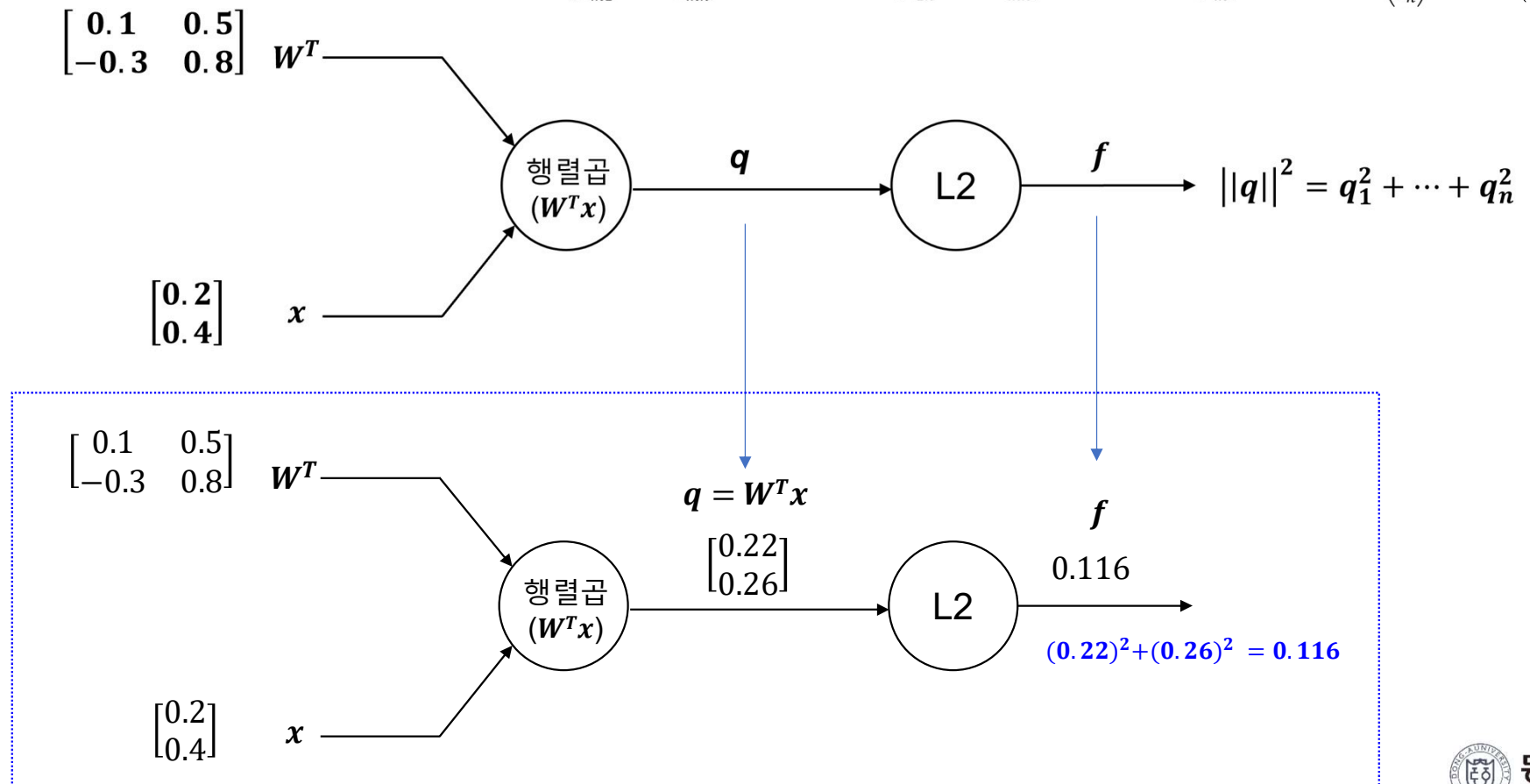
↓ 간소화

$$q = W^T \cdot x = \begin{pmatrix} W_{1,1}x_1 + \cdots + W_{m,1}x_m \\ \vdots \\ W_{1,n}x_1 + \cdots + W_{m,n}x_m \end{pmatrix} = \begin{pmatrix} q_1 \\ \vdots \\ q_n \end{pmatrix}$$



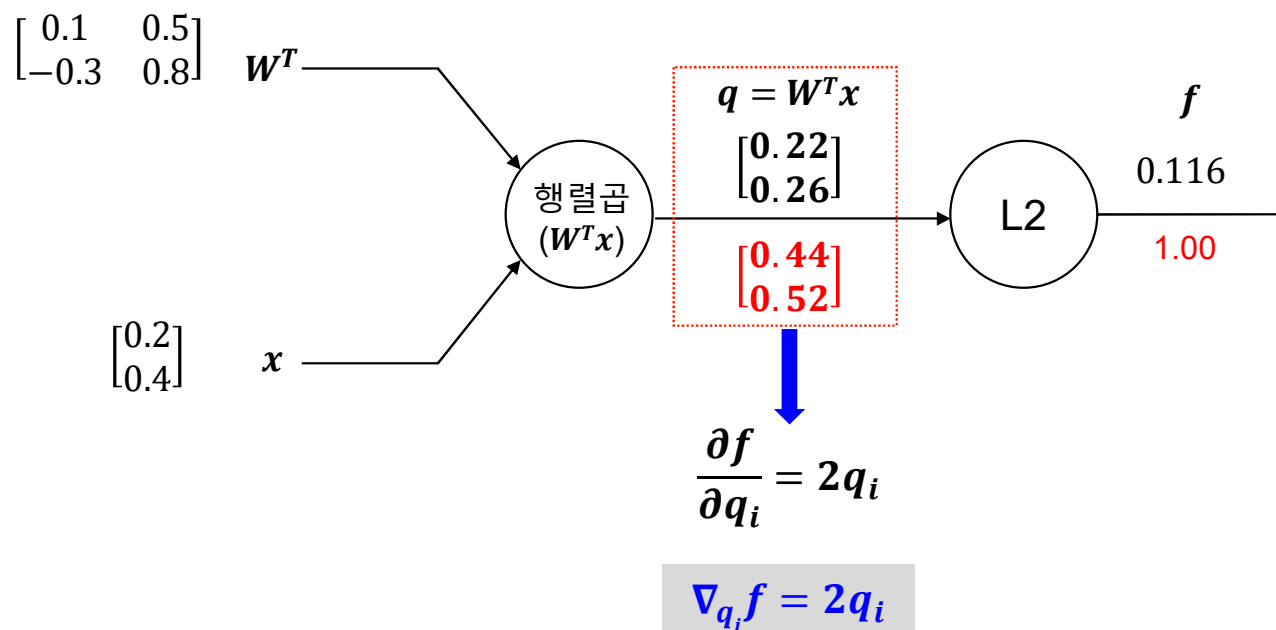
Backpropagation 동작원리

- 행렬의 Backpropagation (Example) $W \in R^{m \times n} = \begin{pmatrix} w_{11} & \cdots & w_{1n} \\ \vdots & \ddots & \vdots \\ w_{m1} & \cdots & w_{mn} \end{pmatrix}, \quad W^T \in R^{n \times m} = \begin{pmatrix} w_{11} & \cdots & w_{m1} \\ \vdots & \ddots & \vdots \\ w_{1n} & \cdots & w_{mn} \end{pmatrix}, \quad x \in R^m = \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix}, \quad b \in R^n = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}, \quad q = \begin{pmatrix} q_1 \\ \vdots \\ q_n \end{pmatrix} \in R^n$



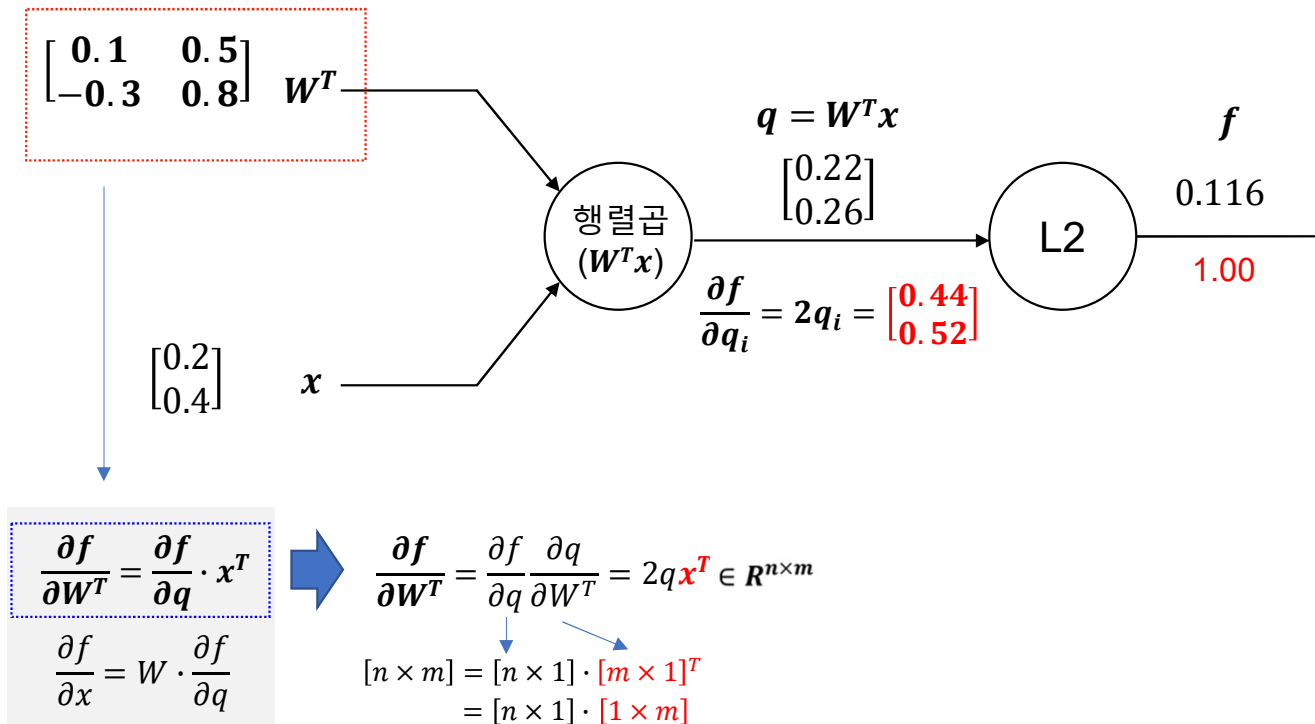
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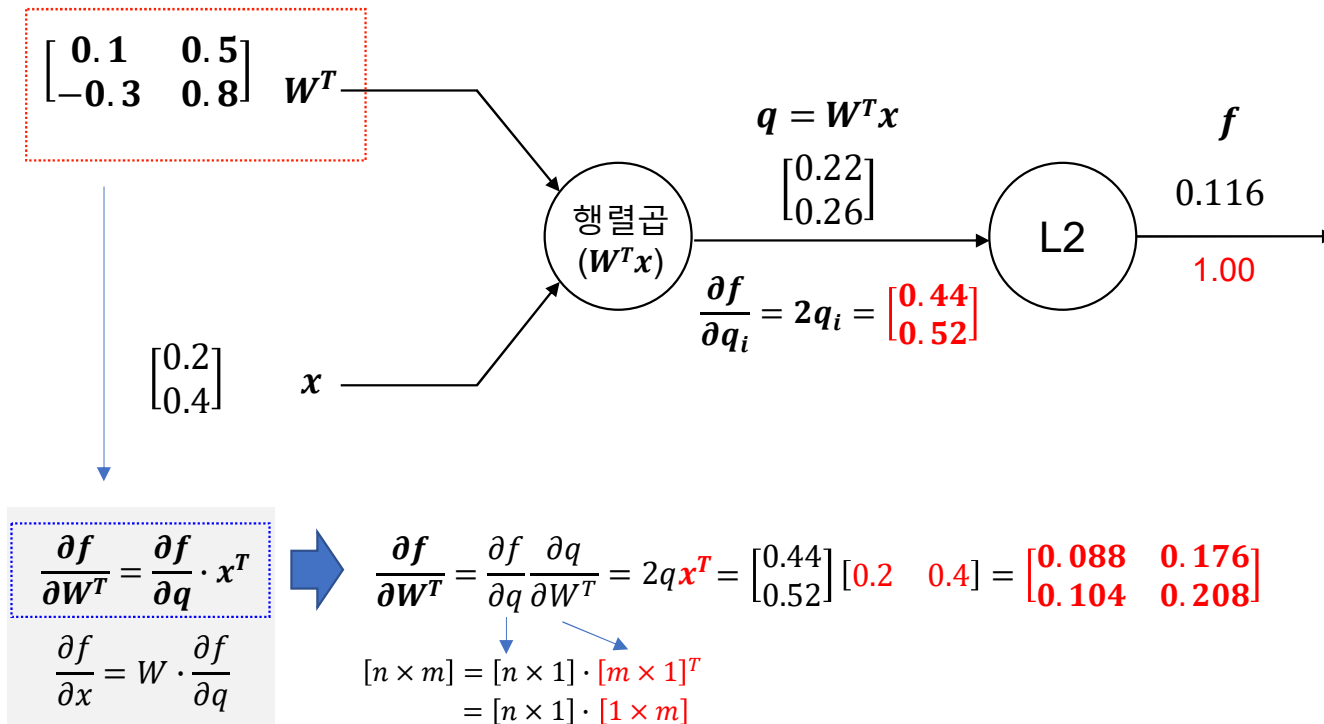
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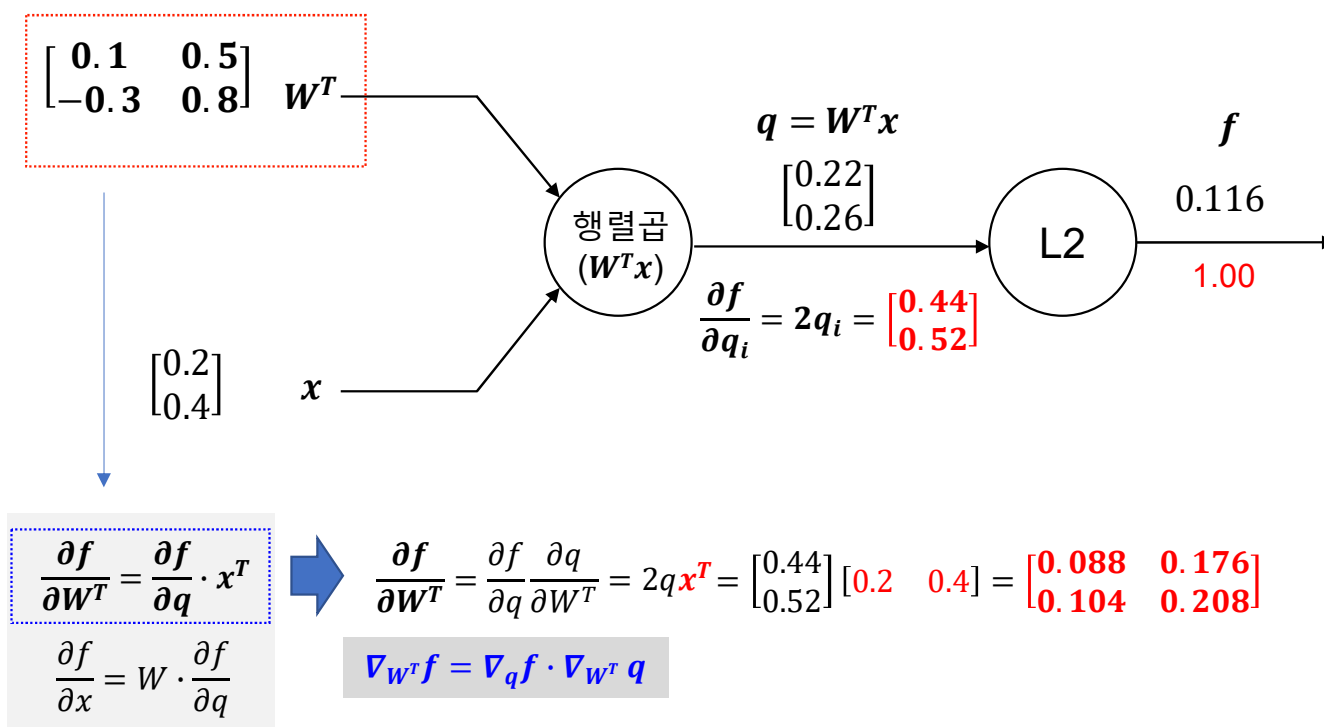
Backpropagation 동작원리

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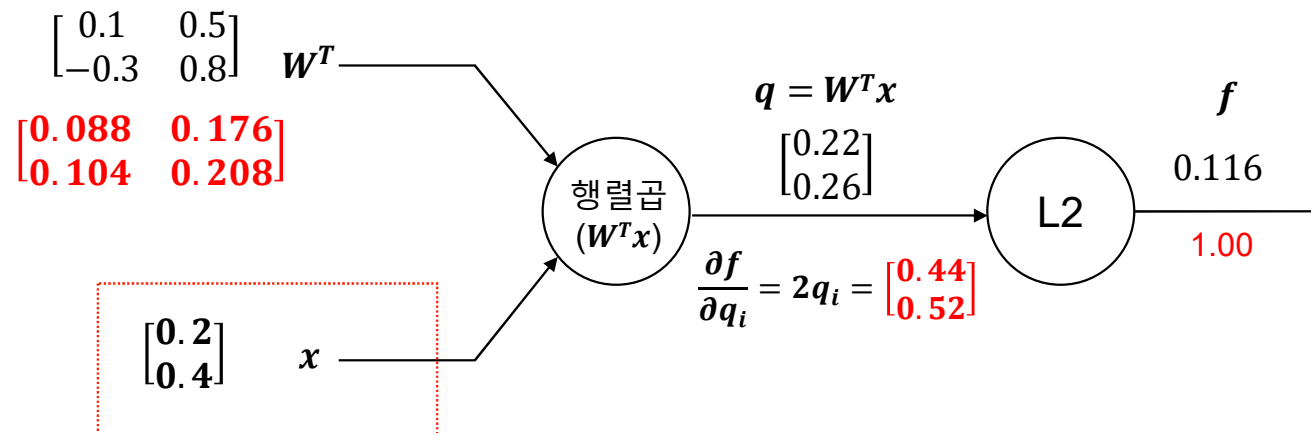
Backpropagation 동작원리

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Backpropagation 동작원리

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$$\frac{\partial f}{\partial W^T} = \frac{\partial f}{\partial q} \cdot x^T$$

$$\frac{\partial f}{\partial x} = W \cdot \frac{\partial f}{\partial q}$$

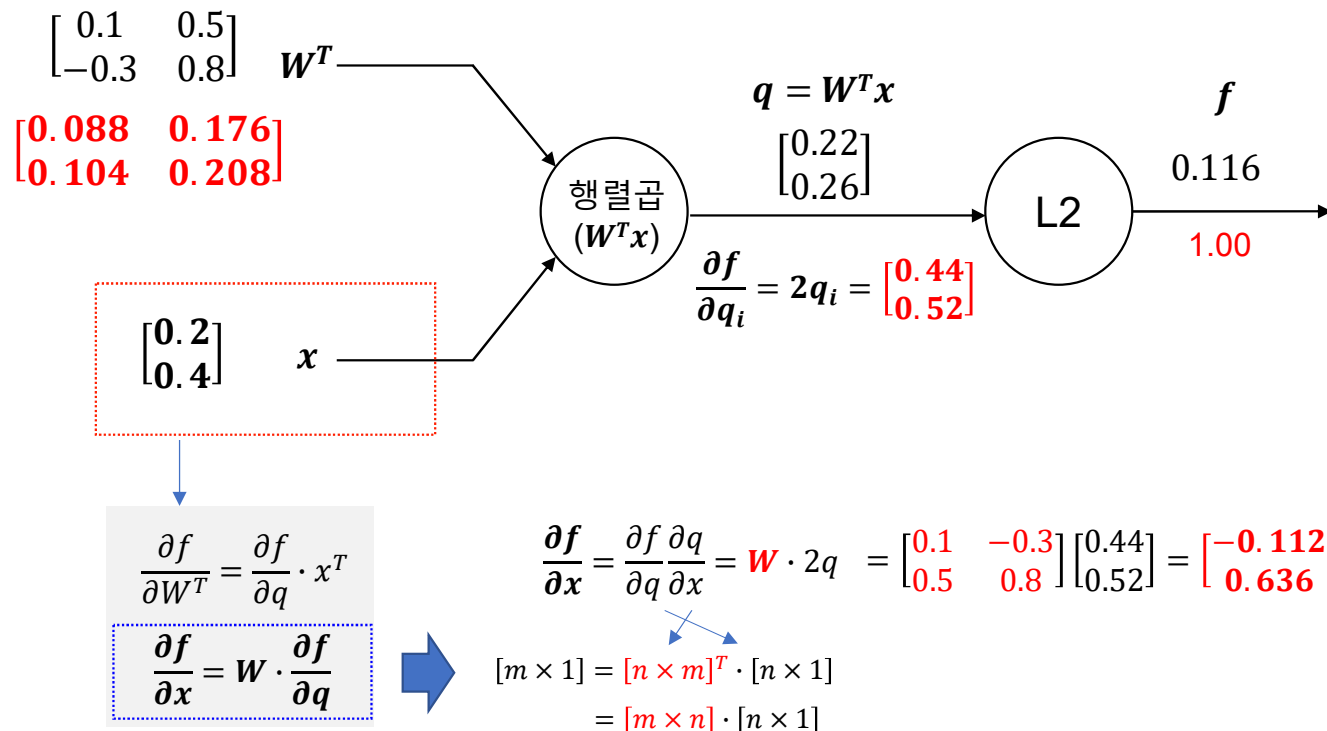
$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial q} \frac{\partial q}{\partial x} = (W^T)^T \cdot 2q \in R^m$$

$$[m \times 1] = [n \times m]^T \cdot [n \times 1]$$

$$= [m \times n] \cdot [n \times 1]$$

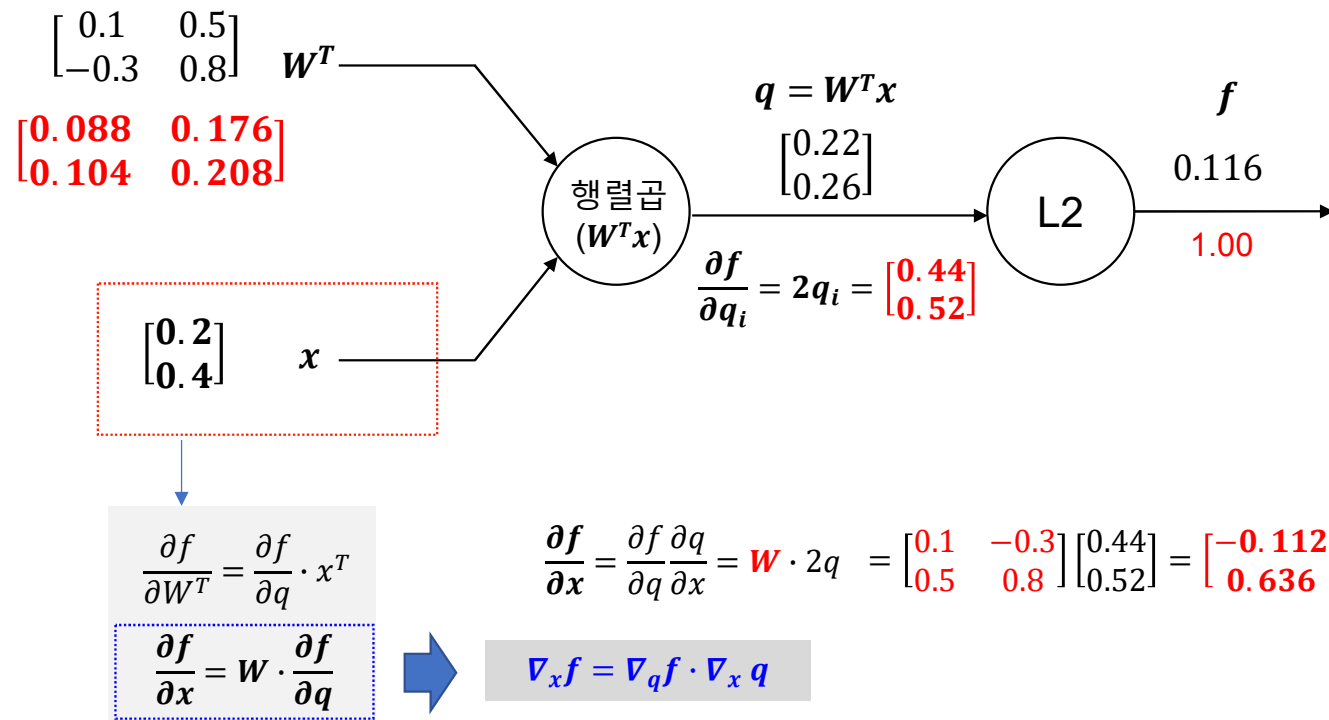
Backpropagation 동작원리

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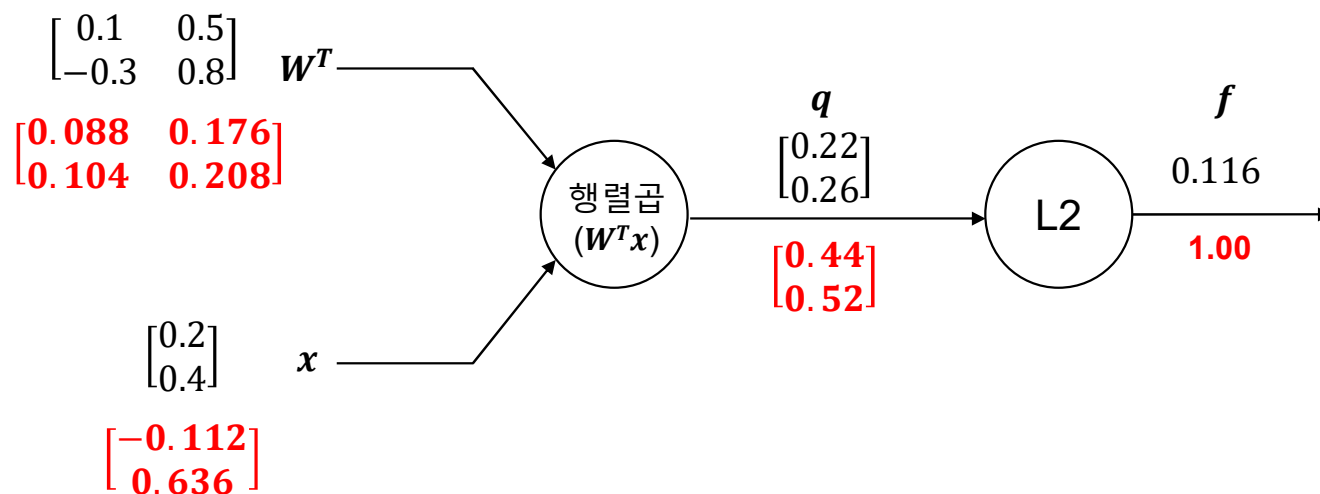
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Backpropagation 동작원리

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$$w^{t+1} \leftarrow w^t - \mu \nabla f(w^t)$$

Descent Method(하강법)

- 주어진 어떤 지점에서부터 오차가 더 작은 곳으로 이동하려는 방법
 - 탐색 방법 Δw 에 따라 하강법이 결정
 - 종류: 경사하강법, 모멘텀 등 → **Optimization(최적화)**

$$w^{t+1} \leftarrow w^t - \underset{\substack{\text{학습률(learning rate, step size)}}}{\mu} \nabla L(w^t), \text{ where } \nabla L(w^t) \Delta w < 0$$

↖
탐색 방향

Questions & Answers

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